جامعة الريادة للعلوم والتكنولوجيا

كلية الهندسة

Faculty of Engineering

اللائحة الداخلية لمرحلة البكالوريوس

بنظام الساعات المعتمدة

2024

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الباب الأول

مقدمة

كليه الهندسة بجامعة الريادة للعلوم والتكنولوجيا هي إحدى كليات الجامعة والتي تقع بالمحور المركزي بمدينة السادات.

مادة (1): رؤية الكلية

تسعى كلية الهندسة بجامعة الريادة للعلوم والتكنولوجيا (RST) أن تكون مركزًا رائدًا في التعليم الهندسي والبحث العلمي، وذات أثر إيجابي مستدام على التنمية والاقتصاد الوطني.

مادة (2): رسالة الكلية

تلتزم كلية الهندسة في جامعة الريادة للعلوم والتكنولوجيا (RST) بتقديم خدمات تعليمية وبحثية متميزة، وتخريج كوادر هندسية محترفة وباحثين مؤهلين، ملتزمين بالأخلاقيات المهنية والقيم المجتمعية، يساهمون في تطوير المجتمع والبيئة المحيطة من خلال تقديم حلول مبتكرة لمشكلات المجتمع والصناعة.

مادة (3): القيم المحورية

تلتزم جامعة الريادة للعلوم والتكنولوجيا وكلياتها المختلفة بمجموعة من القيم المحورية التي تحكم منظومة الأداء وبيئة العمل داخل الجامعة وتستند اليها جميع السياسات والإجراءات والأنشطة الأكاديمية والإدارية والمالية بالجامعة، وأهم القيم ما يلي:

- الأمانة والنزاهة Honesty and Integrity
- الشفافية والمساءلة Transparency and Accountability
- التعاون والتواصل Collaboration and Communication
- التطوير المستمر Continuous Improvement
- حرية الفكر والتعبير Freedom of thought and Expression
- احترام الأفراد Respect for the Individuals

مادة (4): اقسام الكلية : تتضمن كلية الهندسة بجامعة الريادة للعلوم و التكنولوجيا الاقسام العلمية الاتية :

1- قسم الهندسة الميكانيكية (Mechanical Engineering Department)

2- قسم الهندسة الكهربائية (Electrical Engineering Department)

3- قسم العلوم الاساسية (Basic Science Department)

و يقوم كل قسم من الأقسام بإجراء الابحاث العلمية وتدريس المقررات لجميع البرامج و التى تقع فى إطار تخصصة و التى تحمل كود القسم. و القسم مسئول عن المحتوى العلمى للمقرر الدراسى و تحديثة و كذلك ترشيح القائم بالتدريس لكل مقرر دراسى سواء من اعضاء هيئة التدريس بالقسم ، من اعضاء هيئة التدريس بقسم اخر ، او من خارج الكلية . حيث يكون تكليف القائمين بالتدريس من خارج الكلية بعد موافقة لجنة شئون التعليم و الطلاب بالكلية.

مادة (5): برامج الكلية

تطرح كلية الهندسة ثلاثة برامج متنوعة منها برنامج رئيسي متخصص وبرنامجين رئيسين بينيين وهم على النحو التالي:

م	اسم البرنامج	نوع البرنامج
1	برنامج هندسة الأتمتة والروبوتات Automation and Robotics Engineering (ARE) Program	رئيسي بيني Major Inter-Disciplinary
2	برنامج هندسة ميكاترونيات السيارات Automative Mechatronics Engineering (AME) Program	رئيسي متخصص Major Disciplinary
3	برنامج هندسة الطاقة المتجددة والمستدامة Renewable and Sustainable Energy Engineering (RSE) Program	رئيسي بيني Major Inter-Disciplinary

مادة (6): الدرجات العلمية التى تمنحها كلية الهندسة

1. يمنح مجلس جامعة الريادة للعلوم والتكنولوجيا بناء على طلب مجلس كلية الهندسة درجة البكالوريوس في الهندسة بنظام الساعات المعتمدة على النحو التالى :

1- بكالوريوس العلوم فى الهندسة الميكانيكية :

- برنامج هندسة الأتمتة والروبوتات Automation and Robotics Engineering
- برنامج هندسة ميكاترونيات السيارات Automotive Mechatronics Engineering

2- بكالوريوس العلوم فى الهندسة الكهربائية:

- برنامج هندسة الطاقة المتجددة والمستدامة Renewable and Sustainable Energy Engineering

2. للكلية الحق في اضافة درجات علمية مشتركة مع جامعات عربية او أجنبية مع مراعات القواعد القانونية والضوابط الصادرة في هذا الشأن من مجلس الجامعات الخاصة ولجنة القطاع بالمجلس الأعلى للجامعات.

3. للكلية الحق في اضافة برامج علمية خاصة بمرحلة الدراسات العليا وفقاً لم يقرره مجلس الكلية والجامعة (ماجستير العلوم – دكتوراه الفلسفة – درجات مهنية) في ضوء القواعد القانونية والضوابط الصادرة في هذا الشأن من مجلس الجامعات الخاصة ولجنة القطاع بالمجلس الأعلى للجامعات.

الباب الثاني

الهيكل التنظيمي

مادة (7): هيكلة المقررات الدراسية

تهدف هيكلة المقررات الدراسية بكلية الهندسة إلى تنظيم المنهج الدراسي وتوفير تجربة تعليمية متكاملة للطلاب. تم تقسيم المقررات إلى مستويات مختلفة، وتتضمن هذه المستويات متطلبات مختلفة تستند إلى الإطار المرجعي المصري بما يحقق رؤية ورسالة واهداف الكلية وبرامجها المختلفة وتشمل:

المستوى (أ): مقررات دراسية تحقق متطلبات الجامعة.

المستوى (ب): مقررات دراسية تحقق متطلبات الكلية.

المستوى (ج): مقررات دراسية تحقق متطلبات البرامج.

المستوى (د): مقررات دراسية تحقق متطلبات التخصص الدقيق.

تجدر الاشارة الى أهمية أن يكون للطلاب فهم جيد لهيكلة المقررات والمتطلبات الدراسية المطلوبة في كلية الهندسة، حيث يساعدهم ذلك في التخطيط لمسارهم الأكاديمي والتأكد من استيفاء جميع المتطلبات اللازمة للتخرج بنجاح.

مادة (8): متطلبات الجامعة

تشترك كلية الهندسة مع باقي كليات جامعة الريادة للعلوم والتكنولوجيا في عدد (6) مقررات دراسية بواقع (12) ساعة معتمدة وتكافئ (16) نقطة اوروبية معتمدة والتي تمثل متطلبات الجامعة والتي تهدف الى بناء شخصية الخريج الثقافية، وتنمية مهارات الشخصية، والادراك العام بقضايا المجتمع، والتركيز على الهوية والارتباط بالوطن.

مادة (9): متطلبات الكلية

تشترك برامج كلية الهندسة في عدد (12) مقرر دراسي بواقع (31) ساعة معتمدة وتكافئ (51) نقطة اوروبية معتمدة والتي تمثل متطلبات الكلية وهذه المقررات تمثل الحد الأدنى من المعارف المطلوب توافرها لدى الخريج

من العلوم الاساسية، الثقافة الهندسية، والعلوم الهندسية حول كافة التخصصات بحيث تتيح له الحد الأدنى من القدرة على التواصل مع المهندسين في تخصصات اخرى والاندماج في مشروعات متعددة التخصص ليكون عضو فعال.

مادة (10): عدد الساعات المعتمدة لبرامج الكلية

1. يقوم كل برنامج من برامج كلية الهندسة بجامعة الريادة للعلوم والتكنولوجيا بتدريس عدد من المقررات بواقع (102) ساعة معتمدة وتكافئ (173) نقطة اوروبية معتمدة وتنقسم هذه الساعات بين متطلبات التخصص العام ومتطلبات التخصص الدقيق.
2. يحتوي تصميم برنامج هندسة ميكاترونيات السيارات على عدد من المقررات (20 مقرر دراسي) بنسبة 42% من إجمالي الساعات وهو ما يعادل (60) ساعة معتمدة وتكافئ (101) نقطة اوروبية معتمدة كمتطلب التخصص العام وايضا عدد من المقررات (17 مقرر دراسي) بنسبة 29% من إجمالي الساعات وهو ما يعادل (42) ساعة معتمدة وتكافئ (72) نقطة اوروبية معتمدة كمتطلب تخصص دقيق.
3. يحتوي تصميم كلا من برنامجي هندسة الأتمتة والروبوتات وهندسة الطاقة المتجددة والمستدامة على عدد من المقررات (37 مقرر دراسي) وهو يعادل (102) ساعة معتمدة وتكافئ (173) نقطة اوروبية معتمدة، حيث تمثل هذه المقررات متطلبات البرنامج وذلك لكونهما برامج بينية.

مادة (11): متطلبات العلوم الاساسية

كما يحتوي تصميم البرنامج الدراسي على نسبة من مقررات العلوم الأساسية (الرياضيات، الفيزياء، الكيمياء) بنسبة 19% من إجمالي الساعات بواقع (9) مقررات دراسية وهو ما يعادل (27) ساعة معتمدة وتكافئ (45) بالنظام الأوروبي.

ويوضح الجدول التالي توزيع المقررات في كلية الهندسة بجامعة الريادة للعلوم والتكنولوجيا على مستوى المتطلبات المختلفة مع النسب المقررة في المعايير الأكاديمية المرجعية (NARS 2018) والإطار المرجعي لإعداد البرامج الدراسية لمرحلة البكالوريوس بكليات الهندسة (2022) والصادر من لجنة القطاع الهندسي بالمجلس الاعلى للجامعات بتاريخ 2022/12/17.

#	Pro.	NC	Weight			Total Contacts Hours				Requirements Distribution %				BS %	EC %
			CH	ECTS	SWL	Lec	Tut	Lab	TT	UR	FR	DR	PR		
1	ARE	55	145	240	6000	98	58	51	207	8 %	21 %	71 %		19 %	16 %
2	AME	55	145	240	6000	99	62	42	203	8 %	21 %	41 %	30 %	19 %	16 %
3	RSE	55	145	240	6000	100	73	29	202	8 %	21 %	71 %		19 %	16 %

NC	**Total number of Courses**	**UR**	**University Requirement**
CH	**Credit Hour**	**FR**	**Faculty Requirement**
ECTS	**European Credit Transfer System**	**DR**	**Discipline Requirement**
SWL	**Student Workload**	**PR**	**Program Requirement**
Lec	**Lectures**	**BS**	**Basic Sciences Percentage**
Tut	**Tutorials**	**EC**	**Elective Courses Percentage**
Lab	**Laboratory**		
TT	**Total**		

الباب الثالث

القواعد المنظمة للدراسة في البرامج

مادة (12): شروط وقواعد القبول

1. **الطلاب الحاصلين على الثانوية العامة المصرية: ويشترط لقبولهم ما يلى:**

- أن يكون المتقدم حاصلا على الثانوية العامة المصرية (علمي رياضة) أو ما يعادلها من الشهادات الأخرى، في نفس العام الدراسي للالتحاق بالكلية وذلك طبقا لشروط التنسيق بالمجلس الأعلى للجامعات الخاصة والأهلية.
- طلاب الشهادات المعادلة الانجليزي والامريكي والبلاد العربية المعادلة طبقا لقواعد مجلس لجامعات الخاصة والمجلس الاعلى للجامعات.
- يضع مجلس كلية الهندسة قواعد عامة للقبول في البرامج مع مراعاة رغبات الطلاب ومبدأ تكافؤ الفرصة كأساس لقبول الطلاب في البرامج المطروحة.

2. **الطلاب الوافدون:**

تقبل الكلية الطلاب الوافدين الحاصلين على شهادة تعادل شهادة الثانوية العامة المصرية (علمي رياضة) طبقا للفقرة السابقة.

3. **الطلاب المقيدون بإحدى الكليات بالجامعات المصرية أو الأجنبية:**

يجوز للكلية قبول تحويل الطلاب المقيدين بإحدى الكليات المناظرة بالجامعات المصرية، أو بالجامعات أو المعاهد الأجنبية التي يعترف بها المجلس الأعلى للجامعات بشرط:

- أن يكون المتقدم حاصلا على الثانوية العامة المصرية (علمي رياضة) أو ما يعادلها من شهادات أجنبية.
- عمل مقاصة لفحص المحتوي العلمي للمقررات التي درسها الطالب في الكلية المحول منها ونجح فيها بمعدل GPA لا يقل عن 2.

- أن يخضع المتقدم لشروط ومتطلبات التخرج المقررة في كلية الهندسة بجامعة الريادة للعلوم والتكنولوجيا.

4. <u>الطلاب الحاصلين على مؤهلات عليا:</u>

يجوز للكلية قبول الطلاب الحاصلين على دراسات عليا ومؤهلات عليا (بكالوريوس أو ما يعادله) بشرط أن يكون حاصلا عل الثانوية العامة المصرية (علمي رياضة) أو ما يعادلها من شهادات أجنبية وألا يكون مضي على حصوله عليها أكثر من خمس سنوات، وفي جميع الأحوال لا يتم القبول إلا بعد أن يجتاز المتقدم بنجاح اختبارات اللياقة الطبية والبدنية والاختبارات التي تحددها الكلية والجامعة، وفي حالة قبوله يعامل معاملة الطالب العادي.

مادة (13): نظام الدراسة بالبرامج

1. اللغة الإنجليزية هي اللغة المستخدمة في الدراسة والامتحانات والتقارير ببرامج الكلية.
2. تعتمد الدراسة في برامج مرحلة البكالوريوس على:

- نظام الساعات المعتمدة (CH) التي تمثل وحدة قياس دراسية لتحديد وزن كل مقرر بالنسبة للمقررات الأخرى، وهي تعادل محاضرة نظرية مدتها ساعة واحدة أسبوعيا لمدة فصل دراسي واحد، وتعادل تدريبات، أو دروس عملية، أو معملية، أو تطبيقات مدتها ساعتين الى 3 ساعات اسبوعيا لمدة فصل دراسي واحد مدته 15 اسبوعا، تنقسم ساعة الاتصال الواحدة الى 50 دقيقة تدريس فعلى و10 دقائق راحة.
- نظام الساعات المعتمدة اوروبيا (ECTS) والذي يعادل 25-30 ساعة من مجموع ساعات عمل الطالب (Student Workload "SWL") خلال الاسبوع الدراسي الواحد.

3. القيم المتوقعة لكل فصل دراسي:
   - ♦ من 16-19 ساعة معتمدة.
   - ♦ من 25-28 ساعة اتصال في الاسبوع لمدة 15 اسبوعا.
   - ♦ 750 ساعة SWL.
   - ♦ 30 وحدة ECTS.

4. الدراسة بالكلية عملية وتطلب الحضور ويجوز لمجلس الكلية بعد أخذ رأى مجلس البرنامج المختص وحسب طبيعة المقررات الدراسية أن يقرر تدريس مقرر أو أكثر بنمط التعليم الهجين، بحيث تكون الدراسة في المقرر بنسبة تتراوح ما بين 60% و70% وجها لوجه وتتراوح ما بين 30 و40% بنظام التعليم عن بعد، أو بأي نسبة أخري، وعلى أن يتم عرض ذلك على مجلس شنون التعليم والطلاب بالجامعة للموافقة عليه ورفعه إلى مجلس الجامعة لاعتماده وذلك بمالا يتعارض مع ما يقرره مجلس الجامعات الخاصة والأهلية في هذا الشأن.

مادة (14): مستويات الدراسة

يتم نقل الطالب من مستوى الى مستوى اعلى كلما استكمل 25% من متطلبات البرنامج (36 ساعة معتمدة) وتحدد حالة الطالب بنظام الدراسة على أساس نسبة عدد الساعات المعتمدة التي اجتازها الطالب بنجاح، وذلك طبقا للجدول التالي:

المستوى الدراسي	حالة الطالب بنظام الدراسة	نسبة الساعات المعتمدة التي اجتازها الطالب
1	Freshman (المستوى العام)	من 0% الى اقل من 25%
2	Sophomore (المستوى الاول)	من 25% الى اقل من 50%
3	Junior (المستوى الثاني)	من 50% الى اقل من 75%
4	Senior (المستوى الثالث)	من 75% الى اقل من 100%

مادة (15): الفصول الدراسية

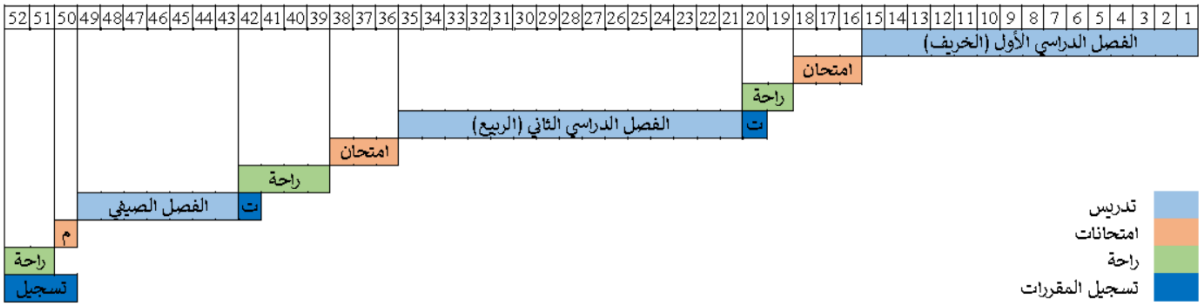
1. تتألف السنة الدراسية من ثلاث فصول دراسية:

- الفصل الدراسي الاول (الخريف)، يبدأ يوم السبت الثالث من شهر سبتمبر حيث يستمر لمدة 15 اسبوعا من التدريس تليها 3 اسابيع من الامتحانات. ويتم تسجيل المقررات في غضون ثلاثة اسابيع قبل بداية الفصل الدراسي بعد استيفاء شروط القيد ودفع الرسوم المقررة.

- الفصل الدراسي الثاني (الربيع)، يبدأ في شهر فبراير ويستمر لمدة 15 اسبوعا من التدريس تليها 3 اسابيع من الامتحانات. ويتم تسجيل المقررات في غضون اسبوع واحد قبل بداية الفصل الدراسي بعد استيفاء شروط القيد ودفع الرسوم المقررة.
- الفصل الدراسي الصيفي، يبدأ في شهر يوليو ويستمر لمدة 7 اسابيع تليها اسبوع واحد من الامتحانات. ويتم تسجيل المقررات في غضون اسبوع واحد قبل بداية الفصل الدراسي بعد استيفاء شروط القيد ودفع الرسوم المقررة.

2. يتم اعتماد التقويم الأكاديمي التفصيلي من مجلس الكلية وإعلانه للطلاب على موقع الكلية وذلك قبل بداية العام الأكاديمي.
3. تسجيل المقررات ليس نهائيا إلا بعد موافقة المرشد الأكاديمي.
4. يمكن التحاق الطلاب الجدد بالبرامج طوال العام، وذلك بعد الوفاء بجميع متطلبات الالتحاق بالبرامج.
5. التسجيل بالفصل الدراسي الصيفي اختياري.

التقويم الأكاديمي



مادة (16): مدة الدراسة

1. يتطلب الحصول على درجة البكالوريوس أن يجتاز الطالب بنجاح 145 ساعة معتمدة في 8 فصول نظامية (خريف وربيع) على الأقل.
2. مدة الدراسة في كلية الهندسة بجامعة الريادة للعلوم والتكنولوجيا هي (4) أعوام جامعية أو (4) مستويات دراسية مقسمة على (8) فصول دراسية أساسية.
3. الحد الاقصى للدراسة في الكلية هي ثماني سنوات دراسية (ستة عشر فصل دراسي نظامي: خريف وربيع) و هو ما لا يشمل الفصول الدراسية المجمدة لاسباب مقبولة من قبل مجلس الكلية ، و بعد هذه المدة يتم فصل الطالب من البرنامج المقيد به.

مادة (17): المرشد الأكاديمي

بناءا على ترشيح مجلس البرنامج المختص، تحدد الكلية لكل مجموعة من الطلاب مرشدا أكاديميا من بين أعضاء هيئة التدريس بالكلية ويمكن الاستعانة بالهيئة المعاونة كمرشدين أكاديميين بعد موافقة مجلس الكلية. وتتلخص مهام المرشد الأكاديمي في النقاط التالية:

- توجيه الطلاب في مسار دراستهم ومساعدتهم على اختيار المقررات الدراسية لكل فصل دراسي.
- يقوم المرشد الأكاديمي بتحديد عدد الساعات المعتمدة التي يمكنهم تسجيلها وفقا للمستوى الأكاديمي للطالب.
- يقوم المرشد الأكاديمي بمساعدة الطلاب على حل المشكلات التي يواجهونها اثناء الدراسة ومراقبة تقدمهم وادائهم اثناء الفصل الدراسي.
- يجوز للمرشد الأكاديمي ان يطلب من الطالب إعادة أحد او بعض المقررات التي اجتازها بالفعل او ان يطلب منه التسجيل في مقررات محددة لرفع معدله التراكمي الى المعدل المطلوب للتخرج.
- يلتزم المرشد الأكاديمي بوضع الطالب تحت الملاحظة لفصلين دراسين مع خفض عدد الساعات المسجلة وبحد اقصى 14 ساعة معتمدة وذلك في حالة حصول الطالب على معدل تراكمي اقل من 2.

مادة (18): شروط تسجيل المقررات الدراسية

يقوم مجلس البرنامج المختص بالكلية بالإعلان عن مواعيد التسجيل في المقررات الدراسية التي يتم فتحها للطلاب قبل كل فصل دراسي، وعلى الطلاب التسجيل في المواعيد المعلنة لهم ويشترط موافقة المرشد الأكاديمي في تسجيل أو حذف المقررات، كما يشترط موافقة مجلس البرنامج المختص على تسجيل المقررات للطلاب المتخلفين عن التسجيل في المواعيد المعلنة. لذا يمكن للطالب تسجيل المقررات الدراسية في الفصول الدراسية بحد اقصى لأجمالي الساعات المعتمدة وفقا للشروط التالية:

1. يجوز للطالب الذي بلغ معدله التراكمي 3.00 أو أكثر بالتسجيل في أكثر من 18 ساعة معتمدة وبحد أقصى 21 ساعة معتمدة في الفصل الدراسي الرئيسي التالي لحصوله على هذا المعدل.
2. يجوز للطالب الذي بلغ معدله التراكمي أكبر من او يساوى 2.00 و اقل من 3 التسجيل بحد اقصى 18 ساعة معتمدة في الفصل الدراسي الرئيسي التالي لحصوله على هذا المعدل.

3. لا يسمح للطالب الذي يقل معدله التراكمي عن 2.00 بالتسجيل في عدد ساعات أكثر من 14ساعة معتمدة في الفصل الدراسي الرئيسي التالي لحصوله على هذا المعدل او 5 مقررات ايهما أكبر بالنسبة للطالب.
4. يسمح للطالب تسجيل المقررات في الفصل الدراسي الصيفي بحد اقصى:
   - 9 ساعات معتمدة للطالب الذي معدلة التراكمي أكبر من أو يساوى 3.00 او ثلاث مقررات دراسية ايهما أكبر بالنسبة للطالب.
   - 8 ساعات معتمدة للطالب الذي معدلة التراكمي اقل من 3.00 او مقررين دراسين ايهما أكبر بالنسبة للطالب.
5. لا يسمح للطالب التسجيل في أي مقررات جديدة لها متطلبات سابقة دون استيفاء شروط النجاح في المقررات طبقا للجداول باللائحة الدراسية، ولا يجوز للطالب ان يسجل أي مقرر ومتطلبه السابق في نفس الفصل الدراسي.
6. في اول فصل دراسي، يتم اتاحة تسجيل المقررات في الخطة الدراسية، ويتم تطبيق القواعد السابقة بدءا من ثاني فصل دراسي للطالب.
7. يمكن للطالب تسجيل مقرر دراسي إضافي واحد عن الحدود المذكورة أعلاه إذا كان هذا سيؤدى الى تخرجه وذلك بعد موافقة المرشد الأكاديمي.
8. يسمح للكلية تحديد المقررات التي يتم طرحها كل فصل دراسي وطرق تسجيل الطلاب بها، عدا المقررات المحددة لمقررات متطلب لدراسة مقررات اخرى او الضرورية للتخرج فيجب اتاحتها للتسجيل في كل فصل.

مادة (19): شروط منح درجة البكالوريوس

للحصول على درجة البكالوريوس في الهندسة، يجب على الطالب اكمال الساعات المعتمدة المطلوبة بنجاح (145 ساعة معتمدة) لجميع البرامج وبمعدل تراكمي لا يقل عن 2.0.

1. يجب على الطالب النجاح في جميع المقررات الدراسية الى لها (0) ساعة معتمدة في برنامجه.
2. مشروع التخرج هو جزء أساسي من متطلبات التخرج في جميع البرامج ولا يتخرج الطالب مالم ينجز متطلبات النجاح بالمشروع ويجب ان يكون الطالب قد اجتاز بنجاح 70% من إجمالي ساعات البرنامج على الاقل حتى يمكنه التسجيل في مقررات مشروع التخرج في المستوى الرابع.
3. يجب ان يقوم الطالب بالتدريب الميداني خلال المدة المحددة لكل برنامج أثناء فترة دراسته.

4. يمكن للطالب ان يدرس عددا من المقررات الدراسية في جامعة اخرى لديها اتفاق تعاون مع جامعة الريادة للعلوم والتكنولوجيا بشأن تحويل الساعات المعتمدة. ويتطلب ذلك موافقة مسبقة من مجلس الكلية، شريطة الا يتجاوز عدد الساعات المعتمدة لهذه المقررات الدراسية نصف الساعات المعتمدة المطلوبة للحصول على درجة البكالوريوس في البرنامج.

مادة (20): التدريب الميداني

1. يجب ان يقوم الطالب بالتدريب الميداني خلال سنوات الدراسة بمدة 8 اسابيع وهي موزعة على تدريبين كل منهما 4 اسابيع في نهاية المستوى الثاني والثالث (كما يسمح للطالب الذى لم يؤدى التدريب الميدانى ان يؤدية خلال مدة الدراسة فى المستوى الرابع لاستيفاء متطلبات التخرج)
2. يكون التدريب في منشأة صناعية او خدمية ذات صلة بتخصص برنامج الطالب ويكون تحت الاشراف الكامل للكلية. ومن الممكن ايضا القيام بالتدريب داخل الكلية في بيئة مماثلة.
3. يقوم مجلس البرنامج الأكاديمي بتعين مرشدا أكاديميا من اعضاء هيئة التدريس لمتابعة التدريب الميداني.
4. يجب تحديد مسئول الاتصال بالشركة التي تقوم بتدريب الطالب.
5. يجب على الطالب ان يقدم تقريرا فنيا في نهاية فترة التدريب ويقيم هذا التقرير بواسطة المرشد الأكاديمي بعد عقد امتحان شفوي للمتدرب.
6. تقدم جهة التدريب التي تقوم بتدريب الطالب نموذجا لتقييم التدريب الذي قام به الطالب وذلك للمرشد الأكاديمي.
7. يقيم التدريب الميداني على اساس ناجح/ راسب ولا يدخل في حساب المعدل التراكمي للطالب.

مادة (21): إضافة وحذف المقررات الدراسية

1. يمكن للطالب ان يضيف مقرر دراسي (بعد موافقة المرشد الأكاديمي) حتى نهاية الاسبوع الثاني للفصلين الدراسيين الاول والثاني او حتى نهاية الاسبوع الاول في الفصل الدراسي الصيفي.
2. يمكن للطالب حذف مقرر دراسي بعد التسجيل (بعد موافقة المرشد الأكاديمي) حتى نهاية الاسبوع الثاني للفصلين الدراسيين الاول والثاني او حتى نهاية الاسبوع الاول في الفصل الدراسي الصيفي.
3. يجب الا يؤدى اضافة او حذف المقررات الدراسية الى مخالفة القواعد المنظمة للحد الاقصى والادنى لعدد الساعات المعتمدة المسجلة لكل فصل دراسي.

مادة (22): الانسحاب من المقررات الدراسية

يمكن للطالب بعد موافقة المرشد الأكاديمي، الانسحاب من مقرر دراسي بدون تسجيل مقرر اخر وبما لا يخالف عدد الساعات المعتمدة عن الحد الأدنى من الساعات المعتمدة الواجب دراستها في هذا الفصل الدراسي، وتحدد القواعد التالية شروط وآليات الانسحاب:

1. يجوز للطالب أن ينسحب من دراسة أي مقرر، وذلك قبل بداية الأسبوع العاشر في الفصل الدراسي الرئيسي وقبل بداية الاسبوع الخامس في الفصل الدراسي الصيفي.
2. لا يرسب الطالب في المقرر المنسحب منه شريطة أن تتم اجراءات الانسحاب من المقرر والموافقة علية خلال الفترة الزمنية المحددة.
3. يحصل الطالب على تقدير W في المقرر المنسحب منه ويسمح له بتسجيل هذا المقرر في الفصول الدراسية لاحقا (مع الحضور الكامل واداء جميع الانشطة بما فيها الامتحانات).
4. يجب الا يؤدى الانسحاب من المقرر الدراسي الى مخالفة القواعد المنظمة للحد الاقصى والادنى لعدد الساعات المعتمدة المسجلة لكل فصل دراسي ولا تدخل المقررات المنسحب منها في حساب المعدل التراكمي للطالب.
5. بالنسبة للمقررات الاختيارية، يسمح للطالب بتغيره في الفصول الدراسية اللاحقة إذا رسب في اجتيازه او قام بالانسحاب منه. وهذا يخضع لموافقة المرشد الأكاديمي ومتطلبات البرنامج.
6. إذا رغب الطالب في الانسحاب من الفصل الدراسي لظروف صحية أو بعذر قهري، عليه التقدم بطلب لإدارة الشئون التعليمية بالكلية للحصول على موافقة مجلس الكلية، وبناء على الموافقة يقوم بإعادة تسجيل هذه المقررات في فصل دراسي لاحق دراسة وامتحانا، وترصد له التقديرات الحاصل عليها في هذه المقررات بالكامل.

مادة (23): المقررات الدراسية الغير مكتملة

إذا لم يحضر الطالب الامتحان النهائي للمقرر الدراسي في فصل دراسي بعذر مقبول من قبل لجنة شئون التعليم والطلاب بالكلية ووافق عليه مجلس الكلية، فإن المقرر يعتبر غير مكتمل ويحصل الطالب فيه على تقدير (I) حتى يتم إجراء الامتحان في هذا المقرر في موعد لاحق تحدده الكلية في اول الفصل الدراسي الرئيسي التالي بحد اقصى الاسبوع الثانى وتضاف درجات هذا الامتحان الى درجات اعمال السنة وذلك لحساب التقدير الكلى

لهذا المقرر. وإذا لم يقم الطالب بإجراء الامتحان النهائي في هذا الموعد، فإنه سوف يحصل على تقدير (F) في هذا المقرر الدراسي.

مادة (24): إعادة المقررات الدراسية

يمكن للطالب، بعد موافقة المرشد الأكاديمي، إعادة مقرر دراسى دراسة و امتحانا بهدف التحسين إذا كان تقديرة فى هذا المقرر الدراسى يستوفى شروط الحد الادنى من النجاح ، وفقا للقواعد التالية:

1. يرصد للطالب التقدير الاعلى في المقرر بعد الإعادة وهذا التقدير هو الذي سيتم احتسابه عند حساب المعدل التراكمي للطالب، شريطة أن يظهر الاعادة في سجل الطالب الاكاديمى.
2. في حالة رسوب الطالب في الاعادة، يلغى تقديره السابق للمقرر ولا يعتد به بعد ذلك ويحصل على تقدير F في هذا المقرر.
3. الحد الاقصى لعدد المرات التي يمكن للطالب تحسين تقديره في المقررات هو خمسة خلال مدة دراسته بالكلية، ويستثنى من ذلك المقررات الدراسية التي يتم التحسين فيها بغرض الخروج من الانذار الأكاديمي او تلبية متطلبات التخرج، ولا يسمح للطالب في تحسين أي مقرر في حالة استكمال الطالب لساعات الدراسة بالبرنامج.
4. يجب على الطالب سداد المصروفات الدراسية كاملة للساعات المعتمدة المناظرة للمقررات الدراسية التي يتم التحسين فيها.
5. يجب على الطالب إعادة التسجيل في كل مقرر دراسي حصل فيه على تقديرF، وطبقا للساعات المسموح بها للتسجيل في الفصل الدراسي وذلك وفقا للقواعد التالية:
   - يعيد الطالب المقرر الذي رسب فيه دراسة وإمتحانا.
   - يحتسب للطالب التقدير الذي حصل عليه في الإعادة بحد أقصي +B ، على أن تذكر جميع التقديرات التي حصل عليها الطالب في سجله الأكاديمي، وعند حساب متوسط النقاط التراكمي يحتسب له التقدير الأخير فقط.
6. إذا قام الطالب بإعادة مقرر دراسي، فإنه يطلب منه ان يعيد جميع متطلبات تقييم المقرر الدراسي حتى يعاد تقييمه بالكامل حيث يعاد احتساب تقدير المقرر الدراسي.

7. يجوز السماح للطالب إذا رسب في مقرر دراسي (حصل على تقدير F)، بإعادة الامتحان النهائي (في ذات الفصل الدراسي) خلال المدة التي تقرها اللائحة وبعد موافقة مجلس الكلية، ولمقرر دراسي واحد فقط، ووفقا للقواعد الاتية:

- ألا تقل درجة الطالب في الامتحان النهائي للمقرر عن 50% من درجة الامتحان، والا تقل نتيجة الطالب في المقرر عن 55% من إجمالي درجات المقرر.
- ألا يزيد تقدير الطالب في المقرر بعد الاعادة عن C+.
- في حالة رسوب الطالب في الامتحان التكميلي علية إعادة المقرر دراسة وامتحانا طبقا لقواعد الاعادة.

8. في حالة عدم اكتمال عدد الساعات المعتمدة المصرح بها في الفصل الدراسي، يجوز للطالب الراسب في متطلب سابق، بتوصية من المرشد الأكاديمي وموافقة لجنة شئون الطلاب والتعليم بالكلية، التسجيل في مقرر بالتزامن مع المتطلب السابق، ويعلق نجاح الطالب في المقرر حتى يجتاز الطالب المتطلب السابق بنجاح.

مادة (25): إيقاف القيد

لمجلس الكلية ان يوقف قيد الطالب إذا تقدم بعذر يمنعه من الانتظام في الدراسة و تم فحصة من لجنة شئون التعليم و الطلاب واعتماده من مجلس الكلية وموافقة مجلس الجامعة.

مادة (26): المواظبة و الحضور

1. يجب على الطالب متابعة المحاضرات والتمارين النظرية والمعامل لكل مقرر، على ألا تقل نسبة حضوره عن 75 % من إجمالي ساعات الاتصال الفعلية للمقرر على حدة ويتم انذاره بالحرمان من أداء الامتحان مرتين: الأولى عند وصول نسبة الغياب إلى 10 % من إجمالي ساعات الاتصال لهذا المقرر، والثانية عند وصول النسبة إلى 20 % من إجمالي ساعات الاتصال لهذا المقرر، وبنا ء على طلب أستاذ المقرر وموافقة مجلس الكلية و بعد أخذ رأى مجلس البرنامج المختص يحرم الطالب من أداء الامتحان النهائي في المقررات التي لم يستوف فيها الطالب النسبة المقررة للحضور، وفى هذه الحالة يعتبر الطالب منسحبا إجباريا (FW) من المقررات التي يحرم من أداء الامتحان فيها ويرصد له تقدير F .

2. إذا قدم الطالب عذرا عن تغيبه في متابعة المحاضرات، أو التمارين النظرية ، أو المعامل وتم قبول العذر من مجلس الكلية بعد أخذ رأى مجلس البرنامج المختص، فيعتبر الطالب غائب بعذر مقبول، ويمكن اعتبار الطالب الغائب بعذر مقبول منسحبا من المقرر دون ان يؤثر ذلك في حساب المعدل التراكمي، ويرصد له تقدير W .

مادة (27): تقييم الطلاب

1. يتم تقييم كل مقرر دراسي من (100) مائة درجة.
2. توزع درجات كل مقرر كنسب مئوية من الدرجة الاجمالية للمقرر، وتنقسم هذه الدرجات الى:
   - درجات انشطة الطالب من تقارير وابحاث وامتحانات قصيرة.
   - درجات امتحان منتصف الفصل الدراسي.
   - درجات الامتحان العملي.
   - درجات مشروع على مستوى المقرر.
   - درجات الامتحان النهائي.
3. يترك توزيع درجات المقرر الدراسي طبقا لتوصيف المقرر. وهذا يعتمد على طبيعة المقرر الدراسي.
4. في حالة مشروع التخرج يخصص نسبة 50 % من الدرجة للمتابعة الدورية و50 % للمناقشة الشفوية.
5. لكي ينجح الطالب في مقرر دراسي، يجب ان يحصل على نسبة لا تقل عن 40 % من الامتحان النهائي وإلا يعتبر راسبا في هذا المقرر الدراسي بغض النظر عن مجموع الدرجات التي تم الحصول عليها في هذا المقرر وسوف يرصد له تقدير (F).
6. يعتبر الطالب راسبا في المقرر الدراسي ويحصل على تقدير (F) إذا حصل على نسبة اقل من 60% من النهاية العظمى لدرجات المقرر.
7. المقررات الدراسية التي لها (0) ساعة معتمدة يكون التقدير فيها راسب او ناجح ولا يدخل في حساب المعدل التراكمي. ولاجتياز هذه المقررات، يجب على الطالب الحصول على نسبة 60% على الاقل من درجات المقرر.

مادة (28): تقدير المقررات الدراسية

1. يتم حساب التقدير وعدد النقاط التي يحصل عليها الطالب في كل مقرر دراسي على النحو المبين بالجدول التالي:

جدول: تقدير المقررات وعدد النقاط المناظر

النسبة المئوية للدرجة	التقدير	عدد النقاط
اكثر من 97%	A+	4
93% الى اقل من 97%	A	
89% الى اقل من 93%	A-	3.7
84% الى اقل من 89%	B+	3.3
80% الى اقل من 84%	B	3.0
76% الى اقل من 80%	B-	2.7
73% الى اقل من 76%	C+	2.3
70% الى اقل من 73%	C	2.0
67% الى اقل من 70%	C-	1.7
64% الى اقل من 67%	D+	1.3
60% الى اقل من 64%	D	1.0
اقل من 60%	F	0.0

2. يجب على الطالب الحصول على الحد الادنى من التقدير (D) لاجتياز أي مقرر دراسي.

3. بالنسبة للمقررات الاخرى التي يسجل فيها الطالب كمستمع، او يطلب منه فقط اجتياز المقرر (المقررات الدراسية ذات عدد الساعات المعتمدة الصفرية) والتي لا تدخل في حساب المعدل التراكمي ستكون التقديرات على النحو المبين بالجدول التالي:

جدول: تقدير المقررات التي لها (0) ساعة معتمدة

التقدير	المدلول
AU	مستمع
P	ناجح
F	راسب
FW	منسحبا إجباريا
W	منسحب
I	مقرر غير مكتمل

مادة (29): الفصل من الدراسة والانذار الأكاديمي

1. يحصل الطالب على إنذار أكاديمي إذا كان معدلة التراكمي في اى فصل دراسي رئيسي اقل من 2.0.
2. إذا تجاوز المعدل التراكمي للطالب 2.0 في اى فصل دراسي رئيسي، فإنه يتم إعادة حساب عدد الانذارات الأكاديمية المتتابعة.
3. يتم فصل الطالب من الدراسة إذا حصل على معدل تراكمي اقل من 2.0 في ستة فصول متتالية باستثناء فصول الصيف.
4. يتم فصل الطالب إذا كان معدلة الأكاديمي اقل من 1.0 بعد اول ثلاثة فصول دراسية بالكلية.
5. يتم فصل الطالب من الدراسة إذا لم يحقق متطلبات التخرج خلال المدة القصوى للدراسة بالكلية وهي 8 سنوات (ستة عشر فصل دراسي أساسي: خريف وربيع).
6. يتم فصل الطالب من الدراسة إذا لم يقم بالتسجيل في اى مقررات لأكثر من فصلين دراسيين رئيسيين متتالين خلال فترة دراسته بالكلية بدون عذر تقبله الكلية.
7. الطالب الذي يتعرض للفصل من الدراسة بسبب عدم تمكنه من رفع المعدل التراكمي الى ما لا يقل عن 2.0، تتاح له فرصة اضافية ونهائية للتسجيل في فصليين رئيسيين متتاليين بالإضافة لفصل صيفي لرفع معدلة التراكمي الى 2.0 وتحقيق متطلبات التخرج، شريطة ان يكون قد انجز بنجاح ما لا يقل عن 80% من العدد الإجمالي للساعات المعتمدة اللازمة لتخرجه وان تكون هناك فرصة للطالب لرفع معدلة التراكمي الى 2.0 على الاقل.

مادة (30): حساب المعدل التراكمي (GPA)

1. تحسب نقاط كل مقرر على أنها عدد الساعات المعتمدة مضروبة في نقاط تقدير المقرر وفقا للجدول المرفق بمادة 27.
2. يحسب مجموع النقاط التي حصل عليها الطالب في أي فصل دراسي على أنها مجموع نقاط كل المقررات التي درسها الطالب في هذا الفصل الدراسي.

3. يحسب المعدل التراكمي لأي فصل دراسي (المعدل الفصلي) GPA ، على أنه ناتج قسمة مجموع النقاط التي حصل عليها الطالب في هذا الفصل مقسوما على مجموع الساعات المعتمدة لهذه المقررات ويكون حساب المعدل الفصلي وفق المعادلة الاتية:

المعدل الفصلي = مجموع النقاط للمقررات التي سجلها الطالب في الفصل الدراسي مقسوما على مجموع عدد الساعات المعتمدة لهذه المقررات.

$$Semester\ GPA = \frac{\sum_{i=1}^{N} Grade_i * Hours_i}{\sum_{i=1}^{N} Hours_i}$$

4. يحسب المعدل التراكمي المجمع/الكلى للطالب عند التخرج (CGPA) علي أنه ناتج قسمة مجموع كل نقاط المقررات التي اجتازها /درسها الطالب علي مجموع الساعات المعتمدة لهذه المقررات مع مراعاة القواعد المتعلقة بإعادة و تحسين المقررات . **ويكون حساب المعدل التراكمي المجمع CGPA وفقا للمعادلة الآتية:**

$$Cumulative\ GPA = \frac{\sum_{Courses} Grade\ Points * Credit\ Hours}{\sum_{Courses} Credit\ Hours}$$

5. المعدل التراكمي للتخرج هو المعدل التراكمي عند التخرج، وذلك بعد اجتياز جميع متطلبات التخرج. ولا يمكن للطالب الحصول على درجة البكالوريوس إلا إذا حقق معدل تراكمي 2.0 على الاقل.

6. يتحدد ترتيب الخريجين على اساس المعدل التراكمي للتخرج. في حالة التساوي في المعدل التراكمي يتم الترتيب طبقا للمجموع التراكمي للدرجات.

7. يجب ان تتضمن شهادة الطالب جميع المقررات الدراسية التي تم دراستها خلال مدة الدراسة، بما في ذلك المقررات الدراسية التي رسب فيها أو انسحب او تم تحسينها.

مادة (31): مرتبة الشرف

تمنح مرتبة الشرف للطالب الذي لا يقل المعدل التراكمي له عن 3.30 خلال جميع الفصول الدراسية الرئيسية، شريطة الا يكون قد حصل على تقدير (F) في اى مقرر خلال فترة دراسته بالبرنامج او لا يكون قد تم توقيع عليه اى عقوبات تأديبية خلال فترة دراسته بالكلية.

مادة (32): طلبات الاستئناف من نتائج المقررات

يمكن للطالب ان يقدم طلب استئناف لمراجعة نتيجة المقرر الدراسي في غضون اسبوع من إعلان النتيجة، وذلك بعد سداد الرسوم المقررة وفقا للوائح الكلية المتعلقة بذلك.

مادة (33): تحويل الطلاب بين البرامج بالكلية

يجوز للطالب، بعد موافقة كلا من مجلسى البرنامج ومجلس الكلية، الانتقال بين البرامج الأكاديمية داخل الكلية، بشرط ألا يكون قد أكمل أكثر من 50% من متطلبات التخرج للبرنامج المحول منه (اى قبل الوصول إلى مستوى Junior) ، و يجوز نقل الساعات المعتمدة المشتركة بين البرنامجين.

مادة (34): التحويل من جامعات اخرى

1. يجوز تحويل الطلاب من كليات الهندسة المعتمدة إلى أحد البرامج التي تطرحها كلية الهندسة في جامعة الريادة للعلوم والتكنولوجيا، بشرط وجود مقاصة بين المقررات التي سبق للطالب إكمالها والمقررات المطلوب منه إكمالها بنجاح و بحد اقصى 50% من الساعات المعتمدة للبرنامج المحول اليه. وعند التحويل من جامعات لا تعمل بنظام الساعات المعتمدة (النظام الفصلي) تحسب التقديرات طبقا للجدول التالي:

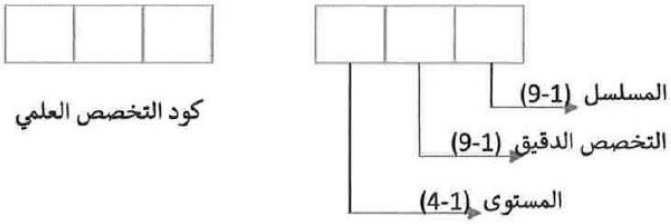
جدول: التقديرات المكافئة عند التحويل من النظام الفصلي الى نظام الساعات المعتمدة

من النظام الفصلي	الى نظام الساعات المعتمدة	
النسبة المئوية للدرجة	التقدير	عدد النقاط
أكثر من 95%	A+	4
90% الى اقل من 95%	A	
85% الى اقل من 90%	A-	3.7
80% الى اقل من 85%	B+	3.3
75% الى اقل من 80%	B	3.0
71% الى اقل من 75%	B-	2.7
68% الى اقل من 71%	C+	2.3
65% الى اقل من 68%	C	2.0
60% الى اقل من 65%	C-	1.7
55% الى اقل من 60%	D+	1.3
50% الى اقل من 55%	D	1.0
اقل من 50%	F	0.0

2. يتم حساب المعدل التراكمي لطلاب محولين لبرامج كلية الهندسة بجامعة الريادة للعلوم والتكنولوجيا، بناء على ما درسه الطالب تحت مظلة البرنامج المحول له فقط، ولا يعتد بما درسه في اى برنامج بنظام الساعات المعتمدة في اى جامعه أخرى او نظام الفصلين الدراسين سواء بجامعته او اى جامعه أخرى، وفى جميع الأحوال يتم اجراء مقاصة لما درسه ليتم حسابه ضمن متطلبات الحصول على الدرجة دون احتسابها في حساب المعدل التراكمي للطالب.

مادة (35): نظام التكويد

يتم تكويد المقررات الدراسية بهذه اللائحة من كود مكون من ثلاثة حروف وثلاثة أرقام، حيث تمثل الحروف كود القسم العلمي المختص بتدريس المقرر وتمثل الأرقام رقم المقرر كالآتي:



Code	Department
UNC	University Requirements (Compulsory)
UNE	University Requirements (Elective)
BSC	Basic Science
ENG	Faculty Requirements
ELE	Electrical Engineering
MEE	Mechanical Engineering

مادة (36): أحكام عامة

1. يطبق فيما لم يرد في شأنه نص في هذه اللائحة أحكام قانون تنظيم الجامعات ولائحته التنفيذية و قانون مجلس الجامعات الخاصة و لائحة التنفيذية والقوانين المعدلة لهم والقرارات الوزارية ذات الصلة.
2. يتحمل الطالب مسئولية علمه بهذه اللائحة وباطلاعه على النشرات والتعليمات الصادرة عن الكلية، وعلى ما يُنشر على لوحة الاعلانات أو بموقع الكلية الإلكتروني فيما يتعلق بهذه اللائحة.
3. يخضع الطالب المقيد بكلية الهندسة بجامعة الريادة للعلوم والتكنولوجيا للنظام العام للجامعة والكلية، وتطبق عليه قواعد الفصل من الجامعة وفرص إعادة القيد والأعذار المقبولة لعدم أداء الامتحان ووقف

القيد الدراسي وكافة القواعد والقوانين واللوائح الخاصة بشأن النظام التأديبي للطلاب المنصوص عليه في قانون تنظيم الجامعات ولائحته التنفيذية.

4. يسمح للكلية تعديل محتويات المقررات جزئيا خلال فترة صلاحية اللائحة شريطة ان تظل المادة العلمية للمقرر متوافقة مع اسم المقرر ولا تحيد عن هدف البرنامج، كما يجوز إضافة او تعديل قائمة المقررات الاختيارية وذلك بموافقة مجلس الكلية ودون الرجوع للقطاع الهندسي.

مادة (37): أحكام انتقالية

1. تطبق احكام هذه اللائحة ابتداء من العام التالي لصدور القرار الوزاري بالموافقة عليها.
2. أي تعديلات تطرأ على هذه اللائحة تطبق على الطلاب المستجدين ابتداء من الفصل الدراسي التالي لإقرار هذه التعديلات.

الباب الرابع

متطلبات الجامعة

University Requirements

1. مقررات متطلبات الجامعة

يحتوي الجدول التالي على بيان بمقررات الجامعة موضحا فيه ساعات المقررات بنظام الساعات المعتمدة الأمريكية ونظام النقاط المعتمدة الأوربية:

University Requirements Courses

Code	Course Title	Weight			Contact Hours			
		CH	ECTS	SWL	LEC	Tut	Lab	TT
UNC01	English Language	2	3	75	2	0	0	2
UNC02	Social Issues	2	3	75	2	0	0	2
UNC03	Technical Writing	2	3	75	2	0	0	2
UNC04	Communication and Presentation Skills	2	3	75	2	0	0	2
--	UR Elective 1	2	2	50	2	0	0	2
--	UR Elective 2	2	2	50	2	0	0	2
Total		12	16	400	12	0	0	12
Pool of UR Electives 1 and 2								
UNE05	Critical Thinking	2	2	50	2	0	0	2
UNE06	Ethics and Professionalism	2	2	50	2	0	0	2
UNE07	Fundamentals of Economics	2	2	50	2	0	0	2
UNE08	Fundamentals of Management	2	2	50	2	0	0	2
UNE09	Marketing & Sales	2	2	50	2	0	0	2
UNE11	Fundamentals of Sociology	2	2	50	2	0	0	2
UNE13	Creativity and Entrepreneurship	2	2	50	2	0	0	2

2. مصفوفة ربط مقررات متطلبات الجامعة بجدارات الجامعة الاكاديمية

إعتمادا على المعايير الأكاديمية القومية المرجعية (NARS 2018) والجدارات المتبناة من قبل جامعة الريادة للعلوم و التكنولوجيا ، فإن الخريج يجب ان يكون قادرا على اكتساب الجدارات التالية .

U1. Act with integrity, ethical reasoning, and social awareness in academic pursuits and community engagement.

U2. Demonstrate an understanding of global perspectives, cultural diversity, and intercultural communication skills.

U3. Demonstrate leadership qualities, take initiative, and adapt to changing environments with resilience and creativity.

U4. Understand sustainable practices and environmental stewardship and apply them to academic work and community engagement.

يوضح الجدول التالي مصفوفة مقررات متطلبات الجامعة وربطها بالجدارات المتبناة من قبل جامعة الريادة للعلوم و التكنولوجيا:

University Requirements Course-Competency Matrix

Code	Course	Univ. Competencies			
		U1	U2	U3	U4
UNC02	Social Issues	■			
UNC01	English Language		■		
UNC03	Technical Writing		■		
UNC04	Communication and Presentation Skills			■	
UNE05	Critical Thinking		■	■	
UNE07	Fundamentals of Economics		■		
UNE08	Fundamentals of Management			■	■
UNE06	Ethics and Professionalism	■			
UNE11	Fundamentals of Sociology		■		
UNE09	Marketing & Sales			■	
UNE13	Creativity and Entrepreneurship			■	■

3. توصيف مقررات متطلبات الجامعة

Code: UNC01			Subject: English Language		Pre. Req.: NA
Credit Hours = 2hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	0	0	50 %	50 %	

Course Syllabus:

Practice in specific points of grammar to consolidate and extend learner's existing knowledge - Analysis of syntax - comprehension - Skimming and scanning exercises develop the learners' skills - Comprehension questions interpretation and implication - the activities and games used develop listening - Speaking and writing skills through a communicative - Functional approach with suggested topics for discussion and exercises in summary writing and composition.

References: Nitin, B. (2010). Communicative English for Engineers and Professionals. Pearson Education India.

Code: UNC02			Subject: Social Issues		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments		
Lecture	Exercises	Practical	Lecture	Exercises	
2	0	0	2	0	

Course Syllabus:

This course aims to introduce the student to the most important social and societal issues. The student discusses the causes of these problems, develops solutions for them, new visions for problems and issues, and develops the student's ability to face problems and issues. This course includes problems resulting from population increase, human rights, transparency and combating corruption, various topics include the impact of poverty, crime, drug addiction and prejudice on the individual and society.

References: Thompson, J. (2008). A history of Egypt: from earliest times to the present. American Univ in Cairo Press.

Code: UNC03			Subject: Technical Writing		Pre. Req.: UNC01
Credit Hours = 2 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	0	0	50 %	50 %	

Course Syllabus:

This technical writing course covers types of reports (summary vs. detailed), report components, and their importance. It includes text and graphic writing techniques, report classification (technical and administrative), essential elements (abstract, summary, objectives, details, conclusions, recommendations, references), and logical report composition. Students will learn about grammar, punctuation, and using software like word processors and multimedia tools for effective communication.

References: David F. Beer and David A. McMurrey (2019). A Guide to Writing as an Engineer, 5th edition, John Wiley & Sons, Inc.

Code: UNC04		Subject: Communication and Presentation Skills		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %

Course Syllabus:

This course develops communication and presentation skills for personal and professional success. Students learn verbal and written communication, including professional documents, presentations, and reports. The course covers technical writing, business letters, resumes, interpersonal communication, cultural sensitivities, active listening, persuasive communication, and interpreting cues. It equips students to handle various communication scenarios effectively, fostering stronger connections and achieving goals.

References: "The College Writer: A Guide to Thinking, Writing, and Researching", 5th Edition. by Randall VanderMey, Verne Meyer, John Van Rys , Patrick Sebranek

Code: UNE05		Subject: Critical Thinking		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %

Course Syllabus:

This course teaches logical and critical thinking, exploring the links between science, society, and daily life. By learning the scientific method and the role of scientific communities, students develop effective persuasive skills.

References: Peter, F., & Gittens, C. A. (2015). Think Critically.

Code: UNE06		Ethics and Professionalism		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %

Course Syllabus:

This course delves into the complexities of ethical matters in the professional and business world. Through its exploration of themes like integrity, civic duty, and ethical behavior, it examines both ethical lapses and responsible conduct. Additionally, it explores the rights and obligations of both employees and corporations, delving into the broader questions of social and economic justice within the global economic landscape.

References:

[1] A Concise Introduction to Ethics Concise Edition", 2019 by Russ Shafer-Landau
[2] Moral Choices: An Introduction to Ethics ", 2018, by Scott Rae

Code: UNE07			Fundamentals of Economics	Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %
Concept of economics – the economic problem – Theory of demand including utility theory – theory of production – theory of cost – theory of firm including pricing theory – Economics of education – Economics of science and technology – Economics of automation including computerization.				
References: Sullivan, W. G., Wicks, E. M., & Koelling, C. P. (2015). *Engineering economy*. Pearson.				

Code: UNE08			Fundamentals of Management	Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %
Course Syllabus: History of Management – planning, fundamentals of planning – making decisions – strategic planning – plans and planning tools – Organizing and managing human resources – Influencing – leadership – controlling – Production management and control – Quality management – Management of service industries				
References: Wren, Daniel A., and Arthur G. Bedeian. *The evolution of management thought*. John Wiley & Sons, 2023.				

Code: UNE09			Marketing & Sales	Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %
Course Syllabus: Introduction to Marketing - Market Research- Marketing Strategy-Product Management Pricing Strategies- Distribution and Logistic- Promotion and Communication- Digital Marketing - Sales Management-Personal selling and the sales process- Customer Service and Retention--Marketing Ethics and Social Responsibility- Ethical issues in marketing- Case Studies				
References: Armstrong, Gary, Stewart Adam, Sara Denize, and Philip Kotler. *Principles of marketing*. Pearson Australia, 2014.				

Code: UNE11		Fundamentals of Sociology		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %

Course Syllabus:

Basic concepts – Basic examination of major theoretical perspectives – Structural functionalism – Symbolic interactionism – conflict theory – Types of Society: Tribal, agrarian, industrial, Post-industrial – Culture – Social networks – Social institutions – Deviance – Education – Religion – Race and ethnicity – Social class – Socialization – Gender identity – Social construction of the family – Community – Health – Social processes – Social change – Social Problems – Social demography

References:

Ferrante, Joan. *Sociology: A global perspective*. Belmont, CA: Wadsworth, 1995.

Code: UNE13		Subject: Creativity and Entrepreneurship		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %

Course Syllabus:

This course covers the basics of innovation and entrepreneurship in engineering. Students learn to identify opportunities, develop solutions, and manage ventures. It includes theory, case studies, guest lectures, and hands-on activities, emphasizing ethical considerations, sustainability, and social impact.

References:

“Innovation and Entrepreneurship. Newark” Wiley, 2015, by Bessant J Tidd J.

الباب الخامس

متطلبات الكلية

Faculty Requirements

1. مقررات متطلبات الكلية

يحتوي الجدول التالي على بيان بمقررات الكلية موضحا فيه ساعات المقررات بنظام الساعات المعتمدة الأمريكية ونظام النقاط المعتمدة الأوربية:

Faculty Requirements Courses

Code	Course Title	Weight			Contact Hours			
		CH	ECTS	SWL	LEC	Tut	Lab	TT
BSC 111	Engineering Mathematics 1	3	5	125	2	2	0	4
BSC 112	Engineering Mathematics 2	3	5	125	2	2	0	4
ELE 171	Introduction to Computer Programming	2	3	75	1	0	2	3
BSC 121	Physics 1	3	5	125	2	2	1	5
BSC 122	Physics 2	3	5	125	2	2	1	5
BSC 131	Engineering Chemistry	3	5	125	2	1	2	5
BSC 113	Statics	3	5	125	2	2	0	4
BSC 114	Dynamics	3	5	125	2	2	0	4
MEE 112	Production Engineering	2	3	75	1	0	2	3
MEE 111	Engineering Drawing	2	4	100	1	2	0	3
--	FR Elective 1 (Engineering Economy and Project Management Elective)	2	3	75	2	0	0	2
--	FR Elective 2 (Structures and Properties of Materials Elective)	2	3	75	2	1	0	3
Total		31	51	1275	21	16	8	45
Pool of Engineering Economy and Project Management Elective Courses								
ENG 111	Project Management for Mechanical Engineering	2	3	75	2	0	0	2
ENG 112	Project Management for Electrical Engineering	2	3	75	2	0	0	2
ENG 113	Project Management and Engineering Economy	2	3	75	2	0	0	2
ENG 114	Engineering Economy and Investments	2	3	75	2	0	0	2
Pool of Structures and Properties of Materials Elective Courses								
ENG 115	Structure and Properties of Materials	2	3	75	2	1	0	3
ENG 116	Properties of Electrical Materials	2	3	75	2	1	0	3
ENG 117	Introduction to Engineering Design	2	3	75	2	1	0	3

2. مصفوفة ربط مقررات متطلبات الكلية بجدارات البرامج

إعتمادا على المعايير الأكاديمية القومية المرجعية (NARS 2018) فإن الخريج يجب أن يكون قادرا على اكتساب الجدارات الأتية والخاصة بالعلوم الهندسية (Level A):

A1. Identify, formulate, and solve complex engineering problems by applying engineering fundamentals, basic science, and mathematics.

A2. Develop and conduct appropriate experimentation and/or simulation, analyze, and interpret data, assess, and evaluate findings, and use statistical analyses and objective engineering judgment to draw conclusions.

A3. Apply engineering design processes to produce cost-effective solutions that meet specified needs with consideration for global, cultural, social, economic, environmental, ethical, and other aspects as appropriate to the discipline and within the principles and contexts of sustainable design and development.

A4. Utilize contemporary technologies, codes of practice, and standards, quality guidelines, health and safety requirements, environmental issues, and risk management principles.

A5. Practice research techniques and methods of investigation as an inherent part of learning.

A6. Plan, supervise, and monitor implementation of engineering projects, taking into consideration other trades requirements.

A7. Function efficiently as an individual and as a member of multi-disciplinary and multicultural teams.

A8. Communicate effectively – graphically, verbally and in writing – with a range of audiences using contemporary tools.

A9. Use creative, innovative, and flexible thinking and acquire entrepreneurial and leadership skills to anticipate and respond to new situations.

A10. Acquire and apply new knowledge, and practice self, lifelong and other learning strategies.

يوضح الجدول التالي مصفوفة ربط مقررات متطلبات الكلية بالجدارات الخاصة بالعلوم الهندسية (Level A) طبقا المعايير الأكاديمية المرجعية (NARS 2018):

Faculty Requirements Course-Competency Matrix

Code	Course	Eng. Competences (Level A)									
		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
BSC 111	Engineering Mathematics 1	■	■								
BSC 112	Engineering Mathematics 2	■	■								
ELE 171	Introduction to Computer Programming	■				■					
BSC 121	Physics 1	■	■								
BSC 122	Physics 2	■	■								
BSC 131	Engineering Chemistry	■	■								
BSC 113	Statics	■	■								
BSC 114	Dynamics	■	■								
MEE 112	Production Engineering	■	■		■						
MEE 111	Engineering Drawing	■		■							
ENG 111	Project Management for Mechanical Engineering				■	■	■		■	■	■
ENG 112	Project Management for Electrical Engineering				■	■	■		■	■	■
ENG 113	Project Management and Engineering Economy				■	■	■		■	■	■
ENG 114	Engineering Economy and Investments				■			■		■	
ENG 115	Structure and Properties of Materials	■	■	■							
ENG 116	Properties of Electrical Materials	■	■	■							
ENG 117	Introduction to Engineering Design	■			■						

3. توصيف مقررات متطلبات الكلية

Code: BSC 111		Subject: Engineering Mathematics 1		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

The course covers functions (definition, domain, range, types, even/odd, composite, inverse), calculus (limits, continuity, differentiation, integration, multivariable), and their applications in engineering. It also includes sequences and series (arithmetic, geometric, convergence, Taylor, Maclaurin), and analytic geometry (parametric equations, vector geometry, curvature, torsion, conic sections in polar coordinates).

References:

[1] Croft, A. (2017). Engineering Mathematics A Foundation for Electronic, Electrical, Communications and Systems Engineers. Pearson Education Limited.

[2] Stroud, K. A., & Booth, D. J. (2020). *Engineering mathematics*. Bloomsbury Publishing.

Code: BSC 112		Subject: Engineering Mathematics 2		Pre. Req.: BSC 111
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

The course covers integration (definite, indefinite, techniques, improper, applications), multiple integrals (double integrals, polar coordinates, engineering applications), vector algebra (vector spaces, linear independence, inner product spaces, cross product, engineering applications), and matrix algebra (matrix operations, inverse matrices, eigenvalues, eigenvectors, engineering applications).

References:

[1] Croft, A. (2017). Engineering Mathematics A Foundation for Electronic, Electrical, Communications and Systems Engineers. Pearson Education Limited.

[2] Stroud, K. A., & Booth, D. J. (2020). *Engineering mathematics*. Bloomsbury Publishing.

Code: ELE 171		Subject: Introduction to Computer Programming		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
1	0	2	50 %	50 %

Course Syllabus:

Basic programming concepts using one of the modern general-purpose programming languages. Basic input/output, Data types, Expressions, Mathematical and logical operators, Mathematical functions, Arrays, Multi-dimensional arrays, Strings, Enumerations, Conditions, Decisions, Loops, Functions, Function-call mechanisms, Recursive functions, Parameter passing, Addresses, Pointers/References, Pointers to pointers, Pointers to functions, Program memory segments, Dynamic allocations, Files, Error handling, Static and dynamic libraries.

References:

[1] D. S. Malik, C++ Programming: From Problem Analysis to Program Design, 8th Edition, Course Technology, 2017.

[2] Paul Deitel and Harvey Deitel, C++ How to Program, 7th Edition, 2010 by Pearson Education, Inc.

Code: BSC 121		Subject: Physics 1		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50%	50%

Course Syllabus:

Mechanics Fundamentals: Work, Energy, and Power (i.e., Fundamental principles, conservation laws, and practical applications)- Forces and Motion (i.e., Newtonian mechanics, force analysis, and vector dynamics)- Potential Gradient (Analysis of potential fields, gradient concepts, and implications). **Vibrations and Oscillations:** Circular Motion- Simple Harmonic Motion (SHM) - Harmonic Superposition- Damped Vibrations -Forced Oscillations and Resonance. **Wave Motion and Optics:** Wave Equation -Wave Types- Sound Waves- Standing Waves-Light Interference and Diffraction.

References:

[1] Serway and Jewett, Physics for scientists and engineers with modern physics, 2014.

[2] Halliday, D., Resnick, R., & Walker, J. (2013). Fundamentals of physics. John Wiley & Sons.

Code: BSC 122		Subject: Physics 2		Pre. Req.: BSC 121
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50%	50%

Course Syllabus:

Electricity: Electric charge, Coulomb's law, Electric field, Electric flux, Gauss Law, Electric potential, Capacitance and Dielectrics, Ohm's law, and RC circuits. **Magnetism**: Magnetic field, Magnetic Force, Sources of Magnetic Fields, Ampere's law, Faraday's Law, Electromagnetic induction, Magnetic properties of materials.

References:

Serway and Jewett, Physics for scientists and engineers with modern physics, 2014.

Code: BSC 131		Subject: Engineering Chemistry		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50%

Course Syllabus:

Gases, Liquids, Solids, Thermochemistry, Thermodynamics, Solutions, Ionic equilibrium, Electrochemistry, Corrosion of metals, Alloys, Water chemistry and its treatment, Chemistry of cements, Chemistry of polymers, Fuels and Combustion, Environmental pollution, and its control.

References:

[1] Sawyer, C. N., McCarty, P. L., & Parkin, G. F. (2003). Chemistry for environmental engineering and science. McGraw-Hill.

[2] Baird, C., & Cann, M. (2005). Environmental chemistry. Macmillan.

Code: MEE 111		Subject: Engineering Drawing		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
1	2	0	50 %	50%

Course Syllabus:

Introduction to Engineering Drawing - Geometric Construction - Orthographic Projection- Dimensioning- Sectional Views - Computer-aided drawing (CAD) of 2D and 3D figures.

References:

Mcgraw-hill Mint, "Mechanical Drawing Board & CAD Techniques", Student Edition, 2011

Code: BSC 113		Subject: Statics		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to Statics - Force Vectors - Equilibrium of a Particle - Force System Resultants- Equilibrium of a Rigid– Trusses and machines – Friction - Centroids and Centers of Mass - Moment of Inertia - Virtual Work and Potential Energy.

References:

[1] R.C. Hibbeler, "Engineering Mechanics: Statics and Dynamics, 14th Edition", Pearson Prentice Hall, New Jersey, 2016.
[2] J. L. Meriam, L. G. Kriage, and J. N. Botton, "Engineering Mechanics: Statics, 8th Edition", John Wiley & Sons, New York, 2016.

Code: BSC 114		Subject: Dynamics		Pre. Req.: BSC 113
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Definition of dynamics- Kinematics of Particles- Newton's laws of motion- Energy and Work - Planar Kinetics of Rigid Bodies- Equations of Motion for Rigid Bodies - Principle of Impulse and momentum- Direct and central impact- Introduction to vibration mechanics.

References:

R.C. Hibbeler, "Engineering Mechanics: Statics and Dynamics, 14th Edition", Pearson Prentice Hall, New Jersey, 2016.

Code: MEE 112		Subject: Production Engineering		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
1	0	2	50 %	50%

Course Syllabus:

Introduction to machining processes, Cutting elements, Cutting with single edge cutting tools, Cutting tool materials and its characteristics, Cutting velocity and feed, Machining time, Power consumption in cutting, Practical machining operations: turning, shaping, drilling, Cutting with multi-edge, Cutting tools: milling, grinding, lapping, Simple dividing and dividing head, Basic elements of machine tools and specifications, Work fixation, Tool fixation, Process sheet, Machining time allowances, Cost elements, Break-even point

References:

[1] Groover, Mikell P. Introduction to manufacturing processes. Wiley Global Education, 2011.
[2] Hitomi, Katsundo. Manufacturing Systems Engineering: A Unified Approach to Manufacturing Technology, Production Management and Industrial Economics. Routledge, 2017.

Code: ENG 111		Subject: Project Management for Mechanical Engineering		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %

Course Syllabus:

This course introduces fundamental project management concepts, offering an overview of organizational strategy, structure, and culture essential for effective project execution. Topics include project planning, scheduling, resource allocation, risk management, team building, cost estimation, project quality management, and supply chain integration.

References:

Siegel, N. G. (2020). *Engineering Project Management*. John Wiley & Sons.

Code: ENG 112			Subject: Project Management for Electrical Engineering		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	0	0	50 %	50 %	

Course Syllabus:

An overview of project management, Terminology commonly used in project management, The various phases of a project and its life cycle, Managing project timelines, Utilizing tools like Gantt charts, Activity on Arrow, and Activity on Node, Understanding the relationships between project activities, Estimating project timelines using probability, Strategies for reducing project time through time crashing, Efficiently managing project resources, Implementing quality control measures to ensure project success.

References:

[1] Gray, C. F., & Larson, E. W. (2011). *Project management the managerial process*. McGraw-Hill.

[2] Rojas, E. M. (Ed.). (2009). *Construction project management: A practical guide for building and electrical contractors*. J. Ross Publishing.

Code: ENG 113			Subject: Project Management and Engineering Economy		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	0	0	50 %	50 %	

Course Syllabus:

The course covers project management basics, stakeholder engagement, project life cycle stages, feasibility studies, project selection methods, iterative planning and control, scope management, work breakdown structures, network diagrams, critical path method, bar charts, program evaluation and review technique, resource management (smoothing and leveling), project duration compression, and earned value analysis for performance measurement.

References:

Zeitoun, A. (2023). *Program Management: Going Beyond Project Management to Enable Value-Driven Change*. John Wiley & Sons.

Code: ENG 114		Subject: Engineering Economy and Investments		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50 %

Course Syllabus:

Introduction to engineering economy covers principles like the time value of money and interest rates. The course includes analyzing cash flows with present and future values, using cash flow diagrams. The course evaluates project options and assesses their worth using methods like payback period and internal rate of return. It also explores financial tools such as bond considerations, break-even points, and asset depreciation. Real-world applications illustrate economic analysis in engineering fields.

References:

Sullivan, W. G., Wicks, E. M., & Koelling, C. P. (2015). *Engineering economy*. Pearson.

Code: ENG 115		Subject: Structure and Properties of Materials		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	0	50 %	50 %

Course Syllabus:

Categorize engineering materials: metals, polymers, ceramics, and composites. Examine internal structures: atomic, micro, and macrostructure. Explore material properties: physical, chemical, mechanical, electrical, thermal, and optical. Understand the relationship between structure and properties. Investigate solidification, phase diagrams, and the iron-carbon diagram. Consider material-environment interactions and study corrosion effects on materials.

References:

Trolier-McKinstry, S., & Newnham, R. E. (2018). *Materials engineering: bonding, structure, and structure-property relationships*. Cambridge University Press.

Code: ENG 116		Subject: Properties of Electrical Materials		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	0	50 %	50 %

Course Syllabus:

Conductive Materials: Study structure, conductivity, alloys, and mechanical properties. Insulating Materials: Analyze ceramic and polymer insulators, focusing on PVC and XLPE in cables. Magnetic Materials: Examine B-H curves, magnetic losses, and ferro-resonance. Battery Materials: Explore materials for primary and secondary batteries. Silicon Semiconductor Industry: Study metallurgical and electronic grade silicon for semiconductor applications.

References:

Callister, W. D., & Rethwisch, D. G. (2000). *Fundamentals of materials science and engineering* (Vol. 471660817). London: Wiley.

Code: ENG 117		Subject: Introduction to Engineering Design		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	0	50 %	50 %

Course Syllabus:

The Design Process– Technical Sketching—Measurement- Modeling Skills– Geometry and Design – Reverse Engineering– Documentation– Advanced Computer Modeling– Design Teams– Design Challenge.

References

[1] Bethune, James D. Engineering design and graphics with SolidWorks. Prentice Hall Press, 2009.
[2] Samuel, Andrew, and John Weir. Introduction to engineering design. Elsevier, 1999.
[3] Elam, Kimberly. Geometry of design: studies in proportion and composition. Princeton Architectural Press, 2001.

الباب السادس

برنامج هندسة الاتمتة والروبوتات

Automation and Robotics Engineering Program

اسم البرنامج	هندسة الأتمتة والروبوتات Automation and Robotics Engineering (ARE)			
الكلية	الهندسة			
الجامعة	جامعة الريادة للعلوم والتكنولوجيا			
الساعات المعتمدة (CH)	145			
النقاط المعتمدة (ECTS)	240			
عدد المقررات	55 مقرر			
عدد ساعات الاتصال	محاضرات	تمارين	معمل	الإجمالي
	98	58	51	207
توزيع المقررات طبقا لمتطلبات مستويات التعلم				
	CH	ECTS	النسب	المقررات
متطلبات الجامعة	12	16	% 8	6
متطلبات الكلية	31	51	% 21	12
متطلبات البرنامج والتخصص الرئيسي	102	173	% 71	37
بيانات إحصائية				
	CH	ECTS	النسب	المقررات
نسبة مقررات العلوم الأساسية	27	45	% 19	9
نسبة المقررات الاختيارية	23	35	% 16	9

1. رسالة البرنامج

يسعى برنامج هندسة الأتمتة والروبوتات الى اعداد خريجين متميزين من خلال تخصصات بينية في هندسة الأتمتة والروبوتات، ليكونوا قادرين على المنافسة فى سوق العمل محليا وإقليميا في التطبيقات العملية، ويكونوا نموذجا يحتذى به في تطوير المجتمع وتنمية موارده.

2. أهداف البرنامج

يهدف برنامج هندسة الأتمتة والروبوتات الى تقديم خريج قادر على:

1. تطبيق المعرفة بالرياضيات الهندسية والعلوم الطبيعية واساسيات الهندسة وعلوم الحاسب وقدرات التفكير التحليلي والنقدي والمنهجي لتحديد وحل المشكلات الهندسية.
2. إكساب الطالب القدرة على التعامل مع أنظمة التحكم والأتمتة والروبوتات لأجل التعامل مع مشكلة تطبيقية محددة او لمتابعة التطوير فيها.
3. تشجيع العمل الجماعي بين التخصصات والرؤى المختلفة والمساهمة في تطوير المهنة والمجتمع وكساب الخريجين الصفات القيادية ومهارات ادارة المشروعات وريادة الاعمال..
4. إعداد كوادر فنية للعمل في شتى مجالات هندسة الأتمتة الصناعية والروبوتات.
5. تمكين الطالب من التعلم الذاتي المستمر لاكتساب القدرة على إجراء البحث العلمي وتطوير القطاع الصناعي وخدمة المجتمع.

3. مواصفات الخريج

يحرص برنامج هندسة الأتمتة والروبوتات على تخريج مهندسين متميزين ومؤهلين لدخول سوق العمل ويراعى فيهم تحقيق المواصفات الأتية:

1. Master a wide spectrum of engineering knowledge and specialized skills and can apply acquired knowledge using theories and abstract thinking in real life situations.
2. Apply analytic critical and systemic thinking to identify, diagnose and solve engineering problems with a wide range of complexity and variation.
3. Behave professionally and adhere to engineering ethics and standards.

4. Work in and lead a heterogeneous team of professionals from different engineering specialties and assume responsibility for own and team performance.
5. Recognize his/her role in promoting the engineering field and contribute in the development of the profession and the community.
6. Value the importance of the environment, both physical and natural, and work to promote sustainability principles.
7. Use techniques, skills and modern engineering tools necessary for engineering practice.
8. Assume full responsibility for own learning and self-development, engage in lifelong learning and demonstrate the capacity to engage in post- graduate and research studies.
9. Communicate effectively using different modes, tools and languages with various audiences; to deal with academic/professional challenges in a critical and creative manner.
10. Demonstrate leadership qualities, business administration and entrepreneurial skills.
11. Design of automation and robotics systems considering the principles of mechanical, manufacturing, electrical/electronic, and computer engineering.
12. Analyze and investigate the inter-disciplinary characteristics of mechanical, electrical, control and robotics systems.
13. Work within a multidisciplinary team during the analysis, design and implementation phases of automation and robotics engineering projects.

4. المعايير الأكاديمية للبرنامج

بالإضافة الى الجدارات لمعظم البرامج الهندسية (Level A) فأن برنامج هندسة الأتمتة والروبوتات له بعض الجدارات الخاصة لطبقا المعايير الأكاديمية المرجعية (NARS 2018) وهي كما يلي:

D1. Evaluate and apply principles of control theory, automation systems and mechanical design to select, model and analysis custom solutions for different engineering problems.

D2. Design, model, and analyze electrical, electronic, mechanical, or digital components for automation and robotics applications, utilizing appropriate optimization tools.

D3. Integrate a wide range of analytical tools, techniques, equipment, and software packages pertaining to design and develop automation and robotics systems.

D4. Write the necessary software for the equipment and the control of the automation and robotics systems.

D5. Distinguish the layout and the key parameters to the field of the concentration within the automation and robotics engineering as listed below:

Concentration	Competence
Bio-Robotics	**D5a**. Design mechanical, electrical, electronic and programming techniques for solving problems related to the life of humans.
Advanced Robotics in Industrial Applications	**D5b**. Design robotic systems to perform tasks used in the synthesis of industrial applications.
Industrial Automation	**D5c**. Design mechanical, electrical, electronic and software for solving problems related to industrial automation systems.

5. المقررات الدراسية

للحصول على درجة بكالوريوس الهندسة في تخصص هندسة الأتمتة والروبوتات ولتحقيق مخرجات التعلم المستهدفة للبرنامج وللتوافق مع مواصفات وجدارات الخريج الموصفة من قبل هيئة ضمان الجودة والاعتماد فإنه يتعين على الطالب أن يجتاز بنجاح المقررات التالية والمقسمة الى (متطلبات الجامعة – متطلبات الكلية – متطلبات البرنامج والتخصص):

List of Automation and Robotics Engineering Program Requirements

Code	Course Title	Weight			Contact Hours			
		CH	ECTS	SWL	LEC	Tut	Lab	TT
--	University Requirements	12	16	400	12	0	0	12
--	Faculty Requirements	31	51	1275	21	16	8	45
Program and specialization Requirements								
BSC 215	Differential Equations	3	5	125	2	2	0	4
BSC 216	Numerical Analysis	3	5	125	2	2	0	4
ELE 211	Electrical Circuits	3	6	150	2	2	1	5
ELE 212	Electronics	3	5	125	2	2	1	5
ELE 213	Electronic Circuits	3	5	125	2	1	2	5
ELE 214	Measurements and Sensors	3	5	125	2	1	2	5
ELE 215	Electromagnetic Fields	3	5	125	2	2	0	4
ELE 321	Electrical Power and Machines	3	5	125	2	2	1	5
ELE 324	Power Electronics and Motor Drives	3	5	125	2	2	1	5
ELE 251	Signals and Systems Fundamentals	3	5	125	2	2	0	4
ELE 374	Digital Signal Processing	3	5	125	2	1	2	5
ELE 252	Modeling of Dynamic Systems	3	5	125	2	2	0	4
ELE 353	Automatic Control	3	5	125	2	2	1	5
ELE 354	Digital Control Systems	3	5	125	2	2	1	5
ELE 273	Logic Design	3	5	125	2	0	2	4
ELE 372	Computer Programming	3	5	125	2	0	2	4
MEE 216	Computer Aided Design	2	4	100	1	0	2	3
MEE 215	Theory of Machines	3	5	125	2	2	0	4
MEE 221	Fluid Mechanics	3	5	125	2	2	0	4
MEE 223	Thermodynamics	3	5	125	2	2	0	4
ELE 463	Mechatronic Systems	3	5	125	2	2	1	5
ELE 361	Embedded Systems	3	5	125	2	0	2	4
ELE 362	Programmable Logic Controllers	3	5	125	1	0	3	4
MEE 371	Industrial Robotics	3	5	125	2	2	1	5
ELE 464	Industrial Automation and Networks	3	6	150	2	0	2	4
MEE 442	Pneumatic and Hydraulic Systems	3	6	150	2	0	2	4
ELE 475	Introduction to Machine Learning	3	4	100	2	0	2	4
MEE 472	Mobile Robots	3	5	125	2	1	1	4
--	ARE Elective (1)	3	5	125	2	2	1	5
--	ARE Elective (2)	3	5	125	2	1	1	4
--	ARE Elective (3)	3	5	125	2	1	1	4
--	ARE Elective (4)	3	5	125	2	1	1	4
--	ARE Elective (5)	3	5	125	2	1	1	4
MEE 491	ARE Graduation Project 1	2	4	100	1	0	2	3
MEE 492	ARE Graduation Project 2	2	3	75	0	0	4	4
MEE x95	ARE Practical Training 1	0	0	0	0	0	0	0
MEE x96	ARE Practical Training 2	0	0	0	0	0	0	0
Total		145	240	6000	98	58	51	207

Code	Course Title	Weight			Contact Hours			
		CH	ECTS	SWL	LEC	Tut	Lab	TT
Pool of ARE Elective (1)								
ELE 455	Process Control Systems	3	5	125	2	2	1	5
MEE 473	Robotic Operating Systems	3	5	125	2	2	1	5
ELE 476	Digital Image Processing	3	5	125	2	2	1	5
ELE 456	Modern Control Systems	3	5	125	2	2	1	5
Pool of ARE Elective (2)								
MEE 474	Embedded Computing for Robotics	3	5	125	2	1	1	4
MEE 444	Selected Topics in Mechatronic Engineering	3	5	125	2	1	1	4
ELE 463	Advanced Programmable Logic Controllers	3	5	125	2	0	2	4
MEE 445	Autotronics	3	5	125	2	1	1	4
Pool of ARE Electives 3,4, and 5 for the Bio-Robotics Concentration								
MEE 479	Rehabilitation Robots	3	5	125	2	1	1	4
ELE 477	Fundamentals of Biomedical Engineering	3	5	125	2	1	1	4
ELE 478	System Physiology	3	5	125	2	1	1	4
MEE 443	Selected Topics in Bio-Mechatronics	3	5	125	2	1	1	4
Pool of ARE Electives 3,4, and 5 for the Advanced Robotics in Industrial Applications Concentration								
MEE 475	Advanced Robotics	3	5	125	2	1	1	4
MEE 476	Robotics Applications in Manufacturing Systems	3	5	125	2	1	1	4
MEE 477	Robot Vision	3	5	125	2	1	1	4
MEE 478	Selected Topics in Advanced Robotics Applications	3	5	125	2	1	1	4
Pool of ARE Electives 3,4, and 5 for the Industrial Automation Concentration								
ELE 457	Introduction to Intelligent Control Systems	3	5	125	2	1	1	4
ELE 467	Internet of Things	3	5	125	2	2	0	4
ELE 466	Wireless Sensors in Industrial Networks	3	5	125	2	2	0	4
ELE 416	Fault Detection and Diagnosis in Industrial Processes	3	5	125	2	2	0	4
ELE 465	Selected Topics in Industrial Automation	3	5	125	2	1	1	4

6. مصفوفة ربط المقررات الدراسية بجدارات البرنامج

يوضح الجدول التالي مصفوفة ربط مقررات متطلب البرنامج بالجدارات الخاصة بالعلوم الهندسية (Level A) والجدارات الخاصة بالبرنامج (Level D):

Program and Specialization Requirements Course-Competency Matrix

Code	Course	Eng. Competences (Level A)										ARE Competences (Level D)						
		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	D1	D2	D3	D4	D5a	D5b	D5c
BSC 215	Differential Equations	■	■															
BSC 216	Numerical Analysis	■	■															
ELE 211	Electrical Circuits		■	■									■					
ELE 212	Electronics		■										■					
ELE 213	Electronic Circuits					■					■		■					
ELE 214	Measurements and Sensors	■	■		■													
ELE 215	Electromagnetic Fields	■						■			■							
ELE 321	Electrical Power and Machines				■	■				■								
ELE 324	Power Electronics and Motor Drives					■	■				■	■						
ELE 251	Signals and Systems Fundamentals	■	■															
ELE 374	Digital Signal Processing			■										■	■			
ELE 252	Modeling of Dynamic Systems	■										■	■					
ELE 353	Automatic Control		■									■		■				
ELE 354	Digital Control Systems										■	■		■				
ELE 273	Logic Design			■			■							■				
ELE 372	Computer Programming						■			■					■			
MEE 216	Computer Aided Design				■									■	■			
MEE 215	Theory of Machines	■		■								■						
MEE 221	Fluid Mechanics	■	■									■						
MEE 223	Thermodynamics	■	■	■								■						
ELE 463	Mechatronic Systems						■					■	■	■				
ELE 361	Embedded Systems						■						■		■			
ELE 362	Programmable Logic Controllers		■												■			■

Code	Course	Eng. Competences (Level A)										ARE Competences (Level D)						
		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	D1	D2	D3	D4	D5a	D5b	D5c
MEE 371	Industrial Robotics		■									■					■	
ELE 464	Industrial Automation and Networks					■	■					■						■
MEE 442	Pneumatic and Hydraulic Systems				■								■					■
ELE 475	Introduction to Machine Learning									■				■	■			
MEE 472	Mobile Robots	■										■					■	
MEE 491	ARE Graduation Project 1							■	■	■	■	■	■	■	■	■	■	■
MEE 492	ARE Graduation Project 2							■	■	■	■	■	■	■	■	■	■	■
MEE x95	ARE Practical Training 1					■	■		■			■	■	■	■			
MEE x96	ARE Practical Training 2					■	■		■			■	■	■	■			
ARE Elective (1)																		
ELE 455	Process Control Systems				■						■	■	■					
MEE 473	Robotic Operating Systems				■									■	■			
ELE 476	Digital Image Processing													■	■	■		
ELE 456	Modern Control Systems	■										■	■					
ARE Elective (2)																		
MEE 474	Embedded Computing for Robotics						■						■		■	■		
MEE 444	Selected Topics in Mechatronic Eng.	Based on the topic																
ELE 463	Advanced Programmable Logic Controllers									■				■	■			■
MEE 445	Autotronics	■									■	■	■					
ARE Elective for (Bio-Robotics) Concentration																		
MEE 479	Rehabilitation Robots											■	■			■		
ELE 477	Fundamentals of Biomedical Engineering		■		■					■						■		
ELE 478	System Physiology	■								■						■		
MEE 443	Selected Topics in Bio-Mechatronics	Based on the topic																

Code	Course	Eng. Competences (Level A)										ARE Competences (Level D)						
		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	D1	D2	D3	D4	D5a	D5b	D5c
ARE Elective for (Advanced Robotics in Industrial Applications) Concentration																		
MEE 475	Advanced Robotics				■								■	■			■	
MEE 476	Robotics Applications in Manufacturing Systems					■						■	■				■	
MEE 477	Robot Vision									■				■	■		■	
MEE 478	Selected Topics in Advanced Robotics Applications	Based on the topic																
ARE Elective for (Industrial Automation) Concentration																		
ELE 457	Introduction to Intelligent Control Systems		■									■	■					■
ELE 467	Internet of Things			■										■				■
ELE 466	Wireless Sensors in Industrial Networks				■									■				■
ELE 416	Fault Detection and Diagnosis in Industrial Processes				■									■				■
ELE 465	Selected Topics in Industrial Automation	Based on the topic																

7. الخطة الدراسية المقترحة

يوضح الجدول التالي النموذج الاسترشادى للخطة الدراسية للطالب المنتظم:

Suggested Study Plan

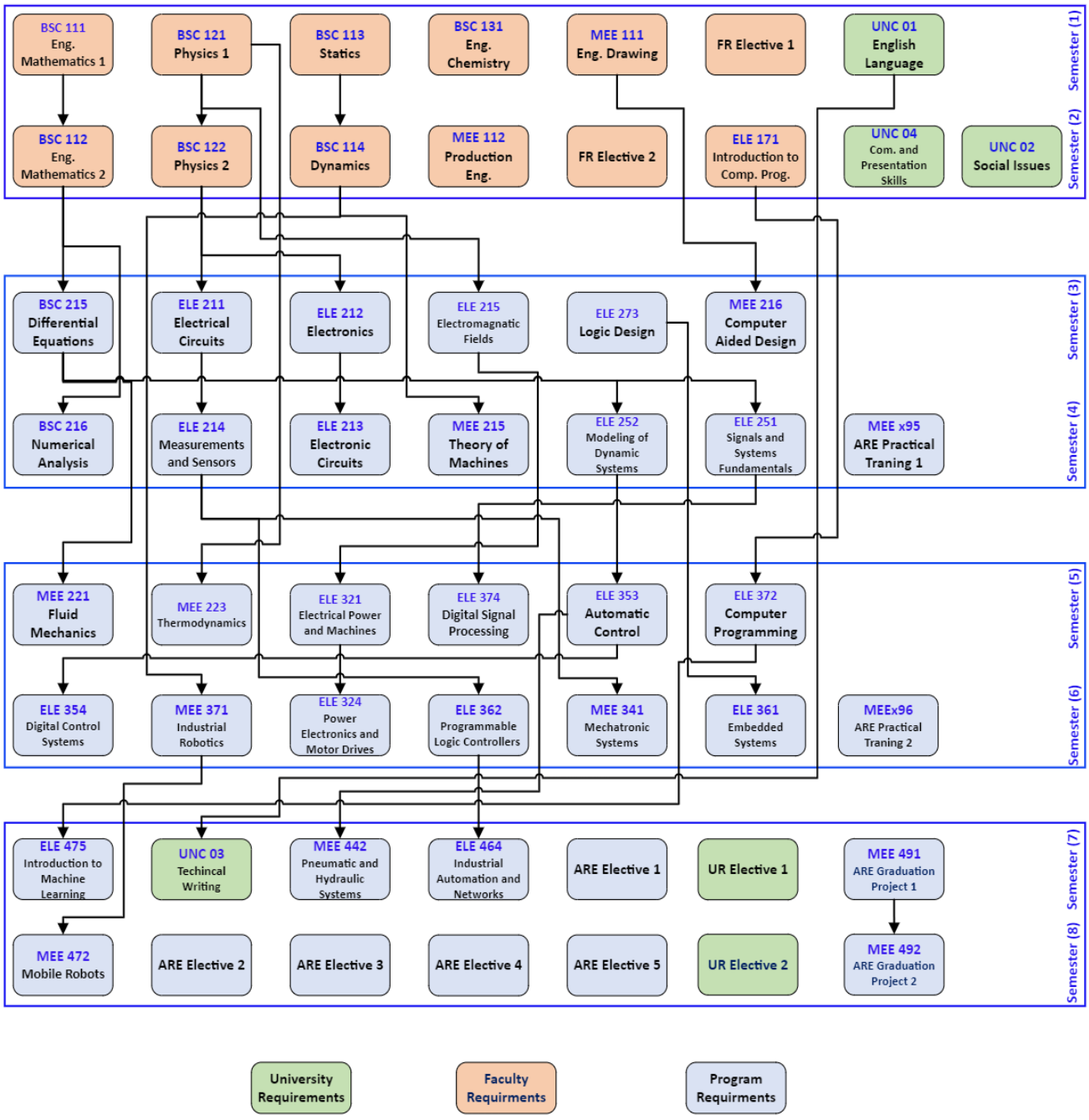
Code	Course Title	Weight			Contact Hours				Pre-requisites
		CH	ECTS	SWL	LEC	Tut	Lab	TT	
Semester (1)									
BSC 111	Engineering Mathematics 1	3	5	125	2	2	0	4	--
BSC 121	Physics 1	3	5	125	2	2	1	5	--
BSC 113	Statics	3	5	125	2	2	0	4	--
BSC 131	Engineering Chemistry	3	5	125	2	1	2	5	--
MEE 111	Engineering Drawing	2	4	100	1	2	0	3	--
--	FR Elective 1	2	3	75	2	0	0	2	--
UNC01	English Language	2	3	75	2	0	0	2	--
Total		18	30	750	13	9	3	25	
Semester (2)									
BSC 112	Engineering Mathematics 2	3	5	125	2	2	0	4	BSC 111
ELE 171	Introduction to Computer Programming	2	3	75	1	0	2	3	--
BSC 122	Physics 2	3	5	125	2	2	1	5	BSC 121
BSC 114	Dynamics	3	5	125	2	2	0	4	BSC 113
MEE 112	Production Engineering	2	3	75	1	0	2	3	--
--	FR Elective 2	2	3	75	2	1	0	3	--
UNC04	Communication and Presentation Skills	2	3	75	2	0	0	2	--
UNC02	Social Issues	2	3	75	2	0	0	2	--
Total		19	30	750	14	7	5	26	
Semester (3)									
BSC 215	Differential Equations	3	5	125	2	2	0	4	BSC 112
ELE 211	Electrical Circuits	3	6	150	2	2	1	5	BSC 122
ELE 212	Electronics	3	5	125	2	2	1	5	BSC 122
ELE 273	Logic Design	3	5	125	2	0	2	4	--
MEE 216	Computer Aided Design	2	4	100	1	0	2	3	MEE 111
ELE 215	Electromagnetic Fields	3	5	125	2	2	0	4	BSC 121
Total		17	30	750	11	8	6	25	
Semester (4)									
BSC 216	Numerical Analysis	3	5	125	2	2	0	4	BSC 112
MEE 215	Theory of Machines	3	5	125	2	2	0	4	BSC 114
ELE 213	Electronic Circuits	3	5	125	2	1	2	5	ELE 212
ELE 251	Signals and Systems Fundamentals	3	5	125	2	2	0	4	BSC 215
ELE 252	Modeling of Dynamic Systems	3	5	125	2	2	0	4	BSC 215
ELE 214	Measurements and Sensors	3	5	125	2	1	2	5	ELE 211
Total		18	30	750	12	10	4	26	

Semester (5)									
MEE 221	Fluid Mechanics	3	5	125	2	2	0	4	BSC 215
ELE 374	Digital Signal Processing	3	5	125	2	1	2	5	ELE 251
MEE 223	Thermodynamics	3	5	125	2	2	0	4	BSC 121
ELE 321	Electrical Power and Machines	3	5	125	2	2	1	5	ELE 215
ELE 372	Computer Programming	3	5	125	2	0	2	4	ELE 171
ELE 353	Automatic Control	3	5	125	2	2	1	5	ELE 252
Total		**18**	**30**	**750**	**12**	**9**	**6**	**27**	
Semester (6)									
ELE 361	Embedded Systems	3	5	125	2	0	2	4	ELE 273
ELE 463	Mechatronic Systems	3	5	125	2	2	1	5	ELE 214
ELE 324	Power Electronics and Motor Drives	3	5	125	2	2	1	5	ELE 321
ELE 362	Programmable Logic Controllers	3	5	125	1	0	3	4	ELE 214
MEE 371	Industrial Robotics	3	5	125	2	2	1	5	BSC 114
ELE 354	Digital Control Systems	3	5	125	2	2	1	5	ELE 353
Total		**18**	**30**	**750**	**11**	**8**	**9**	**28**	
Semester (7)									
ELE 464	Industrial Automation and Networks	3	6	150	2	0	2	4	ELE 362
MEE 442	Pneumatic and Hydraulic Systems	3	6	150	2	0	2	4	ELE 353
--	ARE Elective (1)	3	5	125	2	2	1	5	
ELE 475	Introduction to Machine Learning	3	4	100	2	0	2	4	ELE 372
--	UR Elective 1	2	2	50	2	0	0	2	
UNC03	Technical Writing	2	3	75	2	0	0	2	UNC01
MEE 491	ARE Graduation Project 1	2	4	100	1	0	2	3	
Total		**18**	**30**	**750**	**13**	**2**	**9**	**24**	
Semester (8)									
MEE 472	Mobile Robots	3	5	125	2	1	1	4	MEE 371
--	ARE Elective (2)	3	5	125	2	1	1	4	
--	ARE Elective (3)	3	5	125	2	1	1	4	
--	ARE Elective (4)	3	5	125	2	1	1	4	
--	ARE Elective (5)	3	5	125	2	1	1	4	
--	UR Elective 2	2	2	50	2	0	0	2	
MEE 492	ARE Graduation Project 2	2	3	75	0	0	4	4	MEE 491
Total		**19**	**30**	**750**	**12**	**5**	**9**	**26**	

8. شجرة المقررات الدراسية

يوضح الشكل التالي نموذج شجرة المقررات الدراسية للبرنامج:

Course Map for Automation and Robotics Engineering Program



9. توصيف المقررات الدراسية لبرنامج هندسة الاتمتة والروبوتات

Code: BSC 215		Subject: Differential Equations		Pre. Req.: BSC 112
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Definition and classification of differential equations- First-Order Ordinary Differential Equations- Second-Order Ordinary Differential Equations- Higher-Order Ordinary Differential Equations- Partial Differential Equations- Laplace transform and its applications -Numerical Methods for Differential Equations-Engineering applications

References:

S. A. Wirkus, and R. J. Swifi, "A Course of Ordinary Differential Equations", Taylor & Francis Group, LLC, 2015.

Code: BSC 216		Subject: Numerical Analysis		Pre. Req.: BSC 112
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Numerical Analysis: Sources of error in numerical computation, Round-off error and truncation error, Accuracy, and convergence of numerical algorithms. **Numerical Differentiation and Integration:** Finite difference approximations, Richardson extrapolation, Newton-Cotes formulas: Trapezoidal rule, Simpson's rule, Gaussian quadrature. **Difference equations:** Definition and classification of difference equations, First-order and higher-order difference equations, Solutions of difference equations: homogeneous and nonhomogeneous cases. **Z-Transform. Fourier analysis:** Fourier series representation of periodic functions, Fourier transform and inverse Fourier transform. **Engineering applications**.

References:

[1] Burden, R. L. (2011). Numerical analysis. Brooks/Cole Cengage Learning.
[2] Hamming, R. W. (2012). Introduction to applied numerical analysis. Courier Corporation.

Code: ELE 211		Subject: Electrical Circuits		Pre. Req.: BSC 122
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Elements of electrical circuits - Simple resistive circuits - Analysis of DC circuits - Theories of electrical circuits – Capacitors and Inductors - First-Order Circuits - AC fundamentals -AC Circuits Sinusoidal Steady - State Analysis - Three-Phase Circuits- Power and power factor- Resonance circuits - Three-phase circuits.

References:

[1] Charles K. Alexander and Matthew N. O. Sadiku, Fundamentals of Electric Circuits, McGraw-Hill, Fourth Edition, 2009.
[2] J. W. Nilsson, "Electric Circuits", Pearson; 11th edition, 2018.

Code: ELE 212		Subject: Electronics		Pre. Req.: BSC 122
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Semiconductors: Overview of PN junctions, their biasing, and types of diodes. Bipolar Junction Transistors (BJTs): properties, and applications in DC circuits. Field-Effect Transistors (JFET & MOSFET): properties and applications. Introduction to CMOS technology and its applications.

References:

[1] Thomas L. Floyd David L. Buchla “Electronics Fundamentals Circuits, Devices and Applications” , Eighth Edition, PEARSON, 2014.
[2] T. Floyd, "Electronic Devices", 10th edition, Pearson, 2018

Code: ELE 213		Subject: Electronic Circuits		Pre. Req.: ELE 212
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50%

Course Syllabus:

Multi stages amplifier – Feedback amplifier oscillators – Power amplifier- High frequency amplifiers - Integrated Circuit amplifiers – Wide band amplifier- Operational Amplifiers characteristics- Wave generation and shaping- Application of non-linear circuits- Design of analog electronic circuits- Circuit simulation- Response description and printed circuit building.

References:

[1] Thomas L. Floyd David L. Buchla "Electronics Fundamentals Circuits, Devices and Applications" , Eighth Edition, PEARSON, 2014.
[2] T. Floyd, "Electronic Devices", 10th edition, Pearson, 2018.

Code: ELE 214		Subject: Measurements and Sensors		Pre. Req.: ELE 211
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50%

Course Syllabus:

Introduction to measurement systems – Instrument types and performance characteristics — Measurement errors – Galvanometers - Analog Ammeters, Voltmeters, and Ohmmeters – measurement errors- Bridges- Filters- Sensor performance criteria and selection - Thermocouples - Resistive sensors – Inductive sensors - Capacitive sensors - Piezoelectric sensors - Encoders and tachometers - A/D and D/A converters

References:

[1] A. Morris, "Measurement and Instrumentation Theory and Application", Academic Press; 2nd edition, 2015.
[2] C. de Silva," Sensors and Actuators: Engineering System Instrumentation" , CRC Press; 2nd edition, 2015.

Code: ELE 215		Subject: Electromagnetic Fields		Pre. Req.: BSC 121
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Electrostatic Fields - Electric flux and gauss's law- Electrical Potential – Electric boundary conditions – Dipole moment – Capacitance - Poisson and Laplace equations – Piot & savart law- Amperes law- Vector potentials- Magnetic boundary conditions- Magnetic flux – Force and energy in magnetic field- Coils – Time varying field- Maxwell's equations- Wave propagation – Introduction to transmission line theory- Propagation in losses and lossless media- Reflection and refraction. Electromagnetics in AC electrical machines.

References:

[1] M. N. O. Sadiku, "Elements of Electromagnetics, " 6th Ed, Oxford Press, 2006.
[2] William H. Hayt, Johan A. Buck, "Engineering Electromagnetics" 8th Ed., McGraw Hill, 2012

Code: ELE 321		Subject: Electrical Power and Machines		Pre. Req.: ELE 215
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Power: Electrical power systems - three phase systems - transformers - Transmission line models - Voltage and frequency control - effective and ineffective power - Understand elements, distribution, protection, and quality improvement of power system.
Machines: The theory of operation - Direct Current motors. The speed, torque, and current characteristics - DC motors applications - stepper motors - Permanent-magnet DC motor and Low-inertia DC Motors. Three phase induction motors.

References:

[1] Nilsson, J.W. and S.A. Riedel, Electric circuits. 2015: Pearson Upper Saddle River, NJ.
[2] Slade, P.G., Electrical contacts: principles and applications. 2017: CRC press.

Code: ELE 324		Subject: Power Electronics and Motor Drives		Pre. Req.: ELE 321
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Introduction to Power Electronics – Power Electronic Semiconductor Devices –Firing Circuits – Uncontrolled Rectifiers – Controlled Rectifiers –Three Phase Controlled Rectifiers – Single Phase AC Voltage Regulators – Commutation circuits - Heat sinks specification- Voltage controllers & AC static switches. DC to DC Converters. PWM inverters. DC motor Drives. AC motor Drives.

References:

[1] N. Mohan, T. M. Undeland, and W.P. Robbins Power Electronics: Converters, Applications and - Design, John Wiley, 3rd edition, 2016.
[2] M. H. Rashid, Power electronics: circuits, devices, and applications, Pearson Prentice Hall, 4th edition, 2014.

Code: ELE 251		Subject: Signals and Systems Fundamentals		Pre. Req.: BSC 215
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Definition of signals and systems- Classification of signals- Properties of signals - System classification: (linear vs. nonlinear, time-invariant vs. time-varying, causal vs. non-causal) - Time-Domain Analysis of Continuous-Time Signals and Systems- Frequency Domain Analysis of Continuous-Time Signals and Systems- Sampling and Discrete-Time Signals- Frequency-Domain Analysis of Discrete-Time Signals and Systems.

References:

[1] Phillips, C. L., Parr, J. M., Riskin, E. A., & Prabhakar, T. (2003). Signals, systems, and transforms (p. 209). Upper Saddle River: Prentice Hall.
[2] B. Boashash , "Time-Frequency Signal Analysis and Processing", Academic Press; 2nd edition, 2015.

Code: ELE 374		Subject: Digital Signal Processing		Pre. Req.: ELE 251
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50%

Course Syllabus:

Basics of digital signal processing- Characterization, and classification of Discrete-Time (DT) signals and systems- Typical DT signal processing operations - Introduction to digital filters- Digital Filter Structures –Finite-impulse response (FIR) digital filter design - Infinite-impulse response (IIR) digital filter design – Multi-rate signal processing- Hands-on experiments using DSP hardware and software tools (e.g., MATLAB)- Design and implementation of DSP algorithms- Real-world applications and projects in DSP.

References:

Tan, L., & Jiang, J. (2018). Digital signal processing: fundamentals and applications. Academic press.

Code: ELE 252		Subject: Modeling of Dynamic Systems		Pre. Req.: BSC 215
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to Dynamic Systems- Electrical Systems Modeling- Mechanical Systems Modeling- Thermal Systems Modeling- Electromechanical systems modeling -Hydraulic and Pneumatic Systems Modeling- State-Space Representation- Nonlinear Systems Modeling- Nonlinear systems linearization- Modeling Tools and Software- Applications of Dynamic System Modeling.

References:

[1] Ogata, Katsuhiko. System dynamics. 2004.
[2] Chaturvedi, D. K. (2017). Modeling and simulation of systems using MATLAB and Simulink. CRC press.

Code: ELE 353		Subject: Automatic Control		Pre. Req.: ELE 252
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Introduction to systems and their classifications: open loop, and closed loop systems – Block diagrams models- signal flow graphs- Time domain analysis – Frequency domain analysis -Stability of linear systems – Root locus – Classical control techniques (PID control, lead-lag compensation) –-PID controllers design and tuning –Experiments and simulations using control system software (e.g., MATLAB/Simulink). Design and analysis of control systems through practical exercises and projects.

References:

[1] Kuo, B. C., & Golnaraghi, M. F. (1995). Automatic control systems (Vol. 8). Englewood Cliffs, NJ: Prentice Hall.
[2] Franklin, Gene F., J. David Powell, Abbas Emami-Naeini, and J. David Powell. Feedback control of dynamic systems. Vol. 4. Upper Saddle River: Prentice hall, 2002.
[3] Dorf, Modern Control Systems, 12th edition, Prentice Hall, 2012.

Code: ELE 354		Subject: Digital Control Systems		Pre. Req.: ELE 353
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Review on Control Systems-Continuous-time vs. Discrete-time Systems- Z-Transform and pulse transfer function- The stability of digital control systems. Designing a digital controller from a continuous controller- Discrete state space representation - Design and tuning of digital PID controllers. Simulation of digital control systems using MATLAB.

References:

[1] K. Ogata, " Discrete Time Control Systems ", Prentice-Hall, Inc,Upper Saddle River, Nj, 2nd ed. , 1995.
[2] M. Sami Fadali and Antonio Visioli, "Digital Control Engineering: Analysis and Design", Academic Press (Elsevier) 2nd edition, 2013.

Code: ELE 273		Subject: Logic Design		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50%

Course Syllabus:

Introduction to logic circuits . Logic circuit simplification using Boolean algebra, DeMorgan theorem, and K-Map. Combinational logic circuits like adders, subtractors, decoders, encoders, multiplexers, and de-multiplexers. Sequential logic circuits like Flip-Flops, counters (synchronous and asynchronous), and registers. The course also explores design methodologies, programmable logic devices, and gate arrays. Hands-on practice with state tables, state transition diagrams, and counter design is included.

References:

M Morris, M., & Michael, D. (2013). Digital Design With an Introduction to the Verilog HDL FIFTH EDITION

Code: ELE 372		Subject: Computer Programming		Pre. Req.: ELE 171
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50%

Course Syllabus:

Introduction to advanced programming – Advanced Data Structures-Advanced Sorting and Searching Algorithms-Dynamic Programming- Graph Algorithms - Introduction to Object-oriented programming - Functional Programming- Presenting these topics using C++ and Python.

References:

[1] Mark Lutz, “Learning Python”, O’ Reilly 5th edition, 2013.
[2] Sommerville, software engineering, 10 ed., Pearson India 2018.

Code: MEE 216		Subject: Computer Aided Design		Pre. Req.: MEE 111
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
1	0	2	50 %	50 %

Course Syllabus:

The computer-Aided Design (CAD) course covers CAD software fundamentals, including interface navigation, 2D drawing, and geometric construction. It progresses to 3D modeling, assembly modeling, and rendering techniques. Topics include orthographic projections, dimensioning, advanced modeling techniques, and CAD file management. Students will engage in hands-on projects, exploring real-world engineering applications. The course also introduces CAM integration and future CAD trends, providing a comprehensive understanding of CAD in engineering.

References

[1] Randy H. Shih, "Learning SOLIDWORKS 2023: Modeling, Assembly and Analysis", SDC Publications, 2023.
[2] David Planchard, "Engineering Design with SOLIDWORKS", SDC Publications, 2021.

Code: MEE 215		Subject: Theory of Machines		Pre. Req.: BSC 114
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Identify mechanisms and predict their motion - Calculate the degrees of freedom of mechanisms - Determine the positions, velocities and accelerations of links and points on mechanisms - Calculate dynamic joint forces of mechanisms - Balance simple rotating objects - Analyze and modify existing mechanisms.

References

[1] Skvarenina T. L, ana Dewitt W. E. , “Electrical Power and Controls”,Prentic Hall , London, 2009.
[2] S. K. Bhattacharya, B. Singh, "Control of Machines," revised 2nd edition, New Age International (P) Ltd., 2006.

Code: MEE 221		Subject: Fluid Mechanics		Pre. Req.: BSC 215
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Fluid properties – Fluid statics and kinematics – Flow in pipes – Pumps – Valves – Dimensionless analysis and similitude– Buoyancy – Kinematics of three-dimensional fluid motions – Fluid masses moving with acceleration – Vortex motion – Hydrodynamics – Momentum equation - Euler's and Bernoulli's equations.

References

[1] B. R. Munson, D. F. Young and A. T. Okiishi, "Fundamentals of Fluid Mechanics ," 4th edition, John Wiley and Sons, Inc., New York, 2002.
[2] F. M. White, "Fluid Mechanics," 7th Edition, McGraw-Hill, 2011.

Code: MEE 223		Subject: Thermodynamics		Pre. Req.: BSC 121
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Heat and work, first law of thermodynamics, properties of steam and gases, steam tables and charts, entropy, second law, air standard cycle, compressors, psychrometry of gas mixtures, reversibility, availability and second law efficiency, gas turbine power cycles, steam turbine power cycles, combined gas-vapor power cycles.

References

[1] Yunus A. Cengel and Michael A. Boles, Thermodynamics: an engineering approach, 8th edition, New York: McGraw-Hill.
[2] Borgnakke, Claus, and Richard E. Sonntag, Fundamentals of Thermodynamics, Wiley Global Education, 9th edition.

Code: ELE 463		Subject: Mechatronic Systems		Pre. Req.: ELE 214
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50 %

Course Syllabus:

Introduction to Mechatronics - Mechanisms for motion electronic components for mechatronic systems –Transmission - Mechanical actuator systems (Mechanical elements - Kinematic chains - Cam mechanisms - Gears - Flexible mechanical elements - Brakes) - Electrical actuation systems – Case study and computer simulation.

References

[1] J. Paolo Davim, "Mechatronics," Wiley-ISTE, April 2011.

Code: ELE 361		Subject: Embedded Systems		Pre. Req.: ELE 273
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50%

Course Syllabus:

Introduction to microcontroller - Overview of the Microcontroller families – Addressing modes- Embedded C programming language (fundamentals of Embedded C) –General purpose I/O -Timer modules- Interrupt standard mechanism- analog to digital converter (ADC) and pulse width modulations (PWM)- Serial Communications.

References:

[1] Mazidi, M. A., Naimi, S., & Naimi, S. (2017). The AVR microcontroller and embedded systems: using Assembly and C. MicroDigitalEd. com.
[2] Kevin M. Lynch, Nicholas Marchuk, Matthew L. Elwin, Embedded Computing and Mechatronics with the PIC32 Microcontroller, Elsevier, 2016.

Code: ELE 362		Subject: Programmable Logic Controllers		Pre. Req.: ELE 214
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
1	0	3	50 %	50%

Course Syllabus:

Introduction to classical control - Fundamentals to programmable logic controllers (PLCs) - Input-output devices - I/O processing - Ladder and functional block programming - IL, SFC and ST programming methods – Function and function blocks - Jump and call – Timers – Counters - Shift registers - Applications Using PLCs.

References:

[1] Bolton, William. Programmable logic controllers. Newnes, 2015.
[2] Information and training automation and devices, SIMATIC S7, Version 5.6, 2008.

Code: MEE 371		Subject: Industrial Robotics		Pre. Req.: BSC 114
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50 %

Course Syllabus:

Introduction to robotics - Industrial robotics applications - Types of robot manipulators – Main components of the robot manipulator (electrical – drives – mechanical - grippers) - Homogeneous transformations - Forward and inverse kinematics - Path and trajectory planning – Motion control – Introduction to robot programming.

References

[1] Spong, Mark W., Seth Hutchinson, and Mathukumalli Vidyasagar. "Robot modeling and control. " John Wiley & Sons, 2005.
[2] Niku, Saeed B. "Introduction to Robotics: Analysis, Control, Applications." John Wiley & Sons, 2010.

Code: ELE 464		Subject: Industrial Automation and Networks		Pre. Req.: ELE 362
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50 %

Course Syllabus:

Introduction to industrial automation systems – Supervisory control and data acquisition (SCADA) systems – Distributed control systems (DCS) – Foundation field bus – Data communication – Batch automation systems - Hybrid systems - Cyber security in industrial automation – introduction to networked control systems

References:

[1] B.R. Mehta, Y.J. Reddy, Industrial Process Automation Systems, Butterworth-Heinemann, 2015.
[2] A. Boyer , "SCADA :Suervisory control And Data Acquisition" 2nd edition, ISA, 2004
[3] M. Elshafei, " Modern Distributed Control Systems, 1st Edition, CreateSpace independan tpublishing platform, 2016.

Code: MEE 442		Subject: Pneumatic and Hydraulic Systems		Pre. Req.: ELE 353
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50 %

Course Syllabus:

Introduction – Actuators – Valves - Compressed air - Sensor and monitor equipment's - Piping and connections - Pneumatic systems control – Cylinders – Motors – Valves - Hydraulic Accumulators - Heat Exchanger – Filters - Piping and connections - Hydraulic systems control.

References:

[1] M. Jelali, A. Kroll, "Hydraulic servo-systems: modelling, identification and control," Springer Science & Business Media, 2012.
[2] P. Beater, "Pneumatic drives," Springer-Verlag Berlin Heidelberg, 2007.
[3] J. Takosoglu, P.A. Laski, R.F. Dindorf, "Fuzzy logic positioning system of electro-pneumatic servo-drive," INTECH Open Access Publisher, 2010.

Code: ELE 475		Subject: Introduction to Machine Learning		Pre. Req.: ELE 372
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50 %

Course Syllabus:

Introduction – Machine learning definition and basics - Learning models – Classification - Clustering - Introduction to deep learning – Introduction to reinforcement learning.

References

[1] Gopinath Rebala, Ajay Ravi, Sanjay Churiwala, An Introduction to Machine Learning, Springer, 2019.

Code: MEE 472		Subject: Mobile Robots		Pre. Req.: MEE 371
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

Introduction to mobile robots – Locomotion - Sensors for mobile robots – Mobile robot main components (Electrical – Drives – Mechanical – wheels) - Mobile robot kinematics –- Mobile robot Localization and navigation - Motion control - Introduction to mobile robot programming.

References

[1] Klancar, Gregor, et al. "Wheeled mobile robotics: from fundamentals towards autonomous systems." Butterworth-Heinemann, 2017.
[2] Nonami, Kenzo, et al. "Autonomous flying robots: unmanned aerial vehicles and micro aerial vehicles." Springer Science & Business Media, 2010.

Code: ELE 455		Subject: Process Control Systems		Pre. Req.: ELE 353
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50 %

Course Syllabus:

Fundamental principles of process control - Importance of process control - Closed loop control - Terms process variable setpoint - Measured variables, process variables, and manipulated variables - Interaction between controllers - Sensors and actuators – Controller operating parameters - PID controllers – Controller units - The analysis of systems - Examples of single control loops - Pressure control loops - Flow control loops - Level control loops - Temperature control loops.

References

[1] W. C. Dunn, "Fundamentals of Industrial Instrumentation and Process Control," McGraw-Hill Companies. 2005.

Code: MEE 473		Subject: Robotic Operating Systems		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50 %

Course Syllabus:

Introduction to robotic operating systems (ROS) - ROS file system level - ROS computation graph level - ROS programming – Modeling the robot using ROS – Simulating the robot manipulator – Simulating mobile robot – Programming the robots using ROS.

References

[1] Lentin Joseph , and Jonathan Cacace " Mastering ROS for Robotics Programming - Second Edition: Design, build, and simulate complex robots using the Robot Operating System Paperback". 2018.

Code: ELE 476		Subject: Digital Image Processing		Pre. Req.: ELE 374
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50 %

Course Syllabus:

Introduction to the theory and applications of 2D signal and image processing: 2D signals and systems analysis - 2D sampling and quantization - 2D signals and image transforms - 2D FIR filter design - Image formation - Image enhancement - Image restoration - Image compression - Image reconstruction from projections - Image compression - Image segmentation - Morphological operations - Super resolution - Wavelets.

References

[1] Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Pearson, 2017, Fourth Edition.

Code: ELE 456		Subject: Modern Control Systems		Pre. Req.: ELE 353
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50 %

Course Syllabus:

Introduction - PID controller concept and design- State variables models (i.e., the state variables of dynamic systems and the state differential equations) - Transfer function from the state equations - Time response and state transition matrix - The stability of state variables systems - State feedback control systems (i.e. pole placement method, full state feedback control method, observer design, Kalman filters and observer -based state feedback control) – Case studies.

References

[1] Dorf, Modern Control Systems, 12th edition, Prentice Hall, 2012.
[2] K. Ogata "Modern Control Engineering" (5th Edition), Wiley and Sons, Sep 4, 2009.
[3] F. Golnaraghi and B. CKuo"Automatic Control Systems"9th edition, Wiley and Sons, 2009.

Code: MEE 474		Subject: Embedded Computing for Robotics		Pre. Req.: ELE 361
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

Introduction to embedded computing – Embedded systems - Embedded architectures - Design and programming of advanced microcontrollers for robots - Embedded applications and technologies – Applications for robotics systems.

References:

Bräunl, Thomas, Embedded Robotics, Springer, 2008.

Code: MEE 444		Subject: Selected Topics in Mechatronic Engineering		Pre. Req.: ELE 463
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

This course is designed for specialized and current topics in Mechatronics Engineering. Course topics are chosen so they do not duplicate other courses in the program. The course needs the program approval.

References:

Based on the topic.

Code: ELE 463		Subject: Advanced Programmable Logic Controllers		Pre. Req.: ELE 362
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50 %

Course Syllabus:

Overview to PLCs - Internal relays - Function and function blocks - Jump and call – Timers – Counters - Shift registers - Data handling - Designing systems - Industrial communication protocols of PLCs (modbus RTU, Ethernet/IP, Ethernet TCP/IP, modbus TCP/IP, profibus, profinet) - MPI network - Hardware and software redundancy - Applications of PLCs in Industrial automation engineering.

References

[1] W. Bolton, Programmable Logic Controllers (PLCs), Elsevier Ltd. ,2009.
[2] Information and training automation and devices, SIMATIC S7, Version 5.6, 2008.
[3] Using the manuals for any type of the PLCs which will be used in the laboratory

Code: MEE 445		Subject: Autotronics		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

Ground vehicles types - Vehicle main systems (Propulsion systems, braking systems, suspension systems, steering systems) - Engine starting system - fuel supply system and ignition system - Air conditioning and climate control system - Electric vehicles - Examples of autotronic systems.

References

[1] Bonnick, A. and Newbold, D., 2011. A practical approach to motor vehicle engineering and maintenance.
[2] Routledge. Ribbens, W., 2017. Understanding automotive electronics: an engineering perspective. Butterworth-heinemann
[3] Reif, K., 2014. Automotive Mechatronics. Germany: Springer Fachmedien Wiesbaden.
[4] Levine, W.S. ed., 2018. The Control Systems Handbook: Control System Advanced Methods. CRC press.

Code: MEE 479		Subject: Rehabilitation Robots		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

Introduction to rehabilitation robots - Assistive robotics terminology – Assistive robotic control systems including mechanical design - Sensors & actuators, and user interfaces - Rehabilitative robotics for neurologic injuries such as stroke - Prosthetics to enable mobility function in upper and lower limb amputees - Social robotics for cognitive impairments and developmental disorders (such as autism) - Guidelines for designing assistive robots - Ethical and regulatory considerations in the design of assistive robots.

References

[1] Pedro Encarnação and Albert Cook (Editors), “Robotic Assistive Technologies: Principles and Practice (Rehabilitation Science in Practice Series)” 1st Edition, 2017.

Code: ELE 477		Subject: Fundamentals of Biomedical Engineering		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

The Biomedical Engineering course covers tissue engineering, artificial organs, and prosthetics, including lower and upper limb designs. It includes basics of medical instrumentation, biological signals, and biomedical sensors (e.g., ECG, EEG). Topics include bio signal processing, physiological origins, signal acquisition, frequency domain representation, Fourier and wavelet analysis, sampling, filtering, and EKG analysis. The course also addresses medical imaging techniques.

References

Joseph J. Carr, John M. Brown, "Introduction to Biomedical Equipment Technology” 4th Edition - ISBN-10: 0130104922.

Code: ELE 478		Subject: System Physiology		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

Fundamental concepts and terminologies - Anatomical basis of human and biological systems – Integumentary – Skeletal – Muscular – Cardiovascular – Lymphatic – Nervous – Digestive - Endocrine and respiratory systems - Engineering approach toward understanding biological function.

References:

Atheena Milagi Pandian, " Anatomy and Human Physiology: Biomedical Engineering," Kindle Edition, November 3, 2016.

Code: MEE 443		Subject: Selected Topics in Bio-Mechatronics		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

This course is designed for specialized and current topics in Bio-Mechatronics. Course topics are chosen so they do not duplicate other courses in the program. The course needs the program approval.

References:

Based on the topic.

Code: MEE 475		Subject: Advanced Robotics		Pre. Req.: ELE 463
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

Trajectory generation – Position control of manipulators – Force control of manipulators – Hybrid position - Force control scheme – Robot programming languages - Introduction to the use of homogeneous coordinates - dynamics, and camera orientation - sensors and actuators, control - The vision – Robots connected to computer - Applications of robots in advanced manufacturing systems - Industrial applications.

References

[1] A. Dutta, "Robotic Systems - Applications, Control and Programming," InTech, 2012.
[2] H. Samani, "Cognitive Robotics," CRC Press, 2015.

Code: MEE 476		Subject: Robotics Applications in Manufacturing Systems		Pre. Req.: ELE 463
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

Introduction - Manufacturing system components (feeding-rotating tables-belt conveyer - rolling) - Robot system in manufacturing – Specifications – Programming - Applications.

References

Alex Khang, Vugar Abdullayev, Anuradha Misra, Eugenia Litvinova, Machine Vision and Industrial Robotics in Manufacturing, 2024, CRC Press

Code: MEE 477		Subject: Robot Vision		Pre. Req.: ELE 374
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

Tracking and density propagation. Visual surveillance and activity monitoring. Medical imaging. Product inspection. Color sorting. Surface detection. Automated visual grading. Vision-based quality control. Defect detection. Estimating the size and shape of objects. Applications of machine vision in robotic systems.

References

[1] Stanisic, M.M., 2014. Mechanisms and Machines: kinematics, dynamics, and synthesis. Cengage Learning.
[2] Angeles, J. and Angeles, J., 2002. Fundamentals of robotic mechanical systems (Vol. 2). New York: Springer-Verlag.
[3] Huat, L.K., 2006. Industrial robotics: programming, simulation, and applications.

Code: MEE 478		Subject: Selected Topics in Advanced Robotics Applications		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

This course is designed for specialized and current topics in Advanced Robotics. Course topics are chosen so they do not duplicate other courses in the program. The course needs the program approval.

References:

Based on the topic.

Code: ELE 457		Subject: Introduction to Intelligent Control Systems		Pre. Req.: ELE 353
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

Introduction to intelligent control systems - Fuzzy set theory - Fuzzy operations and relations - Fuzzy rule-based system - Design and structure of fuzzy logic control systems - Types of fuzzy logic control systems - Fuzzy logic control system applications - Introduction to neural networks – Biological and artificial models of neural networks - Neural networks learning rules - Back-propagation neural networks - Neural networks in process modeling and control - Neural networks Applications.

References

[1] Hung T. Nguyen, A First Course in Fuzzy and Neural Control, CRC press, 2002.
[2] Ali Zilouchian Mo Jamshidi , Intelligent Control Systems Using Soft Computing Methodologies, CRC, 2001.

Code: ELE 467		Subject: Internet of Things		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Introduction to IOT - IOT and SCADA – Smart sensors - IOT standards - Components of IOT system – Applications (Intelligent traffic systems - Smart parking – Smart homes) - Challenges in IOT implementation.

References

Vlasios Tsiatsis, Stamatis Karnouskos, Catherine Mulligan, Internet of Things: Technologies and Applications for a New Age of Intelligence Elsevier, 2019.

Code: ELE 466		Subject: Wireless Sensors in Industrial Networks		Pre. Req.: ELE 214
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Introduction to networked Control Systems - Wireless networks, architectures and technologies - Wireless sensor network platforms: Hardware and Software - Communication architecture and protocols for WSN (MAC, Link, Routing) - Energy management - Sensor data acquisition, processing and handling, Signal processing, target localization and tracking, self-organization - Modelling and Simulation of WSN – Applications in Industrial four.

References

[1] Feng Zhao and Leonidas Guibas, “Wireless sensor networks: an information processing approach,” Morgan Kaufmann, 2004.
[2] Gregory J. Pottie and William Kaiser, “Principles of embedded networked systems design,” Cambridge University Press, 2005.

Code: ELE 416		Subject: Fault Detection and Diagnosis in Industrial Processes		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

This course provides engineering undergraduates with a thorough understanding of fault detection and diagnosis in industrial processes. It covers fundamental principles, methodologies, and tools for detecting and diagnosing faults in both static and dynamic processes. Key topics include basic concepts, model-based, data-driven, and machine learning techniques for fault detection and diagnosis. Through lectures, practical exercises, and case studies, students will gain skills to effectively identify and analyze faults in various industrial applications.

References

[1] Ding, Steven X. Advanced methods for fault diagnosis and fault-tolerant control. Springer Berlin Heidelberg, 2021.
[2] Isermann, Rolf. Fault-diagnosis systems: an introduction from fault detection to fault tolerance. Springer Science & Business Media, 2006.
[3] Chiang, Leo H., Evan L. Russell, and Richard D. Braatz. Fault detection and diagnosis in industrial systems. Springer Science & Business Media, 2000.

Code: ELE 465		Subject: Selected Topics in Industrial Automation		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

This course is designed for specialized and current topics in Industrial Automation. Course topics are chosen so they do not duplicate other courses in the program. The course needs the program approval.

References:

Based on the topic.

Code: MEE 491		Subject: ARE Graduation Project 1		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
1	0	2	-	-

Course Syllabus:

An independent project under the supervision of a staff member; to enable the student to understand and apply the knowledge gained throughout his coursework to an engineering problem at large scale.

Code: MEE 492		Subject: ARE Graduation Project 2		Pre. Req.: MEE 491
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
0	0	4	-	--

Course Syllabus:

This project is an extension to graduation project 1. In this part the student under the supervision of his supervisor may make a prototype or introduce an idea to be implemented in the future. The student should submit a report.

الباب السابع

برنامج هندسة ميكاترونيات السيارات

Automative Mechatronics Engineering Program

اسم البرنامج	هندسة ميكاترونيات السيارات Automative Mechatronic Engineering (AME)			
الكلية	الهندسة			
الجامعة	جامعة الريادة للعلوم والتكنولوجيا			
الساعات المعتمدة (CH)	145			
النقاط المعتمدة (ECTS)	240			
عدد المقررات	55 مقرر			
عدد ساعات الاتصال	محاضرات	تمارين	معمل	الإجمالي
	99	62	42	203
توزيع المقررات طبقا لمتطلبات مستويات التعلم				
	CH	ECTS	النسب	المقررات
متطلبات الجامعة	12	16	% 8	6
متطلبات الكلية	31	51	% 21	12
متطلبات البرنامج	59	100	% 41	20
متطلبات التخصص الرئيسي	43	73	% 30	17
بيانات إحصائية				
	CH	ECTS	النسب	المقررات
نسبة مقررات العلوم الاساسية	27	45	% 19	9
نسبة المقررات الاختيارية	23	35	% 16	9

1. رسالة البرنامج

يسعي برنامج هندسة ميكاترونيات السيارات لإعداد مهندسين متميزين ومؤهلين من الناحيتين العلمية والعملية والمهنية في مجال ميكاترونيات السيارات لتلبية احتياجات سوق العمل، قادرين على نقل المعارف، وتوطين التقنية، ومتمكنين من ادوات البحث العلمي، وتقديم الاستشارات الهندسية المتخصصة لتطوير المجتمع الصناعي وخدمة المجتمع المحلي والإقليمي.

2. أهداف البرنامج

يهدف برنامج هندسة ميكاترونيات السيارات الى تقديم خريج قادر على:

1. تطبيق المعرفة بالرياضيات الهندسية والعلوم الطبيعية واساسيات الهندسة وعلوم الحاسب وقدرات التفكير التحليلي والنقدي والمنهجي لتحديد وحل المشكلات الهندسية.
2. تنمية القدرات البشرية لسد حاجة المجتمع الصناعي من مهندسي ميكاترونيات السيارات.
3. المشاركة في تحقيق الخطة الوطنية للتنمية واستخدام المعرفة لخدمتها بهدف تنمية المجتمع.
4. الاسهام في رفع الكفاءة المهنية وتكوين جيل من المهندسين المتميزين والباحثين المؤهلين في مجال هندسة ميكاترونيات السيارات.
5. تمكين الطالب من التعلم الذاتي المستمر لاكتساب القدرة على إجراء البحث العلمي وتطوير القطاع الصناعي وخدمة المجتمع.

3. مواصفات الخريج

يحرص برنامج هندسة ميكاترونيات السيارات على تخريج مهندسين متميزين ومؤهلين لدخول سوق العمل ويراعى فيهم تحقيق المواصفات الأتية:

1. Master a wide spectrum of engineering knowledge and specialized skills and can apply acquired knowledge using theories and abstract thinking in real life situations.
2. Apply analytic critical and systemic thinking to identify, diagnose and solve engineering problems with a wide range of complexity and variation.
3. Behave professionally and adhere to engineering ethics and standards.

4. Work in and lead a heterogeneous team of professionals from different engineering specialties and assume responsibility for own and team performance.
5. Recognize his/her role in promoting the engineering field and contribute in the development of the profession and the community.
6. Value the importance of the environment, both physical and natural, and work to promote sustainability principles.
7. Use techniques, skills and modern engineering tools necessary for engineering practice.
8. Assume full responsibility for own learning and self-development, engage in lifelong learning and demonstrate the capacity to engage in post- graduate and research studies.
9. Communicate effectively using different modes, tools and languages with various audiences; to deal with academic/professional challenges in a critical and creative manner.
10. Demonstrate leadership qualities, business administration and entrepreneurial skills.
11. Design of automative mechatronic systems considering the principles of mechanical, manufacturing, electrical/electronic, and computer engineering.
12. Analyze, program, install, integrate, troubleshoot and diagnose systems including automative mechatronic systems.
13. Solve problems in the areas of integrated mechanics, electronics and computer systems.

4. المعايير الأكاديمية للبرنامج

بالإضافة الى الجدارات لمعظم البرامج الهندسية (Level A) وبالتوافق مع المعايير الأكاديمية المرجعية (NARS 2018) فأن خريجي برنامج هندسة ميكاترونيات السيارات يتعين أن يكونوا قادرين على تحقيق (Level B - Competencies for Mechanical Engineering) على النحو التالي:

B1. Model, analyze and design physical systems applicable to the specific discipline by applying the concepts of: Thermodynamics, Heat Transfer, Fluid Mechanics, solid Mechanics, Material Processing, Material Properties, Measurements, Instrumentation, Control Theory and Systems, Mechanical Design and Analysis, Dynamics and Vibrations.

B2. Plan, manage and carry out designs of mechanical systems and machine elements using appropriate materials both traditional means and computer-aided tools and software contemporary to the mechanical engineering field.

B3. Select conventional mechanical equipment according to the required performance.

B4. Adopt suitable national and international standards and codes; and integrate legal, economic and financial aspects to: design, build, operate, inspect and maintain mechanical equipment and systems.

بالإضافة الى (Level A and Level B) فأن خريجي برنامج هندسة ميكاترونيات السيارات يتعين أن يكونوا قادرين على تحقيق (Level C - Competencies for Automative Mechatronic Engineering) على النحو التالي:

C1. Illustrate the basic concepts and theories of mathematics, sciences, engineering projection and their applications within the field of Automative mechatronics engineering.

C2. Analyze automative mechatronic systems using scientific, mathematical and computer-based models and asses the limitations of particular cases.

C3. Identify and discuss vehicles as complex systems from technical and social perspectives through a broad platform in automotive engineering.

C4. Design, model, and analyze electrical, electronic, mechanical, or digital components for automative mechatronic applications, utilizing appropriate optimization tools.

C5. Exercise creative approaches to the analysis and solution of problems in automative mechatronic engineering.

5. المقررات الدراسية

للحصول على درجة بكالوريوس الهندسة في تخصص هندسة ميكاترونيات السيارات ولتحقيق مخرجات التعلم المستهدفة للبرنامج وللتوافق مع مواصفات وجدارات الخريج الموصفة من قبل هيئة ضمان الجودة والاعتماد فإنه يتعين على الطالب أن يجتاز بنجاح المقررات التالية والمقسمة الى (متطلبات الجامعة – متطلبات الكلية – متطلبات البرنامج – متطلبات التخصص الرئيسي):

List of Automotive Mechatronics Engineering Program Requirements

Code	Course Title	Weight			Contact Hours			
		CH	ECTS	SWL	LEC	Tut	Lab	TT
--	University Requirements	12	16	400	12	0	0	12
--	Faculty Requirements	31	51	1275	21	16	8	45
Basic Mechanical Engineering Requirements courses								
BSC 215	Differential Equations	3	5	125	2	2	0	4
BSC 216	Numerical Analysis	3	5	125	2	2	0	4
ELE 211	Electrical Circuits	3	6	150	2	2	1	5
ELE 212	Electronics	3	5	125	2	2	1	5
ELE 214	Measurements and Sensors	3	5	125	2	1	2	5
ELE 273	Logic Design	3	5	125	2	0	2	4
ELE 372	Computer Programming	3	5	125	2	0	2	4
ELE 251	Signals and Systems Fundamentals	3	5	125	2	2	0	4
ELE 353	Automatic Control	3	5	125	2	2	1	5
ELE 361	Embedded Systems	3	5	125	2	0	2	4
MEE 215	Theory of Machines	3	5	125	2	0	2	4
MEE 223	Thermodynamics	3	5	125	2	2	0	4
MEE 214	Strength of Materials	3	5	125	2	1	1	4
MEE 216	Computer Aided Design	2	4	100	1	0	2	3
MEE 221	Fluid Mechanics	3	5	125	2	2	0	4
MEE 325	Mechanical Vibrations	3	5	125	2	2	0	4
MEE 351	Automotive Engineering	3	5	125	2	2	0	4
MEE 313	Manufacturing Technology	3	5	125	2	1	2	5
MEE 327	Vehicle Dynamics	3	5	125	2	2	0	4
ELE 463	Mechatronic Systems	3	5	125	2	2	1	5
AME Requirements Courses								
MEE 326	Heat Transfer	3	5	125	2	1	1	4
MEE 361	Internal Combustion Engines	3	5	125	2	1	2	5
MEE 331	Electric Vehicles Technology	3	5	125	2	2	0	4
MEE 452	Automotive Mechatronics Systems	3	5	125	2	2	0	4
MEE 362	Automotive Engine and Fuel Systems	3	5	125	2	0	2	4
MEE 453	Automotive Maintenance Engineering	3	5	125	2	0	2	4
MEE 428	Vehicle Design and Simulation	3	6	150	2	1	2	5
MEE 432	Performance of Electric and Hybrid Vehicles	3	5	125	2	2	0	4
--	AME Elective (1)	3	5	125	2	2	0	4
--	AME Elective (2)	3	5	125	2	2	0	4
--	AME Elective (3)	3	5	125	2	2	0	4
--	AME Elective (4)	3	5	125	2	2	0	4
--	AME Elective (5)	3	5	125	2	2	0	4
MEE 492	AME Graduation Project 1	2	4	100	1	0	2	3
MEE 494	AME Graduation Project 2	2	3	75	0	0	4	4
MEE x96	AME Practical Training 1	0	0	0	0	0	0	0
MEE x98	AME Practical Training 2	0	0	0	0	0	0	0
Total		145	240	6000	99	62	42	203

Code	Course Title	Weight			Contact Hours			
		CH	ECTS	SWL	LEC	Tut	Lab	TT
Pool of AME Elective (1) and (2)								
MEE 424	Stress Analysis	3	5	125	2	2	0	4
MEE 466	Race Car Technology	3	5	125	2	2	0	4
ELE 426	Automotive Power Electronics	3	5	125	2	2	0	4
MEE 465	Automotive Pollution Control	3	5	125	2	2	0	4
ELE 458	Automotive Control Systems	3	5	125	2	1	1	4
Pool of AME Elective (3), (4), (5)								
MEE 456	Automotive Air Conditioning	3	5	125	2	2	0	4
MEE 433	Automotive Embedded Systems	3	5	125	2	0	2	4
MEE 434	Intelligent Vehicle Technology	3	5	125	2	2	0	4
MEE 457	Selected Topics in Automotive Mechatronic Engineering	3	5	125	2	2	0	4
MEE 455	Automotive Body Design and Analysis	3	5	125	2	2	0	4
MEE 467	Automotive Powertrain Systems	3	5	125	2	0	2	4
MEE 455	Automotive Safety Systems	3	5	125	2	2	0	4
MEE 463	Alternative Fuels and Emissions Control Systems	3	5	125	2	2	0	4
ELE 469	Automotive Communication Systems	3	5	125	2	2	0	4
MEE 464	Fuel Cell Technology	3	5	125	2	2	0	4
MEE 435	Introduction to Autonomous Vehicle	3	5	125	2	2	0	4

6. مصفوفة ربط المقررات الدراسية بجدارات البرنامج

يوضح الجدول التالي مصفوفة ربط مقررات متطلب البرنامج بالجدارات الخاصة بالعلوم الهندسية (Level A) والجدارات الخاصة بالبرنامج (Level C and Level B):

Program and Specialization Requirements Course-Competency Matrix

Code	Course	Eng. Competences (Level A)										Level B				Level C				
		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	B1	B2	B3	B4	C1	C2	C3	C4	C5
Basic Mechanical Engineering Requirements courses																				
BSC 215	Differential Equations	■	■																	
BSC 216	Numerical Analysis	■	■																	
ELE 211	Electrical Circuits		■	■																
ELE 212	Electronics		■																	
ELE 214	Measurements and Sensors	■	■		■															
ELE 273	Logic Design			■			■												■	
ELE 372	Computer Programming						■			■			■							
ELE 251	Signals and Systems Fundamentals	■	■																	
ELE 353	Automatic Control		■									■								
ELE 361	Embedded Systems						■										■			
MEE 215	Theory of Machines	■		■								■								
MEE 223	Thermodynamics	■	■	■								■			■					
MEE 214	Strength of Materials			■	■						■	■								
MEE 216	Computer Aided Design				■							■	■							
MEE 221	Fluid Mechanics	■	■									■								
MEE 325	Mechanical Vibrations		■	■								■								
MEE 351	Automotive Engineering	■										■				■				
MEE 313	Manufacturing Technology			■									■	■						
MEE 327	Vehicle Dynamics	■										■								
ELE 463	Mechatronic Systems						■						■	■	■					

Code	Course	Eng. Competences (Level A)										Level B				Level C				
		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	B1	B2	B3	B4	C1	C2	C3	C4	C5
AME Requirements Courses																				
MEE 326	Heat Transfer																			
MEE 361	Internal Combustion Engines																			
MEE 331	Electric Vehicles Technology																			
MEE 452	Automotive Mechatronics Systems																			
MEE 362	Automotive Engine and Fuel Systems																			
MEE 453	Automotive Maintenance Engineering																			
MEE 428	Vehicle Design and Simulation																			
MEE 432	Performance of Electric and Hybrid Vehicles																			
MEE 492	AME Graduation Project 1																			
MEE 494	AME Graduation Project 2																			
MEE x96	AME Practical Training 1																			
MEE x98	AME Practical Training 2																			
AME Elective (1) and (2)																				
MEE 424	Stress Analysis																			
MEE 456	Automotive Air Conditioning																			
ELE 426	Automotive Power Electronics																			
ELE 458	Automotive Control Systems																			
MEE 465	Automotive Pollution Control																			

Code	Course	Eng. Competences (Level A)										Level B				Level C				
		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	B1	B2	B3	B4	C1	C2	C3	C4	C5
AME Elective (3), (4), (5)																				
MEE 455	Automotive Body Design and Analysis																			
MEE 433	Automotive Embedded Systems																			
MEE 434	Intelligent Vehicle Technology																			
MEE 457	Selected Topics in Automotive Mechatronic Engineering	Based on the topic																		
MEE 466	Race Car Technology																			
MEE 467	Automotive Powertrain Systems																			
MEE 455	Automotive Safety Systems																			
MEE 463	Alternative Fuels and Emissions Control Systems																			
ELE 469	Automotive Communication Systems																			
MEE 464	Fuel Cell Technology																			
MEE 435	Introduction to Autonomous Vehicle																			

7. الخطة الدراسية المقترحة

يوضح الجدول التالي النموذج الاسترشادى للخطة الدراسية للطالب المنتظم:

Suggested Study Plan

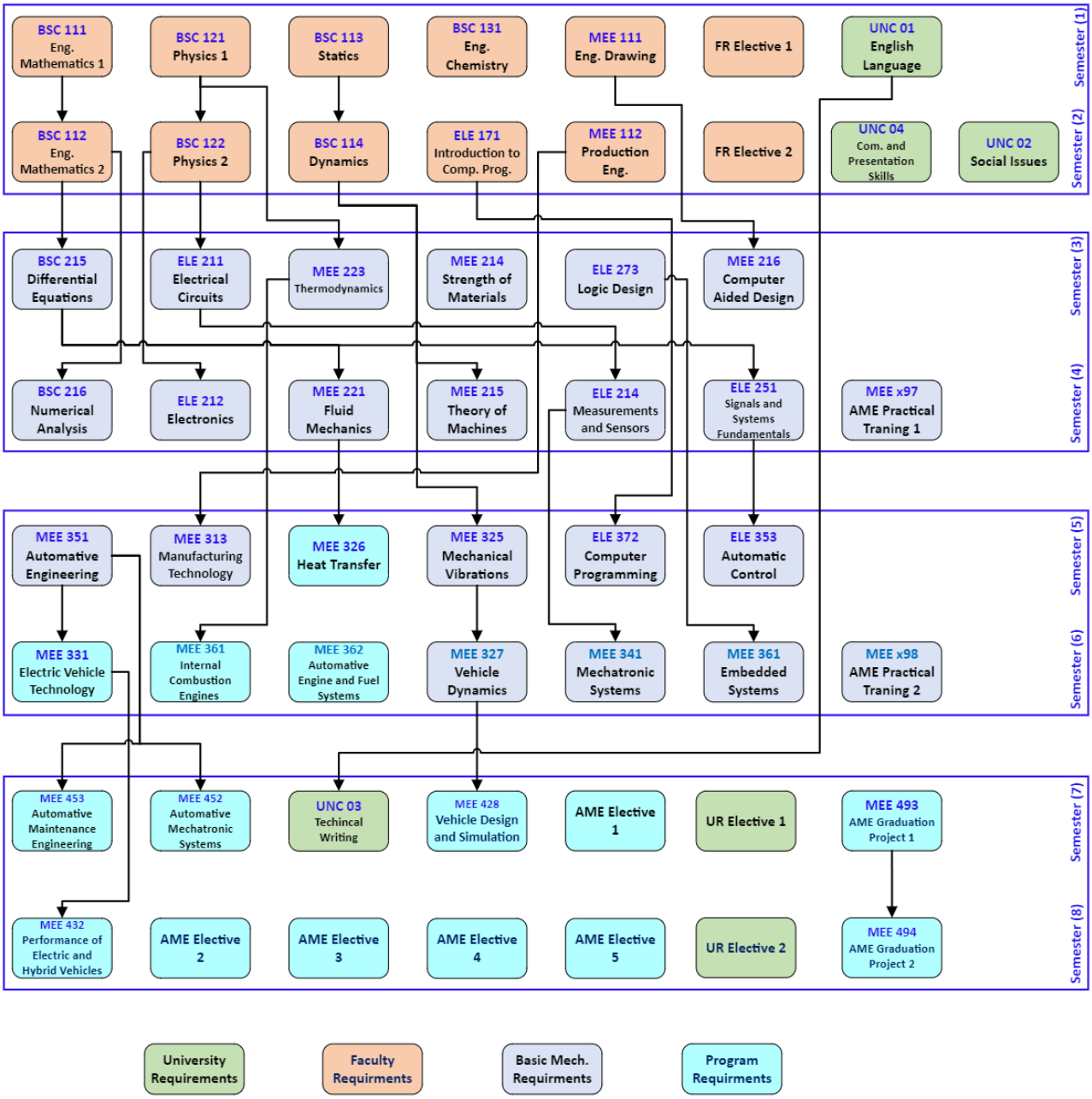
Code	Course Title	Weight			Contact Hours				Pre-requisites
		CH	ECTS	SWL	LEC	Tut	Lab	TT	
Semester (1)									
BSC 111	Engineering Mathematics 1	3	5	125	2	2	0	4	--
BSC 121	Physics 1	3	5	125	2	2	1	5	--
BSC 113	Statics	3	5	125	2	2	0	4	--
BSC 131	Engineering Chemistry	3	5	125	2	1	2	5	--
MEE 111	Engineering Drawing	2	4	100	1	2	0	3	--
--	FR Elective 1	2	3	75	2	0	0	2	--
UNC01	English Language	2	3	75	2	0	0	2	--
Total		18	30	750	13	9	3	25	
Semester (2)									
BSC 112	Engineering Mathematics 2	3	5	125	2	2	0	4	BSC 111
ELE 171	Introduction to Computer Programming	2	3	75	1	0	2	3	--
BSC 122	Physics 2	3	5	125	2	2	1	5	BSC 121
BSC 114	Dynamics	3	5	125	2	2	0	4	BSC 113
MEE 112	Production Engineering	2	3	75	1	0	2	3	--
--	FR Elective 2	2	3	75	2	1	0	3	--
UNC04	Communication and Presentation Skills	2	3	75	2	0	0	2	--
UNC02	Social Issues	2	3	75	2	0	0	2	--
Total		19	30	750	14	7	5	26	
Semester (3)									
BSC 215	Differential Equations	3	5	125	2	2	0	4	BSC 112
ELE 211	Electrical Circuits	3	6	150	2	2	1	5	BSC 122
MEE 223	Thermodynamics	3	5	125	2	2	0	4	BSC 121
MEE 214	Strength of Materials	3	5	125	2	1	1	4	--
ELE 273	Logic Design	3	5	125	2	0	2	4	--
MEE 216	Computer Aided Design	2	4	100	1	0	2	3	MEE 111
Total		17	30	750	11	7	6	24	
Semester (4)									
BSC 216	Numerical Analysis	3	5	125	2	2	0	4	BSC 112
ELE 212	Electronics	3	5	125	2	2	1	5	BSC 122
MEE 221	Fluid Mechanics	3	5	125	2	2	0	4	BSC 215
MEE 215	Theory of Machines	3	5	125	2	0	2	4	BSC 114
ELE 214	Measurements and Sensors	3	5	125	2	1	2	5	ELE 211
ELE 251	Signals and Systems Fundamentals	3	5	125	2	2	0	4	BSC 215
Total		18	30	750	12	9	5	26	

Semester (5)									
ELE 353	Automatic Control	3	5	125	2	2	1	5	ELE 251
ELE 372	Computer Programming	3	5	125	2	0	2	4	ELE 171
MEE 325	Mechanical Vibrations	3	5	125	2	2	0	4	BSC 114
MEE 351	Automotive Engineering	3	5	125	2	2	0	4	--
MEE 313	Manufacturing Technology	3	5	125	2	1	2	5	MEE 112
MEE 326	Heat Transfer	3	5	125	2	1	1	4	MEE 221
Total		**18**	**30**	**750**	**12**	**9**	**5**	**26**	
Semester (6)									
MEE 327	Vehicle Dynamics	3	5	125	2	2	0	4	MEE 325
ELE 463	Mechatronic Systems	3	5	125	2	2	1	5	ELE 214
ELE 361	Embedded Systems	3	5	125	2	0	2	4	ELE 273
MEE 361	Internal Combustion Engines	3	5	125	2	1	2	5	MEE 223
MEE 362	Automotive Engine and Fuel Systems	3	5	125	2	0	2	4	--
MEE 331	Electric Vehicles Technology	3	5	125	2	2	0	4	MEE 351
Total		**18**	**30**	**750**	**12**	**7**	**5**	**24**	
Semester (7)									
MEE 452	Automotive Mechatronics Systems	3	5	125	2	2	0	4	MEE 351
MEE 453	Automotive Maintenance Engineering	3	5	125	2	0	2	4	MEE 351
MEE 428	Vehicle Design and Simulation	3	6	150	2	1	2	5	MEE 327
--	AME Elective (1)	3	5	125	2	2	0	4	--
UNC03	Technical Writing	2	3	75	2	0	0	2	UNC01
--	UR Elective 1	2	2	50	2	0	0	2	--
MEE 492	AME Graduation Project 1	2	4	100	1	0	2	3	--
Total		**18**	**30**	**750**	**13**	**4**	**8**	**25**	
Semester (8)									
MEE 432	Performance of Electric and Hybrid Vehicles	3	5	125	2	2	0	4	MEE 331
--	AME Elective (2)	3	5	125	2	2	0	4	--
--	AME Elective (3)	3	5	125	2	2	0	4	--
--	AME Elective (4)	3	5	125	2	2	0	4	--
--	AME Elective (5)	3	5	125	2	2	0	4	--
--	UR Elective 2	2	2	50	2	0	0	2	--
MEE 492	AME Graduation Project 2	2	3	75	0	0	4	4	MEE 492
Total		**19**	**30**	**750**	**12**	**10**	**4**	**24**	

8. شجرة المقررات الدراسية

يوضح الشكل التالي نموذج شجرة المقررات الدراسية للبرنامج:

Course Map for Automative Mechatronic Engineering Program



9. توصيف المقررات الدراسية لبرنامج هندسة ميكاترونيات السيارات

Code: BSC 215		Subject: Differential Equations		Pre. Req.: BSC 112
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Definition and classification of differential equations- First-Order Ordinary Differential Equations- Second-Order Ordinary Differential Equations- Higher-Order Ordinary Differential Equations- Partial Differential Equations- Laplace transform and its applications -Numerical Methods for Differential Equations- Engineering applications

References:

S. A. Wirkus, and R. J. Swifi, "A Course of Ordinary Differential Equations", Taylor & Francis Group, LLC, 2015.

Code: BSC 216		Subject: Numerical Analysis		Pre. Req.: BSC 112
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Numerical Analysis: Sources of error in numerical computation, Round-off error and truncation error, Accuracy, and convergence of numerical algorithms. **Numerical Differentiation and Integration:** Finite difference approximations, Richardson extrapolation, Newton-Cotes formulas: Trapezoidal rule, Simpson's rule, Gaussian quadrature. **Difference equations:** Definition and classification of difference equations, First-order and higher-order difference equations, Solutions of difference equations: homogeneous and nonhomogeneous cases. **Z-Transform. Fourier analysis:** Fourier series representation of periodic functions, Fourier transform and inverse Fourier transform. **Engineering applications**.

References:

[1] Burden, R. L. (2011). Numerical analysis. Brooks/Cole Cengage Learning.
[2] Hamming, R. W. (2012). Introduction to applied numerical analysis. Courier Corporation.

Code: ELE 211		Subject: Electrical circuits		Pre. Req.: BSC 122
Credit Hours = 4 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
3	2	1	50 %	50%

Course Syllabus:

Elements of electrical circuits - Simple resistive circuits - Analysis of DC circuits - Theories of electrical circuits – Capacitors and Inductors - First-Order Circuits - AC fundamentals -AC Circuits Sinusoidal Steady - State Analysis - Three-Phase Circuits- Power and power factor- Resonance circuits - Three-phase circuits.

References:

[1] Charles K. Alexander and Matthew N. O. Sadiku, Fundamentals of Electric Circuits, McGraw-Hill, Fourth Edition, 2009.
[2] J. W. Nilsson, "Electric Circuits", Pearson; 11th edition, 2018.

Code: ELE 212		Subject: Electronics		Pre. Req.: BSC 122
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Semiconductors: Overview of PN junctions, their biasing, and types of diodes. Bipolar Junction Transistors (BJTs): properties, and applications in DC circuits. Field-Effect Transistors (JFET & MOSFET): properties and applications. Introduction to CMOS technology and its applications.

References:

[1] Thomas L. Floyd David L. Buchla “Electronics Fundamentals Circuits, Devices and Applications” , Eighth Edition, PEARSON, 2014.
[2] T. Floyd, "Electronic Devices", 10th edition, Pearson, 2018

Code: ELE 214		Subject: Measurements and Sensors		Pre. Req.: ELE 211
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50%

Course Syllabus:

Introduction to measurement systems – Instrument types and performance characteristics -Measurement errors – Galvanometers - Analog Ammeters, Voltmeters, and Ohmmeters – measurement errors- Bridges- Filters- Sensor performance criteria and selection - Thermocouples - Resistive sensors – Inductive sensors - Capacitive sensors - Piezoelectric sensors - Encoders and tachometers - A/D and D/A converters

References:

[1] A. Morris, "Measurement and Instrumentation Theory and Application", Academic Press; 2nd edition, 2015.
[2] C. de Silva," Sensors and Actuators: Engineering System Instrumentation" , CRC Press; 2nd edition, 2015.

Code: ELE 273		Subject: Logic Design		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50%

Course Syllabus:

Introduction to logic circuits - Logic circuit simplification using Boolean algebra, DeMorgan theorem, and K-Map. Combinational logic circuits like adders, subtractors, decoders, encoders, multiplexers, and de-multiplexers. Sequential logic circuits like Flip-Flops, counters (synchronous and asynchronous), and registers. The course also explores design methodologies, programmable logic devices, and gate arrays. Hands-on practice with state tables, state transition diagrams, and counter design is included.

References:

M Morris, M., & Michael, D. (2013). Digital Design With an Introduction to the Verilog HDL FIFTH EDITION

Code: ELE 372		Subject: Computer Programming		Pre. Req.: ELE 171
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50%

Course Syllabus:

Introduction to advanced programming – Advanced Data Structures-Advanced Sorting and Searching Algorithms-Dynamic Programming- Graph Algorithms- Introduction to Object-oriented programming — Functional Programming- Presenting these topics using C++ and Python.

References:

[1[Mark Lutz, “Learning Python”, O’ Reilly 5th edition, 2013
[2] Sommerville, software engineering, 10 ed., Pearson India 2018

Code: ELE 251		Subject: Signals and Systems Fundamentals		Pre. Req.: BSC 215
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Definition of signals and systems- Classification of signals- Properties of signals - System classification: (linear vs. nonlinear, time-invariant vs. time-varying, causal vs. non-causal) - Time-Domain Analysis of Continuous-Time Signals and Systems- Frequency Domain Analysis of Continuous-Time Signals and Systems- Sampling and Discrete-Time Signals- Frequency-Domain Analysis of Discrete-Time Signals and Systems.

References:

[1] Phillips, C. L., Parr, J. M., Riskin, E. A., & Prabhakar, T. (2003). Signals, systems, and transforms (p. 209). Upper Saddle River: Prentice Hall.
[2] B. Boashash , "Time-Frequency Signal Analysis and Processing", Academic Press; 2nd edition, 2015.

Code: ELE 353		Subject: Automatic Control		Pre. Req.: ELE 252
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Introduction to systems and their classifications: open loop, and closed loop systems – Block diagrams models- signal flow graphs- Time domain analysis – Frequency domain analysis -Stability of linear systems – Root locus – Classical control techniques (PID control, lead-lag compensation) –-PID controllers design and tuning –Experiments and simulations using control system software (e.g., MATLAB/Simulink). Design and analysis of control systems through practical exercises and projects.

References:

[1] Kuo, B. C., & Golnaraghi, M. F. (1995). Automatic control systems (Vol. 8). Englewood Cliffs, NJ: Prentice Hall.
[2] Franklin, Gene F., J. David Powell, Abbas Emami-Naeini, and J. David Powell. Feedback control of dynamic systems. Vol. 4. Upper Saddle River: Prentice hall, 2002.
[3] Dorf, Modern Control Systems, 12th edition, Prentice Hall, 2012.

Code: ELE 361		Subject: Embedded Systems		Pre. Req.: ELE 273
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50%

Course Syllabus:

Introduction to microcontroller - Overview of the Microcontroller families – Addressing modes- Embedded C programming language (fundamentals of Embedded C) –General purpose I/O -Timer modules- Interrupt standard mechanism- analog to digital converter (ADC) and pulse width modulations (PWM)- Serial Communications.

References:

[1] Mazidi, M. A., Naimi, S., & Naimi, S. (2017). The AVR microcontroller and embedded systems: using Assembly and C. MicroDigitalEd. com.
[2] Kevin M. Lynch, Nicholas Marchuk, Matthew L. Elwin, Embedded Computing and Mechatronics with the PIC32 Microcontroller, Elsevier, 2016.

Code: MEE 215		Subject: Theory of Machines		Pre. Req.: BSC 114
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Identify mechanisms and predict their motion - Calculate the degrees of freedom of mechanisms - Determine the positions, velocities and accelerations of links and points on mechanisms - Calculate dynamic joint forces of mechanisms - Balance simple rotating objects - Analyze and modify existing mechanisms.

References

[1] Skvarenina T. L, ana Dewitt W. E. , “Electrical Power and Controls”,Prentic Hall , London, 2009.
[2] S. K. Bhattacharya, B. Singh, "Control of Machines," revised 2nd edition, New Age International (P) Ltd., 2006.

Code: MEE 223		Subject: Thermodynamics		Pre. Req.: BSC 121
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Heat and work, first law of thermodynamics, properties of steam and gases, steam tables and charts, entropy, second law, air standard cycle, compressors, psychrometry of gas mixtures, reversibility, availability and second law efficiency, gas turbine power cycles, steam turbine power cycles, combined gas-vapor power cycles.

References

[1] Yunus A. Cengel and Michael A. Boles, Thermodynamics: an engineering approach, 8th edition, New York: McGraw-Hill.
[2] Borgnakke, Claus, and Richard E. Sonntag, Fundamentals of Thermodynamics, Wiley Global Education, 9th edition.

Code: MEE 214			Subject: Strength of Materials		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	1	1	50 %	50 %	

Course Syllabus:

This course covers loads on mechanical parts, equilibrium analysis, and types of forces like axial, shear, bending, and torsion. It includes stress, strain, and Hooke's Law, as well as design stress, safety factors, and stress concentration. Topics include thermal stresses, bearing stresses, and shear stresses in beams, along with beam deflection, two-dimensional stress analysis using Mohr's Circle, power transmission shafts, eccentric loads, column buckling, and thin-walled vessels.

References

[1] Beer, Ferdinand, John DeWolf, E. Russell Johnston Jr, and David Mazurek. ("Mechanics of materials." 2014.

Code: MEE 216			Subject: Computer Aided Design		Pre. Req.: MEE 111
Credit Hours = 2 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
1	0	2	50 %	50 %	

Course Syllabus:

The computer-Aided Design (CAD) course covers CAD software fundamentals, including interface navigation, 2D drawing, and geometric construction. It progresses to 3D modeling, assembly modeling, and rendering techniques. Topics include orthographic projections, dimensioning, advanced modeling techniques, and CAD file management. Students will engage in hands-on projects, exploring real-world engineering applications. The course also introduces CAM integration and future CAD trends, providing a comprehensive understanding of CAD in engineering.

References

[1] Randy H. Shih, "Learning SOLIDWORKS 2023: Modeling, Assembly and Analysis", SDC Publications, 2023.
[2] David Planchard, "Engineering Design with SOLIDWORKS", SDC Publications, 2021.

Code: MEE 221			Subject: Fluid Mechanics		Pre. Req.: BSC 215
Credit Hours = 3 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	2	0	50 %	50 %	

Course Syllabus:

Fluid properties – Fluid statics and kinematics – Flow in pipes – Pumps – Valves – Dimensionless analysis and similitude– Buoyancy – Kinematics of three-dimensional fluid motions – Fluid masses moving with acceleration – Vortex motion – Hydrodynamics – Momentum equation - Euler's and Bernoulli's equations.

References

[1] B. R. Munson, D. F. Young and A. T. Okiishi, "Fundamentals of Fluid Mechanics ," 4th edition, John Wiley and Sons, Inc., New York, 2002.
[2] F. M. White, "Fluid Mechanics," 7th Edition, McGraw-Hill, 2011.

Code: 324			Subject: Mechanical Vibrations		Pre. Req.: BSC 114
Credit Hours = 3 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	2	0	50 %	50 %	

Course Syllabus:

Reciprocating motion - Free vibration - Forced vibration with hormonic forces - Transient vibration - Vibration analysis for two degrees of freedom - Vibration characteristics of systems with multiple degrees of freedom - Modelling systems using Lagrange equation - Vibration measurement and analysis - Diagnosis of machine errors using vibration analysis - Vibration control.

References

[1] Daniel J. Inman, "Engineering Vibration", 4th Edition, published by Pearson, 2014.
[2] Michel Geradin, Daniel J. Rixen," Mechanical Vibrations: Theory and Application to Structural Dynamics" published by Wiley, 2014.
[3]. S. Graham Kelly," Mechanical vibrations: theory and applications" published by Cengage Learning 2012.

Code: MEE 351		Subject: Automotive Engineering		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Tires (Tire forces and moments, tire rolling resistance, Tractive effort, Tire radii, Cornering properties) - Ground vehicle dynamics (Road resistance, Traction and tractive effort, performance curves, equations of motion, Dynamic certificate, acceleration and time with distance, grades and overtaking) - Braking Systems (brake system requirements, antilock braking system theory)

References

[1] Wong, J.Y., 2008. Theory of ground vehicles. John Wiley & Sons.
[2] Abe, M., 2015. Vehicle handling dynamics: theory and application. Butterworth-Heinemann.
[3] Reza N. Jazar, 2017. Vehicle dynamics: Theory and application. Springer.

Code: MEE 313		Subject: Manufacturing Technology		Pre. Req.: MEE 112
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50 %

Course Syllabus:

Advanced Forming Technologies (Thread rolling, explosive, electromagnetic forming processes, etc.) - Non-Conventional Machining Processes (ECM, EDM, WEDM, WJM, AWJM, etc.) - CNC technology (machine construction; machine preparation, tool setting, control system, CNC programming) - Additive Manufacturing Technology (Contact and Non-Contact 3D Scanning, AM Techniques, etc.).

References

[1] S. Kalpakjiam Manufacturing Engineering and Technology 7th Edition Pearson Prentice Hall 2017.
[2] Andrew Y. C. Nee Handbook of Manufacturing Engineering and Technology.

Code: MEE 327			Subject: Vehicle Dynamics		Pre. Req.: MEE 325
Credit Hours = 3 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	2	0	50 %	50 %	

Course Syllabus:

Steering Systems (Steering mechanisms, equivalent steering gear ratio, design, steering of dependent and independent suspension) - Concepts of neutral, oversteer and understeer vehicles - vehicle directional stability - Vehicle comfort criteria: sources of vibration inputs - International standards limit of human response to in-vehicle vibrations.

References

[1] Wong, J.Y., 2008. Theory of ground vehicles. John Wiley & Sons.
[2] Abe, M., 2015. Vehicle handling dynamics: theory and application. Butterworth-Heinemann.

Code: ELE 463			Subject: Mechatronic Systems		Pre. Req.: ELE 214
Credit Hours = 3 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	2	1	50 %	50 %	

Course Syllabus:

Introduction to Mechatronics - Mechanisms for motion electronic components for mechatronic systems –Transmission - Mechanical actuator systems (Mechanical elements - Kinematic chains - Cam mechanisms - Gears - Flexible mechanical elements - Brakes) - Electrical actuation systems – Case study and computer simulation.

References

[1] J. Paolo Davim, "Mechatronics," Wiley-ISTE, April 2011.

Code: MEE 424			Subject: Stress Analysis		Pre. Req.: MEE 214
Credit Hours = 3 hrs.			Assessments		
Lecture	Exercises	Practical	Final Exam	Coursework	
2	1	1	50 %	50 %	

Course Syllabus:

Stress relation - Stress and strain transformation in 2D and 3D - Principal stresses and strains - Mohr's circle for 2D and 3D stresses and strains - Stress and strain analysis using finite element programs - Stress and strain measuring apparatus - Electrical resistance strain gauges - Photoelasticity methods - Moire method - Coating methods.

References

M. Hetenyi, "Handbook of Experimental Stress Analysis," Forgotten Books, United States, 2018.

Code: MEE 361		Subject: Internal Combustion Engines		Pre. Req.: MEE 223
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50 %

Course Syllabus:

The course explores internal combustion engine classification, the Fuel-Air Standard Cycle, and variations. It covers combustion chambers, fuel properties' impact on performance, and topics like diesel and gasoline engine combustion, detonation, friction, and lubrication. It discusses engine operating effects on friction and cooling losses, performance at different speeds, lubricating oil properties, oil testing, filters, cooling systems, and engine thermal cycles.

References

[1] Konrad Reif Diesel Engine Management Systems Springer Fachmedin Wiesbaden, 2014
[2] C. Baumgarten Mixture Formation in Internal Combustion Engines Springer Verleg Berlin Heidburg 2006.

Code: MEE 331		Subject: Electric Vehicles Technology		Pre. Req.: MEE 351
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Electric vehicles (configurations, performance, energy consumption) - Hybrid electric vehicles (architectures of drivetrains, series and parallel designs) - Electric propulsion systems - Power sources and energy storage - Regenerative braking. Fuel cells (operation principles, hybrid electric drivetrain).

References

[1] Chen, Yong. New Energy Vehicle Powertrain Technologies and Applications. Springer Nature, 2023.
[2] Hayes, John G., and G. Abas Goodarzi. "Electric powertrain: energy systems, power electronics and drives for hybrid, electric and fuel cell vehicles." (2018).

Code: MEE 452		Subject: Automotive Mechatronics Systems		Pre. Req.: MEE 351
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Introduction to Mechatronics - Sensors, transducers, and Actuators in automotive systems - Basics of Applied electronics in modern vehicles (signal condition of sensors, drive circuits of actuators) - Automotive systems - smart chassis - X-By-Wire - Electric drive vehicles - Autonomous vehicles and its infrastructure.

References

[1] Nise N. S., 2015. Control Systems Engineering. Wiley, 2015. Bosch, R., 2014.
[2] Bosch Automotive Electrics and Automotive Electronics: Systems and Components. In Networking and Hybrid Drive. Springer Vieweg.
[3] Reif, Konrad. Automotive Mechatronics. Germany: Springer Fachmedien Wiesbaden, 2014.

Code: MEE 362		Subject: Automotive Engine and fuel Systems		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50 %

Course Syllabus:

Introduces the types of fuels and air/fuel ratio requirements of the various fuels used in current automotive engines, along with the construction and operation of fuel delivery systems. Discusses fuel pumps, fuel lines, fuel filters, and storage tanks. Fuel system types include returnable electronic fuel injection, returnless electronic fuel injection, and gasoline direct fuel injection. Also discusses elements of mechanical and electronic automotive emissions.

References

[1] Reif K. 2015. Gasoline engine management. Springer Gabler.
[2] Robert Bosch GmbH. 2004. Diesel engine management Vol. 112 . Brill Academic Publishers.

Code: MEE 453		Subject: Automotive Maintenance Engineering		Pre. Req.: MEE 351
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50 %

Course Syllabus:

Maintenance - Test required to evaluate engine performance - Compression test - Vacuum test - Leakage test - Homogeneity test - Bypass gases test and exhaust gas analysis - Periodic maintenance - Fault diagnosis and testing for Engine systems as mechanical system - Fuel systems components - Ignition systems - Cooling system - Oil lubricated system - Charging, starting systems - Braking systems - Steering systems – Tires - Air conditioning system and refrigeration cycle.

References

[1] Erjavec, J., & Thompson, R. (2014). Automotive technology: a systems approach. Cengage learning
[2] Gilles, T. (2018). Automotive engines: diagnosis, repair, and rebuilding. Cengage Learning.

Code: MEE 428		Subject: Vehicle Design and Simulation		Pre. Req.: MEE 327
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50 %

Course Syllabus:

Vehicle components - Basic ergonomics rules in vehicle design - Static and dynamic Vehicle loads - Design of Suspension System - Design of steering System - Dynamic performance of the designed vehicle systems and components and the effects on the dynamics of the vehicle - Applications on systems design using simulation- and finite-elements- software.

References

[1] Genta, G. and Morello, L., 2019. The automotive chassis: volume 2: system design. Springer Nature.
[2] Barton, D.C. and Fieldhouse, J.D., 2018. Automotive Chassis Engineering (pp. 215-254). Berlin: Springer.

Code: MEE 432		Subject: Performance of Electric and Hybrid Vehicles		Pre. Req.: MEE 331
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Introduction - Concept of Hybridization of the Automobile - HEV Fundamentals - Advanced HEV Architectures and Dynamics of HEV Powertrain - Plug-In Hybrid Electric Vehicles - Special Hybrid Vehicles - Power Electronics in HEVs - Modelling and Simulation of Electric and Hybrid Vehicles.

References

Chris Mi, M. Abul Masrur, Hybrid Electric Vehicles, 2nd Edition , Wiley, 2017

Code: MEE 326		Subject: Heat Transfer		Pre. Req.: MEE 221
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Steady and unsteady conduction with heat generation – Heat transfer for finned and un-finned surfaces – Forced convection through pipes and over flat plate – Flow across tube banks – Heat exchangers – Natural convection – Radiation laws.

References

[1] Cengel, Y. A., Ghajar, A. J., & Kanoglu, M. (2011). Heat and mass transfer: fundamentals and applications. *(No Title)*..
[2] Theodore, L. (2011). *Heat transfer applications for the practicing engineer*. John Wiley & Sons.

Code: MEE 455		Subject: Automotive Body Design and Analysis		Pre. Req.: MEE 424
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Introduction to Automotive Body Design -Fundamentals of Automotive Body Engineering- Vehicle Safety Systems and Crashworthiness Analysis- Aerodynamics in Automotive Body Design- Interior Design Considerations and Ergonomic Principles- Sustainability Aspects in Automotive Body Engineering- Integration of Vehicle Systems and Packaging Constraints- Advanced Manufacturing Techniques in Automotive Body Production.

References

[1] Omar, M. A. (2011). The automotive body manufacturing systems and processes. John Wiley & Sons.
[2] Gillespie, T. (Ed.). (2021). Fundamentals of vehicle dynamics. SAE international.
[3] Stone, R., & Ball, J. K. (2004). Automotive engineering fundamentals. SAE International.

Code: ELE 426		Subject: Automotive Power Electronics		Pre. Req.: ELE 212
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Automotive power semiconductor devices - Automotive power electronic converters (DC-DC Converters - AC-DC Rectifiers - DC/AC Inverters - AC/AC Converters) - Power Electronics and Control for Hybrid and Fuel Cell Vehicles - Automotive motor drives - Power Electronics Applications in Vehicle.

References

Ali Emadi, Handbook of Automotive Power Electronics and Motor Drives, CRC Press, 2005.

Code: MEE 456		Subject: Automotive Air Conditioning		Pre. Req.: MEE 326
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50

Course Syllabus:

Introduction to Automotive Air Conditioning Systems - Air-conditioning fundamentals - Air-conditioning components - Air-conditioning electrical and electronic control - Diagnostics and troubleshooting - Environmental Regulations and Refrigerant Management.

References:

Steven Daly, Automotive Air Conditioning and Climate Control Systems, Elsevier, 2006.

Code: MEE 465		Subject: Automotive Pollution Control		Pre. Req.: MEE 361
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Introduction - Pollutant Formation in Engines - Control of Emissions from Engines - Noise Pollution from Automobiles - Measurement Techniques Emission Standards and Test Procedure.

References

[1] Paul Degobert – Automobiles and Pollution – SAE International ISBN-1-56091-563- 3, 1991.
[2] Ganesan, V- "Internal Combustion Engines"- Tata McGraw-Hill Co.- 2003.

Code: ELE 458		Subject: Automotive Control Systems		Pre. Req.: ELE 455
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50 %

Course Syllabus:

State-space (controller design, controllability, observer design, observability)- Digital control systems (z-transform, stability, transient response on the z-plane, cascade compensation via the s-plane) - Applications to automotive case studies.

References

[1] Dorf, R.C. and Bishop, R.H., 2011. Modern control systems. Pearson. Ellis, G., 2002.
[2] Observers in control systems: a practical guide. Elsevier.

Code: MEE 433		Subject: Automotive Embedded Systems		Pre. Req.: ELE 361
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50 %

Course Syllabus:

Introduction to Automative embedded system – Automative embedded applications - Automotive product development life cycle - Building Blocks of Automotive Product Interfacing and Drivers for a Microcontroller - Use of CAN & LIN Automotive protocol - Vehicle Control - Car Communication based on Bluetooth Link - Applied Automotive Projects.

References

[1] Francoise Simonot-Lion, Nicolas Navet, "Automotive Embedded Systems Handbook", CRC Press, 2017
[2] Richard Zurawski, "Embedded Systems Handbook: Networked Embedded Systems", CRC Press, 2017

Code: MEE 434		Subject: Intelligent Vehicle Technology		Pre. Req.: MEE 452
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Introduction to Intelligent Vehicle Technology- Sensors and Perception Systems- Vehicle-to-Vehicle (V2V) Communication Systems- Vehicle-to-Infrastructure (V2I) Communication Systems- Advanced Driver Assistance Systems -Artificial Intelligence and Machine Learning in Intelligent Vehicles- Future Trends and Challenges in Intelligent Vehicle Technology.

References

[1] Ghosh, S., & Lee, T. S. (2010). Intelligent Transportation systems: smart and green infrastructure design. CRC press.
[2] Ljubo Vlacic, Michel Parent and Fumio Harashima, Intelligent Vehicle Technologies: Theory and Applications, Elseiver, 2001.

Code: MEE 457		Subject: Selected Topics in Automotive Mechatronic Engineering		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

This course is designed for specialized and current topics in Automotive Mechatronic Engineering. Course topics are chosen so they do not duplicate other courses in the program. The course needs the program approval.

References Based on the topic.

Code: MEE 466		Subject: Race Car Technology		Pre. Req.: MEE 351
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Race cars (requirements, design constraints, and specification, tire combined slip characteristics, applied aerodynamics) Ride and roll rates – suspension – dynamics - steering and braking - practice and testing - Race track events and performance evaluation.

References

[1] Pütz, R. and Serné, T., 2022. Race Car Handling Optimization: Magic Numbers to Better Understand a Race Car. Springer Nature.
[2] Seward, D., 2017. Race car design. Bloomsbury Publishing.

Code: MEE 467		Subject: Automotive Powertrain Systems		Pre. Req.: MEE 351
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50 %

Course Syllabus:

Vehicle Powertrain Concepts - Power Generation Characteristics - Vehicle Longitudinal Dynamics - different transmission types design - Standard tests of vehicles and vehicle subsystems (Brakes, Engine, Suspension and other systems) - Society of Automotive Engineering (SAE) testing standards for different systems and different types of vehicles.

References: Hilier V. A. W. (2006) Hillier's Fundamentals of Motor Vehicle Technology: Powertrain Electronics, Fifth Edition

Code: MEE 455		Subject: Automotive Safety Systems		Pre. Req.: MEE 327
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Active and passive safety - Vehicle structures for crash worthiness - Ergonomics in Automotive safety - Vehicle Safety Systems - Types of Crash and Roll-Over - Momentum conservation - Energy conservation - Types of Impact - Coefficient of Restitution - Accident Reconstruction Techniques - Accidents simulation methods and software - Safety Regulations - Active safety systems.

References

[1] Struble, D.E., 2013. Automotive accident reconstruction: practices and principles. CRC Press.
[2] Jazar, R.N., 2017. Vehicle dynamics: theory and application. Springer.

Code: MEE 463		Subject: Alternative Fuels and Emissions Control Systems		Pre. Req.: MEE 362
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Types of alternative fuels (Biofuel – Ethanol – Electricity - Natural Gas – Hydrogen – Propane – Methanol) - Modification to engines for operation on alternative fuels - Fuel alternatives emissions and methods of reduction - Effect of alternative fuels on engine service life - Sources of pollution - Exhaust gas - Types of pollution -Effect of engine performance characteristics on pollutants levels in both diesel and petrol engines - Catalytic Converters - PCV Valve - EGR Valve.

References

[1] Ferguson, Colin R. Internal Combustion Engines. John Wiley and Sons, 2015
[2] Faiz, Asif, Christopher S. Weaver, and S. C. Samuel. Air Pollution from Motor Vehicles: Standards and Technologies for Controlling Emissions. World Bank Publication, 2000

Code: ELE 469		Subject: Automotive Communication Systems		Pre. Req.: MEE 351
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

The course explores communication technologies in modern vehicles, offering a deep understanding of automotive communication systems, including essential principles, protocols, and applications. Students will study various technologies such as wired protocols (Controller Area Network [CAN] bus, Local Interconnect Network [LIN] bus, and Flex Ray) and emerging technologies like automotive Ethernet, crucial for advancing automotive technology.

References

Matheus, K., & Königseder, T. (2021). *Automotive ethernet*. Cambridge University Press.

Code: MEE 464		Subject: Fuel Cell Technology		Pre. Req.: MEE 362
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Introduction - History of Fuel Cells - Fuel Cell Vs. Internal Combustion Engine – Thermodynamics - PEM Fuel Cell Components and Performance Analysis - The Fueling Problem - Fuel Cell Systems - Fuel Cell Catalysis - Prospects of the DMFC – Applications - Stationary Power Generation - Portable Systems - Fuel Cell Vehicles - Competing Technologies for Transportation - Energy Efficiency Of Fuel Cell Systems - The Current Market Situation.

References

Gregor Hoogers, Fuel Cell Technology Handbook, CRC Press, 2003.

Code: MEE 435		Subject: Introduction to Autonomous Vehicle		Pre. Req.: MEE 327
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Introduction to Autonomous Driving - Autonomous Vehicle Localization - Perception in Autonomous Driving - Deep Learning in Autonomous Driving Perception - Prediction and Routing - Decision, Planning, and Control

References

Shaoshan Liu, etc. Creating Autonomous Vehicle Systems, Morgan & Claypool, 2020.

Code: MEE 492		Subject: AME Graduation Project 1		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
1	0	2	-	-

Course Syllabus:

An independent project under the supervision of a staff member; to enable the student to understand and apply the knowledge gained throughout his coursework to an engineering problem at large scale.

Code: MEE 494		Subject: AME Graduation Project 2		Pre. Req.: MEE 492
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
0	0	4	-	--

Course Syllabus:

This project is an extension to graduation project 1. In this part the student under the supervision of his supervisor may make a prototype or introduce an idea to be implemented in the future. The student should submit a report.

الباب الثامن

برنامج هندسة الطاقة المتجددة والمستدامة

Renewable and Sustainable Energy Engineering Program

اسم البرنامج	هندسة الطاقة المتجددة والمستدامة Renewable and Sustainable Energy Engineering (RSE)			
الكلية	الهندسة			
الجامعة	جامعة الريادة للعلوم والتكنولوجيا			
الساعات المعتمدة (CH)	145			
النقاط المعتمدة (ECTS)	240			
عدد المقررات	55 مقرر			
عدد ساعات الاتصال	محاضرات	تمارين	معمل	الإجمالي
	100	73	29	202
توزيع المقررات طبقا لمتطلبات مستويات التعلم				
	CH	ECTS	النسب	المقررات
متطلبات الجامعة	12	16	% 8	6
متطلبات الكلية	31	51	% 21	12
متطلبات البرنامج والتخصص الرئيسي	102	173	% 71	37
بيانات إحصائية				
	CH	ECTS	النسب	المقررات
نسبة مقررات العلوم الاساسية	27	45	19%	9
نسبة المقررات الاختيارية	23	35	% 16	9

1. رسالة البرنامج

يسعى برنامج هندسة الطاقة المتجددة والمستدامة إلى تخريج مهندسين يمتلكون المعرفة ويتقنون المهارات المتخصصة المطلوبة في مجال هندسة الطاقة الجديدة والمستدامة، وقادرين على المنافسة فى سوق العمل المحلى والإقليمى مع التصدي بفعالية للتحديات الملحة مثل تغير المناخ وأمن الطاقة والاستدامة البيئية.

2. أهداف البرنامج

يعمل برنامج هندسة الطاقة المتجددة والمستدامة على تقديم تعليم متميز المستوى بما يتماشى مع المعايير الدولية والاقليمية في مجال هندسة الطاقة المتجددة والمستدامة مع تشجيع أعضاء هيئة التدريس والطلاب على المشاركة بنشاط في خدمة المجتمع وتنمية البيئة. يمكن تلخيص اهداف البرنامج فيما يلي:

1. تطبيق المعرفة بالرياضيات الهندسية والعلوم الطبيعية واساسيات الهندسة وعلوم الحاسب وقدرات التفكير التحليلي والنقدي والمنهجي لتحديد وحل المشكلات الهندسية.
2. تزويد الخريجين بالمهارات اللازمة للتعامل مع أنظمة الطاقة الكهربائية، مع التركيز على أنظمة الطاقة المتجددة والمستدامة.
3. تعزيز أداء الخريجين من خلال إكسابهم المفاهيم المتقدمة والمهارات المميزة في أساسيات الطاقة المتجددة والمستدامة ليكونوا قادرين على المنافسة في الأسواق المحلية والإقليمية والعالمية.
4. غرس القيم المهنية والأخلاقية في نفوس الخريجين وتمكينهم من أن يصبحوا قادة في مختلف قطاعات صناعة الطاقة.
5. تعزيز ثقافة التعلم مدى الحياة، وتمكين الخريجين من تطوير معارفهم ومهاراتهم بشكل مستمر من خلال التعلم الذاتي المستمر.
6. إعداد الخريجين للدراسات العليا والبحث العلمي من خلال تعزيز التفكير الإبداعي، وتحليل المشكلات، ومهارات حل المشكلات بشكل منهجي.

3. مواصفات الخريج

يحرص برنامج هندسة الطاقة المتجددة والمستدامة على تخريج مهندسين متميزين ومؤهلين لدخول سوق العمل ويراعى فيهم تحقيق المواصفات الأتية:

1. Master a wide spectrum of engineering knowledge and specialized skills and can apply acquired knowledge using theories and abstract thinking in real life situations.
2. Apply analytic critical and systemic thinking to identify, diagnose and solve engineering problems with a wide range of complexity and variation.
3. Behave professionally and adhere to engineering ethics and standards.
4. Work in and lead a heterogeneous team of professionals from different engineering specialties and assume responsibility for own and team performance.
5. Recognize his/her role in promoting the engineering field and contribute to the development of the profession and the community.
6. Value the importance of the environment, both physical and natural, and work to promote sustainability principles.
7. Use techniques, skills, and modern engineering tools necessary for engineering practice.
8. Assume full responsibility for own learning and self-development, engage in lifelong learning and demonstrate the capacity to engage in post- graduate and research studies.
9. Communicate effectively using different modes, tools, and languages with various audiences; to deal with academic/professional challenges in a critical and creative manner.
10. Demonstrate leadership qualities, business administration and entrepreneurial skills.
11. Integrate knowledge of renewable energy sciences with various engineering disciplines.
12. Develop models and designs for integrated energy systems that incorporate multiple disciplines.

13. Design, implement, operate, and maintain renewable energy facilities, while also conducting specialized research and studies within the energy sector.

4. المعايير الأكاديمية للبرنامج

بالإضافة الى الجدارات لمعظم البرامج الهندسية (Level A) فأن برنامج هندسة الطاقة المتجددة والمستدامة له بعض الجدارات الخاصة وهي كما يلي:

D1. Evaluate and apply principles of thermodynamics, heat transfer, fluid mechanics, control theory, and mechanical design to select, model, and analyze renewable energy systems tailored to specific disciplines.

D2. Design, model, and analyze electrical, electronic, mechanical, or digital components for renewable energy applications, utilizing appropriate optimization tools.

D3. Assess and measure the performance of electrical, electronic, mechanical, or digital systems and circuits under defined input conditions to determine their suitability for renewable energy utilization.

D4. Analyze, model, and select electrical power systems relevant to renewable and sustainable energy applications, integrating principles of power generation, transmission, and distribution.

D5. Incorporate relevant national and international standards and codes into the design, construction, operation, inspection, and maintenance of electrical, electronic, mechanical, or digital equipment and systems for renewable energy applications.

5. المقررات الدراسية

للحصول على درجة بكالوريوس الهندسة في تخصص هندسة الطاقة المتجددة والمستدامة ولتحقيق مخرجات التعلم المستهدفة للبرنامج وللتوافق مع مواصفات وجدارات الخريج الموصفة من قبل هيئة ضمان الجودة والاعتماد فإنه يتعين على الطالب أن يجتاز بنجاح المقررات التالية والمقسمة الى (متطلبات الجامعة – متطلبات الكلية – متطلبات البرنامج والتخصص):

List of Renewable and Sustainable Energy Engineering Program Requirements

Code	Course Title	Weight			Contact Hours			
		CH	ECTS	SWL	LEC	Tut	Lab	TT
--	University Requirements	12	16	400	12	0	0	12
--	Faculty Requirements	31	51	1275	21	16	8	45
Program and specialization Requirements								
BSC 215	Differential Equations	3	5	125	2	2	0	4
BSC 216	Numerical Analysis	3	5	125	2	2	0	4
ELE 211	Electrical Circuits	3	6	150	2	2	1	5
ELE 212	Electronics	3	5	125	2	2	1	5
MEE 223	Thermodynamics	3	5	125	2	2	0	4
ELE 215	Electromagnetic Fields	3	6	150	2	2	0	4
MEE 221	Fluid Mechanics	3	5	125	2	2	0	4
ELE 231	Electrical Power Engineering	3	5	125	2	2	0	4
ELE 214	Measurement and Sensors	3	5	125	2	1	2	5
ELE 251	Principles of Energy Conversion	2	4	100	2	0	0	2
ELE 222	Electric Machines (1)	3	6	150	2	1	2	5
ELE 323	Electric Machines (2)	3	5	125	2	1	2	5
ELE 332	Power System Analysis	3	5	125	2	2	0	4
ELE 342	Fundamental of PV	3	5	125	2	1	1	4
ELE 252	Modeling of Dynamic Systems	3	5	125	2	2	0	4
ELE 388	Energy Efficiency in Buildings	3	5	125	2	2	0	4
ELE 337	Electrical Distribution System Installation	3	5	125	2	1	1	4
ELE 353	Automatic Control	3	5	125	2	2	1	5
ELE 343	Solar Energy	3	5	125	2	2	0	4
ELE 333	Power System Protection	3	5	125	2	1	1	4
ELE 327	Power Electronics and Electric Drives	3	5	125	2	2	1	5
MEE 326	Heat transfer	3	5	125	2	2	0	4
ELE 344	Fundamentals of Wind Energy	3	5	125	2	2	0	4
ELE 445	Energy Storage Systems	3	5	125	2	2	0	4
MEE 447	Refrigeration and Air Conditioning	3	4	100	2	2	0	4
ELE 447	Introduction to Biomass Energy	3	5	125	2	2	0	4
ELE 434	Power System Control and Operation	3	5	125	2	2	0	4
ELE 482	Power Plants	3	5	125	2	1	2	5
--	RSE Elective (1)	3	5	125	2	2	0	4
--	RSE Elective (2)	3	5	125	2	2	0	4
--	RSE Elective (3)	3	5	125	2	2	0	4
--	RSE Elective (4)	3	5	125	2	2	0	4
--	RSE Elective (5)	3	5	125	2	2	0	4
ELE 491	RSE Graduation Project 1	2	4	100	1	0	2	3
ELE 492	RSE Graduation Project 2	2	3	75	0	0	4	4
ELE x93	RSE Practical Training 1	0	0	0	0	0	0	0
ELE x94	RSE Practical Training 2	0	0	0	0	0	0	0
Total		145	240	6000	100	73	29	202

Code	Course Title	Weight			Contact Hours			
		CH	ECTS	SWL	LEC	Tut	Lab	TT
Pool of RSE Elective (1) and (2)								
ELE 362	Programmable Logic Controllers	3	5	125	1	0	3	4
ELE 435	Electrical Power Quality	3	5	125	2	2	0	4
ELE 448	Fundamentals of Electric and Hybrid Vehicles	3	5	125	2	2	0	4
ELE 484	Illumination Technology	3	5	125	2	2	0	4
Pool of RSE Elective (3), (4), (5)								
ELE 436	SCADA in Power System	3	5	125	2	0	2	4
ELE 449	Hybrid Renewable Energy Systems	3	5	125	2	2	0	4
ELE 485	Smart Grid Technologies	3	5	125	2	2	0	4
ELE 483	Climate Change and Energy	3	5	125	2	2	0	4
ELE 486	Water and Wastewater Treatment Technologies	3	5	125	2	2	0	4
ELE 487	Energy Markets and Regulation	3	5	125	2	2	0	4
MEE 446	Selected Topics in Energy Generation Engineering	3	5	125	2	2	0	4
RSE 446	Tidal and Wave Energy Engineering	3	5	125	2	2	0	4

6. مصفوفة ربط المقررات الدراسية بجدارات البرنامج

يوضح الجدول التالي مصفوفة ربط مقررات متطلب البرنامج بالجدارات الخاصة بالعلوم الهندسية (Level A) والجدارات الخاصة بالبرنامج (Level D):

Program and Specialization Requirements Course-Competency Matrix

Code	Course	Eng. Competences (Level A)										RSE Competences (Level D)				
		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	D1	D2	D3	D4	D5
BSC 215	Differential Equations	■	■													
BSC 216	Numerical Analysis	■	■													
ELE 211	Electrical Circuits	■	■	■												
ELE 212	Electronics	■	■										■			
MEE 223	Thermodynamics	■	■	■								■				
ELE 215	Electromagnetic Fields	■	■													
MEE 221	Fluid Mechanics	■	■									■				
ELE 231	Electrical Power Engineering	■														■
ELE 214	Measurement and Sensors	■	■		■											
ELE 251	Principles of Energy Conversion											■	■		■	
BSC 215	Electric Machines (1)	■											■			
BSC 216	Electric Machines (2)	■											■			
ELE 332	Power System Analysis												■		■	
ELE 342	Fundamental of PV												■		■	
ELE 252	Modeling of Dynamic Systems		■				■							■		
ELE 388	Energy Efficiency in Buildings	■		■							■					
ELE 337	Electrical Distribution System Installation											■		■		■
ELE 353	Automatic Control	■		■								■				
ELE 343	Solar Energy											■			■	
ELE 333	Power System Protection				■								■			■
ELE 327	Power Electronics and Electric Drives			■									■		■	
MEE 326	Heat Transfer			■	■							■				
ELE 344	Fundamentals of Wind Energy	■										■	■			

Code	Course	Eng. Competences (Level A)										RSE Competences (Level D)				
		A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	D1	D2	D3	D4	D5
ELE 445	Energy Storage Systems											■	■			■
MEE 447	Refrigeration and Air Conditioning				■							■		■		
ELE 447	Introduction to Biomass Energy											■	■			
ELE 434	Power System Control and Operation											■		■	■	■
ELE 482	Power Plants											■		■	■	■
ELE x93	RSE Practical Training 1					■	■	■	■		■	■	■	■	■	■
ELE x94	RSE Practical Training 2					■	■	■	■		■	■	■	■	■	■
ELE 491	RSE Graduation Project 1							■		■	■	■	■	■	■	■
ELE 492	RSE Graduation Project 2							■		■	■	■	■	■	■	■
Elective Courses (1 and 2)																
ELE 362	Programmable Logic Controllers		■									■	■			
ELE 435	Electrical Power Quality												■	■		■
ELE 483	Climate Change and Energy			■												■
ELE 484	Illumination Technology			■									■			■
Elective Courses (3, 4, and 5)																
ELE 436	SCADA in Power System											■	■	■		
ELE 449	Hybrid Renewable Energy Systems											■	■	■	■	■
ELE 485	Smart Grid Technologies											■	■	■	■	■
ELE 448	Fundamentals of Electric and Hybrid Vehicles											■	■	■	■	
ELE 486	Water and Wastewater Treatment Technologies	■		■	■							■				
ELE 487	Energy Markets and Regulation			■												■
MEE 446	Selected Topics in Energy Generation Eng.	Based on Topic														
RSE 446	Tidal and Wave Energy Engineering											■	■			

7. الخطة الدراسية المقترحة

يوضح الجدول التالي النموذج الاسترشادى للخطة الدراسية للطالب المنتظم:

Suggested Study Plan

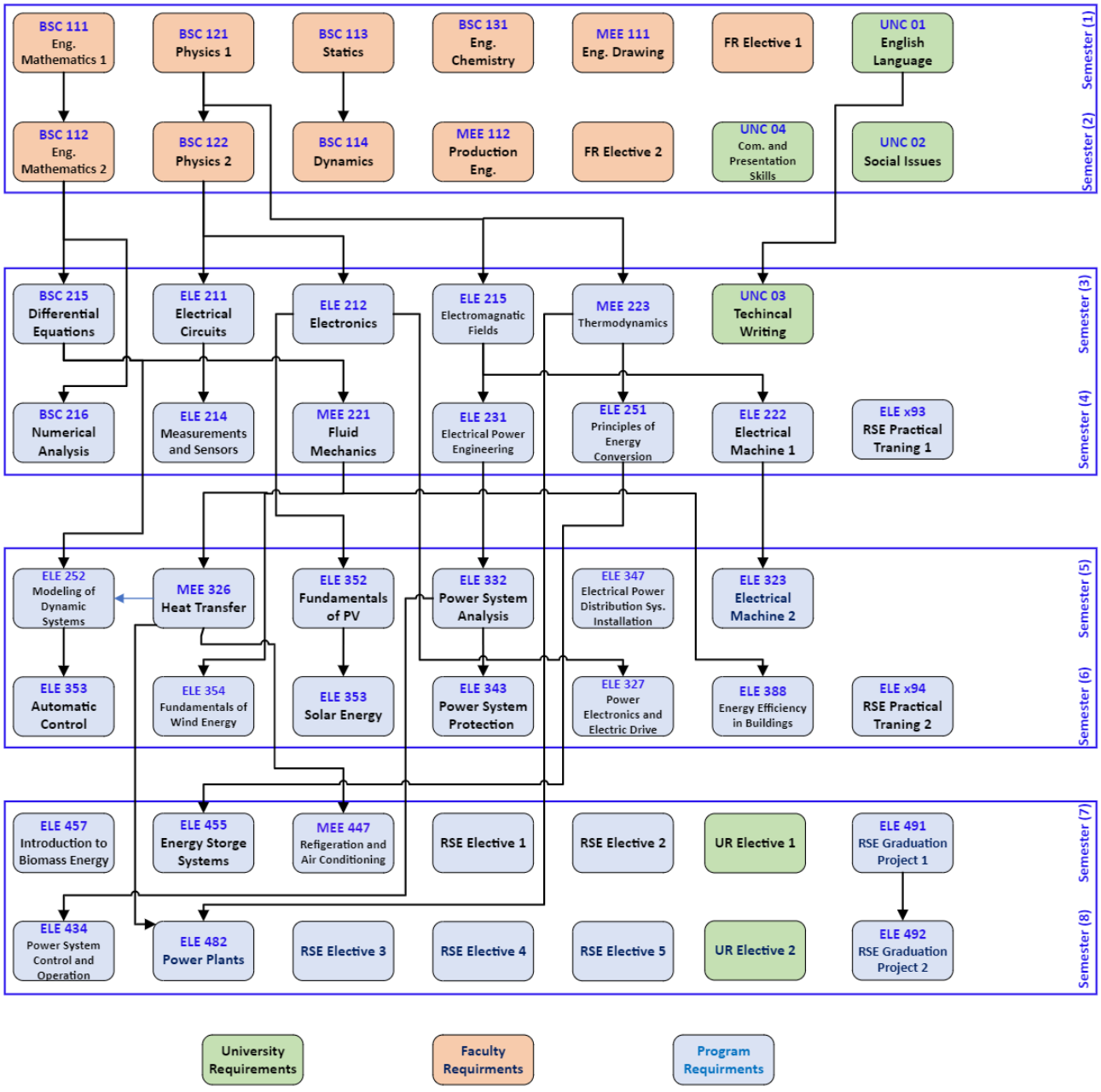
Code	Course Title	Weight			Contact Hours				Pre-requisites
		CH	ECTS	SWL	LEC	Tut	Lab	TT	
Semester (1)									
BSC 111	Engineering Mathematics 1	3	5	125	2	2	0	4	--
BSC 121	Physics 1	3	5	125	2	2	1	5	--
BSC 113	Statics	3	5	125	2	2	0	4	--
BSC 131	Engineering Chemistry	3	5	125	2	1	2	5	--
MEE 111	Engineering Drawing	2	4	100	1	2	0	3	--
--	FR Elective 1	2	3	75	2	0	0	2	--
UNC01	English Language	2	3	75	2	0	0	2	--
Total		18	30	750	13	9	3	25	
Semester (2)									
BSC 112	Engineering Mathematics 2	3	5	125	2	2	0	4	BSC 111
ELE 171	Introduction to Computer Programming	2	3	75	1	0	2	3	--
BSC 122	Physics 2	3	5	125	2	2	1	5	BSC 121
BSC 114	Dynamics	3	5	125	2	2	0	4	BSC 113
MEE 112	Production Engineering	2	3	75	1	0	2	3	--
--	FR Elective 2	2	3	75	2	1	0	3	--
UNC04	Communication and Presentation Skills	2	3	75	2	0	0	2	--
UNC02	Social Issues	2	3	75	2	0	0	2	--
Total		19	30	750	14	7	5	26	
Semester (3)									
BSC 215	Differential Equations	3	5	125	2	2	0	4	BSC 112
ELE 211	Electrical Circuits	3	6	150	2	2	1	5	BSC 122
ELE 212	Electronics	3	5	125	2	2	1	5	BSC 122
MEE 223	Thermodynamics	3	5	125	2	2	0	4	BSC 121
ELE 215	Electromagnetic Fields	3	6	150	2	2	0	4	BSC 121
UNC03	Technical Writing	2	3	75	2	0	0	2	UNC01
Total		17	30	750	12	10	2	24	
Semester (4)									
BSC 216	Numerical Analysis	3	5	125	2	2	0	4	BSC 112
MEE 221	Fluid Mechanics	3	5	125	2	2	0	4	BSC 215
ELE 231	Electrical Power Engineering	3	5	125	2	2	0	4	ELE 215
ELE 214	Measurement and Sensors	3	5	125	2	1	2	5	ELE 211
ELE 251	Principles of Energy Conversion	2	4	100	2	0	0	2	MEE 223
ELE 222	Electric Machines (1)	3	6	150	2	1	2	5	ELE 215
Total		17	30	750	12	8	4	24	

Semester (5)									
ELE 323	Electric Machines (2)	3	5	125	2	1	2	5	ELE 222
ELE 332	Power System Analysis	3	5	125	2	2	0	4	ELE 231
ELE 342	Fundamental of PV	3	5	125	2	1	1	4	ELE 212
ELE 252	Modeling of Dynamic Systems	3	5	125	2	2	0	4	BSC 215
MEE 326	Heat Transfer	3	5	125	2	2	0	4	MEE 221
ELE 337	Electrical Distribution System Installation	3	5	125	2	1	1	4	--
Total		**18**	**30**	**750**	**12**	**9**	**4**	**25**	
Semester (6)									
ELE 353	Automatic Control	3	5	125	2	2	1	5	ELE 252
ELE 343	Solar Energy	3	5	125	2	2	0	4	ELE 342
ELE 333	Power System Protection	3	5	125	2	1	1	4	ELE 332
ELE 327	Power Electronics and Electric Drives	3	5	125	2	2	1	5	ELE 212
ELE 388	Energy Efficiency in Buildings	3	5	125	2	2	0	4	ELE 251
ELE 344	Fundamentals of Wind Energy	3	5	125	2	2	0	4	MEE 221
Total		**18**	**30**	**750**	**12**	**11**	**3**	**26**	
Semester (7)									
ELE 445	Energy Storage Systems	3	5	125	2	2	0	4	ELE 251
MEE 447	Refrigeration and Air Conditioning	3	4	100	2	2	0	4	MEE 326
ELE 447	Introduction to Biomass Energy	3	5	125	2	2	0	4	--
--	RSE Elective (1)	3	5	125	2	2	0	4	--
--	RSE Elective (2)	3	5	125	2	2	0	4	--
--	UR Elective 1	2	2	50	2	0	0	2	--
ELE 491	RSE Graduation Project 1	2	4	100	1	0	2	3	--
Total		**19**	**30**	**750**	**13**	**10**	**2**	**25**	
Semester (8)									
ELE 434	Power System Control and Operation	3	5	125	2	2	0	4	ELE 332
ELE 482	Power Plants	3	5	125	2	1	2	5	MEE 223 - MEE 326
--	RSE Elective (3)	3	5	125	2	2	0	4	--
--	RSE Elective (4)	3	5	125	2	2	0	4	--
--	RSE Elective (5)	3	5	125	2	2	0	4	--
--	UR Elective 2	2	2	50	2	0	0	2	--
ELE 492	RSE Graduation Project 2	2	3	75	0	0	4	4	ELE 491
Total		**19**	**30**	**750**	**12**	**9**	**6**	**27**	

8. شجرة المقررات الدراسية

يوضح الشكل التالي نموذج شجرة المقررات الدراسية للبرنامج:

Course Map for ReneWable and Sustainable Energy Engineering Program



9. توصيف المقررات الدراسية لبرنامج هندسة الطاقة المتجددة والمستدامة

Code: BSC 215		Subject: Differential Equations		Pre. Req.: BSC 112
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Definition and classification of differential equations- First-Order Ordinary Differential Equations- Second-Order Ordinary Differential Equations- Higher-Order Ordinary Differential Equations- Partial Differential Equations- Laplace transform and its applications -Numerical Methods for Differential Equations- Engineering applications

References:

S. A. Wirkus, and R. J. Swifi, "A Course of Ordinary Differential Equations", Taylor & Francis Group, LLC, 2015.

Code: BSC 216		Subject: Numerical Analysis		Pre. Req.: BSC 112
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Numerical Analysis: Sources of error in numerical computation, Round-off error and truncation error, Accuracy, and convergence of numerical algorithms. **Numerical Differentiation and Integration:** Finite difference approximations, Richardson extrapolation, Newton-Cotes formulas: Trapezoidal rule, Simpson's rule, Gaussian quadrature. **Difference equations:** Definition and classification of difference equations, First-order and higher-order difference equations, Solutions of difference equations: homogeneous and nonhomogeneous cases. **Z-Transform. Fourier analysis:** Fourier series representation of periodic functions, Fourier transform and inverse Fourier transform. **Engineering applications**.

References:

[1] Burden, R. L. Numerical analysis. Brooks/Cole Cengage Learning, 2011
[2] Hamming, R. W. Introduction to applied numerical analysis. Courier Corporation, 2012.

Code: ELE 211		Subject: Electrical Circuits		Pre. Req.: BSC 122
Credit Hours = 4 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
3	2	1	50 %	50%

Course Syllabus:

Elements of electrical circuits - Simple resistive circuits - Analysis of DC circuits - Theories of electrical circuits – Capacitors and Inductors - First-Order Circuits - AC fundamentals -AC Circuits Sinusoidal Steady - State Analysis - Three-Phase Circuits- Power and power factor- Resonance circuits - Three-phase circuits.

References:

[1] Charles K. Alexander and Matthew N. O. Sadiku, Fundamentals of Electric Circuits, McGraw-Hill, Fourth Edition, 2009.
[2] J. W. Nilsson, "Electric Circuits", Pearson; 11th edition, 2018.

Code: ELE 212		Subject: Electronics		Pre. Req.: BSC 122
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Semiconductors: Overview of PN junctions, their biasing, and types of diodes. Bipolar Junction Transistors (BJTs): properties, and applications in DC circuits. Field-Effect Transistors (JFET & MOSFET): properties and applications. Introduction to CMOS technology and its applications.

References:

[1] Thomas L. Floyd David L. Buchla “Electronics Fundamentals Circuits, Devices and Applications” , Eighth Edition, PEARSON, 2014.
[2] T. Floyd, "Electronic Devices", 10th edition, Pearson, 2018

Code: ELE 388		Subject: Energy Efficiency in Buildings		Pre. Req.: ELE 251
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

The course covers energy efficiency in buildings, focusing on lighting, heat and moisture transport, and hygrothermal properties of materials. It includes U-value and thermal capacity calculations, heat loss estimation, insulation practices, and thermal comfort. Topics also cover HVAC systems, renewable energy integration, ventilation, air quality, and airtightness. Principles of bioclimatic design and energy performance certification are also addressed to ensure comprehensive understanding and application of sustainable building practices.

References

Sweeney, James L. *Energy efficiency: building a clean, Secure Economy*. Hoover Press, 2016.

Code: MEE 221		Subject: Fluid Mechanics		Pre. Req.: BSC 215
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Fluid properties – Fluid statics and kinematics – Flow in pipes – Pumps – Valves – Dimensionless analysis and similitude– Buoyancy – Kinematics of three-dimensional fluid motions – Fluid masses moving with acceleration – Vortex motion – Hydrodynamics – Momentum equation - Euler's and Bernoulli's equations.

References

[1] B. R. Munson, D. F. Young and A. T. Okiishi, "Fundamentals of Fluid Mechanics ," 4th edition, John Wiley and Sons, Inc., New York, 2002.
[2] F. M. White, "Fluid Mechanics," 7th Edition, McGraw-Hill, 2011.

Code: MEE 223		Subject: Thermodynamics		Pre. Req.: BSC 121
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Heat and work, first law of thermodynamics, properties of steam and gases, steam tables and charts, entropy, second law, air standard cycle, compressors, psychrometry of gas mixtures, reversibility, availability and second law efficiency, gas turbine power cycles, steam turbine power cycles, combined gas-vapor power cycles.

References

[1] Yunus A. Cengel and Michael A. Boles, Thermodynamics: an engineering approach, 8th edition, New York: McGraw-Hill.
[2] Borgnakke, Claus, and Richard E. Sonntag, Fundamentals of Thermodynamics, Wiley Global Education, 9th edition.

Code: ELE 215		Subject: Electromagnetic Fields		Pre. Req.: BSC 121
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Electrostatic Fields - Electric flux and gauss's law- Electrical Potential – Electric boundary conditions – Dipole moment – Capacitance - Poisson and Laplace equations – Piot & savart law- Amperes law- Vector potentials- Magnetic boundary conditions- Magnetic flux – Force and energy in magnetic field- Coils – Time varying field- Maxwell's equations- Wave propagation – Introduction to transmission line theory- Propagation in losses and lossless media- Reflection and refraction. Electromagnetics in AC electrical machines.

References:

[1] M. N. O. Sadiku, “Elements of Electromagnetics, “ 6th Ed, Oxford Press, 2006.
[2] William H. Hayt, Johan A. Buck, “Engineering Electromagnetics” 8th Ed., McGraw Hill, 2012.

Code: ELE 231		Subject: Electrical Power Engineering		Pre. Req.: ELE 215
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to electric power systems - definition of active and reactive power - transmission lines components - characteristics and performance of transmission lines - Overhead lines - mechanical design of overhead transmission lines - underground cables - design of electrical distribution systems- generation of high voltage- measurement of high voltage - electric insulation types- corona- earthing and safety- Introduction to power system planning.

References:

[1] Sadhu, P.K. and S. Das, Elements of Power Systems. CRC Press, 2015.
[2] Weedy, B.M., et al., Electric power systems. John Wiley & Sons, 2012.

Code: ELE 214		Subject: Measurements and Sensors		Pre. Req.: ELE 211
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50%

Course Syllabus:

Introduction to measurement systems – Instrument types and performance characteristics — Measurement errors – Galvanometers - Analog Ammeters, Voltmeters, and Ohmmeters – measurement errors- Bridges- Filters- Sensor performance criteria and selection - Thermocouples - Resistive sensors - Inductive sensors - Capacitive sensors - Piezoelectric sensors - Encoders and tachometers - A/D and D/A converters

References:

[1] A. Morris, "Measurement and Instrumentation Theory and Application", Academic Press; 2nd edition, 2015.
[2] C. de Silva," Sensors and Actuators: Engineering System Instrumentation" , CRC Press; 2nd edition, 2015.

Code: ELE 222		Subject: Electrical Machines (1)		Pre. Req.: ELE 215
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50%

Course Syllabus:

Basic principles of operation of DC Machines- DC generators- DC Motors - -Single Phase Transformers- Three Phase Transformers-Special Purpose Transformers.

References:

[1] Sen P. C. Principles of Electric Machines and Power Electronics 3rd editions J. Wiley Publishers, 2014.
[2] J. Chapman, S. Electric machinery fundamentals. McGraw-hill, 2014.

Code: ELE 323		Subject: Electrical Machines (2)		Pre. Req.: ELE 222
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50%

Course Syllabus:

Basic Concepts of Rotating Electric Machine-Synchronous Generators- Synchronous Motors-Induction Motors-Three phase Induction Motors- Single-phase induction motors -Induction Generators.

References:

[1] Pyrhonen, J., T. Jokinen, and V. Hrabovcova, Design of rotating electrical machines. John Wiley & Sons, 2013.
[2] J. Chapman, S. Electric machinery fundamentals. McGraw-hill, 2004.

Code: ELE 251		Subject: Principles of Energy Conversion		Pre. Req.: MEE 223
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	0	50 %	50%

Course Syllabus:

Exploration of Energy Conversion Matrix- Renewable Energy (Solar, Wind, Biofuels, Hydro-power) and Traditional Methods (Fossil Fuel, Nuclear Fuel)-Introduction to Direct Thermo-Electric Generators, Peltier Cooling, Photovoltaic Cells, and Fuel Cells for Electrical Energy Production.

References:

[1] D. Yogi Goswami, F. Kreith, Energy Conversion, 2nd Edition, CRC Press, 2017
[2] B. Gupta , A Text book of Energy Conversion System, Dhanpat Rai Publishing Company Ltd, January 2008.

Code: ELE 332		Subject: Power System Analysis		Pre. Req.: ELE 231
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

The basic concepts for system analysis-Power system modelling- Steady state and dynamic behavior of power systems- Network matrices and power flow analysis- Power system fault calculations- Power system stability.

References:

[1] Olle. I. Elgerd, 'Electric Energy Systems Theory – An Introduction', McGraw Hill, 2003.
[2] J. Chapman, S. Electric machinery fundamentals. McGraw-hill, 2004.

Code: ELE 342		Subject: Fundamental of PV		Pre. Req.: ELE 212
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50%

Course Syllabus:

Introduction to Photovoltaics -Semiconductor Physics for PV- Solar Cell Operation- Solar Cell Technologies- PV System Components - grid connected -hybrid solar PV system - large scale PV systems. **PV performance enhancement**

References:

[1] Mertens, Konrad. Photovoltaics: fundamentals, technology, and practice. John Wiley & Sons, 2018.
[2] Wenham, S.R., Green, M.A., Watt, M.E., Corkish, R. and Sproul, A., Applied photovoltaics. Routledge, 2013.

Code: ELE 252		Subject: Modeling of Dynamic Systems		Pre. Req.: BSC 215
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to Dynamic Systems- Electrical Systems Modeling- Mechanical Systems Modeling- Thermal Systems Modeling- Electromechanical systems modeling -Hydraulic and Pneumatic Systems Modeling- State-Space Representation- Nonlinear Systems Modeling- Nonlinear systems linearization- Modeling Tools and Software- Applications of Dynamic System Modeling.

References:

[1] Ogata, Katsuhiko. System dynamics. 2004.

[2] Chaturvedi, D. K. Modeling and simulation of systems using MATLAB and Simulink. CRC press, 2017.

Code: ELE 353		Subject: Automatic Control		Pre. Req.: ELE 252
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Introduction to systems and their classifications: open loop, and closed loop systems – Block diagrams models- signal flow graphs- Time domain analysis – Frequency domain analysis -Stability of linear systems – Root locus – Classical control techniques (PID control, lead-lag compensation) –-PID controllers design and tuning –Experiments and simulations using control system software (e.g., MATLAB/Simulink). Design and analysis of control systems through practical exercises and projects.

References:

[1] Kuo, B. C., & Golnaraghi, M. F. (1995). Automatic control systems (Vol. 8). Englewood Cliffs, NJ: Prentice Hall.
[2] Franklin, Gene F., J. David Powell, Abbas Emami-Naeini, and J. David Powell. Feedback control of dynamic systems. Vol. 4. Upper Saddle River: Prentice hall, 2002.
[3] Dorf, Modern Control Systems, 12th edition, Prentice Hall, 2012.

Code: ELE 343		Subject: Solar Energy		Pre. Req.: ELE 342
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to solar energy systems -Solar radiation principles- Measurement and estimation of solar irradiance and insolation - Solar thermal collectors- Thermal energy storage- Solar heating and cooling systems, Solar thermal power generation-Solar water heating- Solar space heating and cooling- Markets for solar thermal energy- Applications in a range of relevant fields, industrial process heating, solar desalination, and materials processing.

References:

[1] Kalogirou, S. A. Solar energy engineering: processes and systems. Elsevier, 2023.
[2] Duffie, J. A., & Beckman, W. A. Solar engineering of thermal processes. John Wiley & Sons, 2013.

Code: ELE 333		Subject: Power System Protection		Pre. Req.: ELE 332
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50%

Course Syllabus:

Fundamentals of switchgear - Fundamentals of power systems protection - Circuit interrupting devices (fuse, circuit breakers) - Relays and Protection Devices - Overcurrent Protection- Transformer Protection-- Distance protection - Transmission Line Protection- Digital protection - Generator and motor Protections.

References:

[1] Ram, B. Power system protection and switchgear. Tata McGraw-Hill Education, 2011.
[2] Blackburn, J. L., & Domin, T. J. Protective relaying: principles and applications. CRC press, 2006.
[3] Gonen, T. Electrical power transmission system engineering: analysis and design. CRC press, 2015.

Code: ELE 327		Subject: Power Electronics and Electric Drives		Pre. Req.: ELE 212
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	1	50 %	50%

Course Syllabus:

Introduction to Power Electronics- Power Semiconductor Devices-Rectifiers- DC-DC Converters- Inverters and DC-AC Converters - Resonant Converters – Cycloconverters - Power electronics for renewable energy.

References:

[1] Mohan, N., Undeland, T. M., & Robbins, W. P. Power electronics: converters, applications, and design. John wiley & sons, 2003.
[2] Akagi, H., Watanabe, E. H., & Aredes, M. Instantaneous power theory and applications to power conditioning. John Wiley & Sons, 2017.

Code: ELE 337		Subject: Electrical Distribution System Installations		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	1	50 %	50%

Course Syllabus:

Introduction to Electrical Distribution Systems- National and international electrical codes and standards- Lighting Systems- Distribution Panels and Substations- Grounding and Bonding- Distribution system earthing- Voltage drop calculations- Electrical Maintenance and Troubleshooting- Power Factor Corrections- Energy Efficiency and Renewable Energy Integration.

References:

[1] Gonen, T. Electric power distribution engineering. CRC press, 2015.
[2] The Egyptian Code of practice for electrical installations.

Code: MEE 326		Subject: Heat Transfer		Pre. Req.: MEE 221
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Steady and unsteady conduction with heat generation – Heat transfer for finned and un-finned surfaces – Forced convection through pipes and over flat plate – Flow across tube banks – Heat exchangers – Natural convection – Radiation laws.

References

[1] Cengel, Y. A., Ghajar, A. J., & Kanoglu, M. (2011). Heat and mass transfer: fundamentals and applications. *(No Title)*..

[2] Theodore, L. (2011). *Heat transfer applications for the practicing engineer*. John Wiley & Sons.

Code: ELE 344		Subject: Fundamentals of Wind Energy		Code: MEE 221
Credit Hours = 3 hrs.			Assessments	
Exercises	Practical	Exercises	Final Exam	Coursework
2	2	0	50 %	50 %

Course Syllabus:

Overview of wind energy as a renewable energy source- Principles of wind energy conversion: wind speed, direction, and power density- Basic components of a wind turbine - Wind turbine types - Wind turbine aerodynamics- Dynamics and control of Wind turbines - Wind energy conversion systems (WECS)- Power electronics and control systems in wind turbines- Wind Energy Economics - Integration of wind energy system with the electrical grid- Applications to wind energy.

References:

[1] Manwell, J. F., McGowan, J. G., & Rogers, A. L. Wind energy explained: theory, design and application. John Wiley & Sons, 2010.

[2] Tong, W. Wind power generation and wind turbine design. WIT Press, 2010.

[3] Adaramola, M. Ed. Wind turbine technology: Principles and design. CRC Press, 2014.

Code: ELE 445		Subject: Energy Storage Systems		Pre. Req.: ELE 251
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Overview of energy storage technologies-Basic principles of energy storage-Classification of energy storage technologies- Battery Energy Storage Systems (BESS)- Supercapacitors- Thermal Energy Storage-Pumped Hydroelectric Storage-Compressed Air Energy Storage (CAES)- Hydrogen Energy Storage -Flywheel Energy Storage Systems- Integration of the Energy storage systems.

References:

[1] A.G.Ter-Gazarian, "Energy Storage for Power Systems", Second Edition, The Institution of Engineering and Technology (IET) Publication, UK, (ISBN – 978-1-84919-219-4), 2011.
[2] Rufer, A. Energy storage: systems and components. CRC Press, 2017.

Code: MEE 447		Subject: Refrigeration and Air Conditioning		Pre. Req.: MEE 326
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

The course covers refrigeration and air conditioning basics, including refrigerants, the vapor compression cycle, and system components and configurations. It addresses cooling loads for cold stores, equipment performance, and system control. For air conditioning, it explores system types, components, properties of moist air, psychrometric charts, and load calculations. Additionally, it includes maintenance and troubleshooting techniques for both refrigeration and air conditioning systems.

References:

[1] Whitman, B., Johnson, B., Tomczyk, J., & Silberstein, E. Refrigeration and air conditioning technology. Cengage Learning, 2012.
[2] Haines, R. W., & Hittle, D. C. Control systems for heating, ventilating, and air conditioning. Springer Science & Business Media, 2006.

Code: ELE 447		Subject: Introduction to Biomass Energy		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	0	50 %	50%

Course Syllabus:

Introduction to biomass as a renewable energy source - Types of biomass feedstocks - Thermochemical conversion processes- Biochemical conversion processes – Bioenergy production process – Biofuels Production process - Biogas for electricity generation –Modeling and optimization for biogas and Biofuels process- Design and Control of bioenergy and biofuel generation -- Economic, and environmental Impacts.

References:

[1] Clark, J. H., & Deswarte, F. Introduction to chemicals from biomass. John Wiley & Sons, 2014.
[2] Dahiya, A. Bioenergy: Biomass to Biofuels and Waste to Energy. Academic Press, 2020.
[3] Cheng, J. Biomass to renewable energy processes. CRC press, 2017.

Code: ELE 434		Subject: Power System Control and Operation		Pre. Req.: ELE 332
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Types of power plants - Fundamental concepts of economic operation and control of power systems- Power System Modelling- Steady-State Power Flow Analysis - Real-Time Generation Control - Real-Time Monitoring – State Estimation -Load Flow Control- Voltage Stability and Control- Wide-Area Monitoring and Control- Power System Operation and Market.

References:

[1] Grigsby, L. L. Power system stability and control. CRC press, 2007.
[2] Machowski, J., Lubosny, Z., Bialek, J. W., & Bumby, J. R. Power system dynamics: stability and control. John Wiley & Sons, 2020.
[3] Vittal, V., McCalley, J. D., Anderson, P. M., & Fouad, A. A. Power system control and stability. John Wiley & Sons, 2019.

Code: ELE 482		Subject: Power Plants		Pre. Req.: MEE 223 - MEE 326
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	1	2	50 %	50%

Course Syllabus:

Overview of power plants- Steam Power Plants- Gas Turbine Power Plants- Basics of nuclear power plants- Internal Combustion Engine - Power Station Design and Construction- Operation and Maintenance of power plants-Hydro- power plants- Environmental aspects of power plants.

References:

Nag, P. K. Power plant engineering. New Delhi, India: Tata McGraw-Hill, 2008.

Code: ELE 362		Subject: Programmable Logic Controllers		Pre. Req.: ELE 214
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
1	0	3	50 %	50%

Course Syllabus:

Introduction to classical control - Fundamentals to programmable logic controllers (PLCs) - Input-output devices - I/O processing - Ladder and functional block programming - IL, SFC and ST programming methods – Function and function blocks - Jump and call – Timers – Counters - Shift registers - Applications Using PLCs.

References:

[1] Bolton, William. Programmable logic controllers. Newnes, 2015.
[2] Information and training automation and devices, SIMATIC S7, Version 5.6, 2008.

Code: ELE 435		Subject: Electrical Power Quality		Pre. Req.: ELE 327
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to Electrical Power Quality and Reliability- Standards and Regulations for Electrical Power Quality- -Causes and Effects of Power Quality Issues-- Voltage Fluctuations - Active Shunt Compensation - Active Series Compensation- Harmonics in Electrical Power Systems- Passive and Active Power Conditioning Devices- Measurement and Monitoring Techniques for Power Quality Assessment- Analysis and Interpretation of Power Quality Data.

References:

[1] Sankaran, C. Power quality. CRC Press, 2017.
[2] Von Meier, A. Electric power systems: a conceptual introduction. John Wiley & Sons, 2006.

Code: ELE 483		Subject: Climate Change and Energy		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to Climate Change and Environmental Processes- Greenhouse Effect and Global Warming- Energy Production and Climate Change- Climate monitoring and Modeling- Expected and Actual Consequences of Climate Change- Negative Carbon Emissions - Sustainable Development Goals- Actions to Adapt to and Mitigate Climate Change.

References:

[1] Hardy, J. T. Climate change: causes, effects, and solutions. John Wiley & Sons, 2003.
[2] MacKay, D. J. Sustainable Energy-without the hot air. Bloomsbury Publishing, 2016.

Code: ELE 484		Subject: Illumination Technology		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Electric Light Sources and Luminaires- Lighting Controls and Daylighting- Photometry and Calculation Methods- Lighting Codes and Standards- Energy Management in Lighting- Sustainability Principles in Lighting Design- Design Processes and Techniques- Lighting Safety and Maintenance- Advanced lighting technologies.

References:

[1] Robert Karlicek et al., Handbook of Advanced Lighting Technology, Springer, Cham, 2017.
[2] L. Bloch, The Science of Illumination; an Outline of the Principles of Artificial Lighting, HardPress Publishing, 2012.

Code: ELE 436		Subject: SCADA in Power System		Pre. Req.: ELE 362
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	0	2	50 %	50%

Course Syllabus:

Definition and history of Supervisory Control and Data Acquisition (SCADA) systems- SCADA system components and architecture -SCADA Communication Protocols - SCADA System Configuration- Real-Time Data Acquisition and Visualization- SCADA Applications in Power System Monitoring and Control (i.e., Monitoring electrical grids, substations, and generation facilities, Remote control and operation of power system equipment using SCADA systems)- Advanced SCADA Functionalities

References:

[1] Knapp, E. D., & Langill, J. T. Industrial Network Security: Securing critical infrastructure networks for smart grid, SCADA, and other Industrial Control Systems. Syngress, 2014.
[2] Mini S. Thomas, John Douglas McDonald, Power System SCADA and Smart Grids, 1st Edition, CRC Press; 2015.

Code: ELE 449		Subject: Hybrid Renewable Energy Systems		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to Renewable Energy Technologies--Hybridization Concepts and System Components-Control and optimization in Hybrid Systems-Real-world examples of hybrid systems-Challenges and Solutions in Hybrid Systems-Regulatory Framework and Policy Considerations-Maintenance and Operations of Hybrid Systems-Future Trends and Innovations in Hybrid Systems.

References:

[1] Masters, G. M. Renewable and efficient electric power systems. John Wiley & Sons, 2013.
[2] Hossain, Jahangir, Mahmud, Apel, Renewable Energy Integration: Challenges and Solutions, Series: Green Energy and Technology, springer, 2014.

Code: ELE 485		Subject: Smart Grid Technologies		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to Smart Grids and Sustainable Energy Systems- Renewable Energy Integration in Smart Grids -Technical Fundamentals of Sustainable Energy Systems- Smart Metering Infrastructure- Demand Side Management, Demand Response and Demand Pricing- Smart Meters - IT Networks and Security & Telecommunication Systems in Smart Grids- Smart Grid Standards- Economics of Smart Grids- Applications.

References:

[1] Uslar, M., et al., Standardization in smart grids: introduction to IT-related methodologies, architectures and standards, Springer Science & Business Media, 2012.
[2] Gellings, C. W. The smart grid: enabling energy efficiency and demand response. CRC press, 2020.

Code: ELE 448		Subject: Fundamentals of Electric and Hybrid Vehicles		Pre. Req.: ELE 327
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Electric vehicle principles and design: architecture, components, integration- Hybrid vehicle configurations and drivetrain architectures- Battery technology: types, characteristics, integration, cooling techniques- Charging infrastructure: standards, protocols, smart charging solutions- Electric propulsion systems: motors, power electronics, controllers, integration- Power sources and energy storage- - Regenerative braking- Fuel cells in electric and hybrid vehicles- Socio-economic and environmental impacts.

References:

[1] Husain, I. Electric and hybrid vehicles: design fundamentals. CRC Press, 2021.
[2] Mi, C., & Masrur, M. A. Hybrid electric vehicles: principles and applications with practical perspectives. John Wiley & Sons, 2017.

Code: ELE 486		Subject: Water and Wastewater Treatment Technologies		Pre. Req.: BSC 131
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Water Treatment: Raw and Treated Water Quality- Water Treatment Processes- Sedimentation and Filtration- Disinfection and Water Storage- Wastewater Treatment Processes: Wastewater quality parameters and characterization- Preliminary treatment methods - Primary treatment methods - Secondary Treatment and Tertiary Treatment- Sludge Treatment and Disposal- Wastewater and sludge treatment as an energy source.

References:

[1] Davis, M. L. Water and wastewater engineering. McGraw-Hill, 2010.
[2] C. Tchobanoglous, G., Stensel, H. D., Tsuchihashi, R., Burton, F., Abu-Orf, M., Bowden, G., & Pfrang, W. Wastewater engineering: treatment and resources recovery. Metcalf and Eddy Inc, 2014.

Code: ELE 487		Subject: Energy Markets and Regulation		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

Introduction to Energy Markets - Classical Energy Markets- Specialized Renewable Energy Markets- Economics of Renewable Energy- Integration of Renewable Energy-Policy Instruments and Regulations- Institutional Frameworks.

References:

[1] Tom James, Energy Markets: Price Risk Management and Trading, John Wiley & Sons Pte Ltd, 2008
[2] Davis W. Edwards, Energy Trading and Investing: Trading, Risk Management and Structuring Deals in the Energy Market, 1st Edition, McGraw-Hill Education; 2009.
[3] Jones, L. E. Renewable energy integration: practical management of variability, uncertainty, and flexibility in power grids. Academic Press, 2017.

Code: ELE 446		Subject: Selected Topics in Renewable and Sustainable Energy Engineering		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

This course covers recent and emerging topics in renewable and sustainable energy engineering.

Code: RSE 446		Subject: Tidal and Wave Energy Engineering		Pre. Req.: NA
Credit Hours = 3 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
2	2	0	50 %	50%

Course Syllabus:

This course delves into the principles, technologies, and applications of tidal and wave energy as renewable resources for sustainable energy production. Students will explore the fundamentals of tidal energy, including tidal energy principles, analysis of tidal energy resources, and the interplay between tidal and wind energy. Additionally, the course covers tidal energy classification, with a focus on Egypt's tidal energy potential and utilization prospects.

References:

[1] Greaves, D., Iglesias, G., & Wiley, J. Wave and tidal energy. John Wiley & Sons Limited, 2018.
[2] Khare, V., Khare, C., Nema, S., & Baredar, P. Tidal Energy Systems: design, optimization and control. Elsevier, 2018.

Code: ELE 491		Subject: ARE Graduation Project 1		Pre. Req.: NA
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
1	0	2	-	-

Course Syllabus:

An independent project under the supervision of a staff member; to enable the student to understand and apply the knowledge gained throughout his coursework to an engineering problem at large scale.

Code: ELE 492		Subject: ARE Graduation Project 2		Pre. Req.: ELE 491
Credit Hours = 2 hrs.			Assessments	
Lecture	Exercises	Practical	Final Exam	Coursework
0	0	4	-	--

Course Syllabus:

This project is an extension to graduation project 1. In this part the student under the supervision of his supervisor may make a prototype or introduce an idea to be implemented in the future. The student should submit a report.